AGHRYMET RCC-WAS

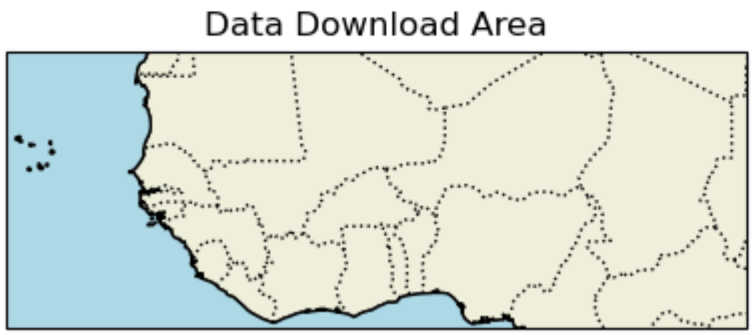
Library

Set forecast working directory

Set Climatological Years

Download Observation and Process

Load Downloader

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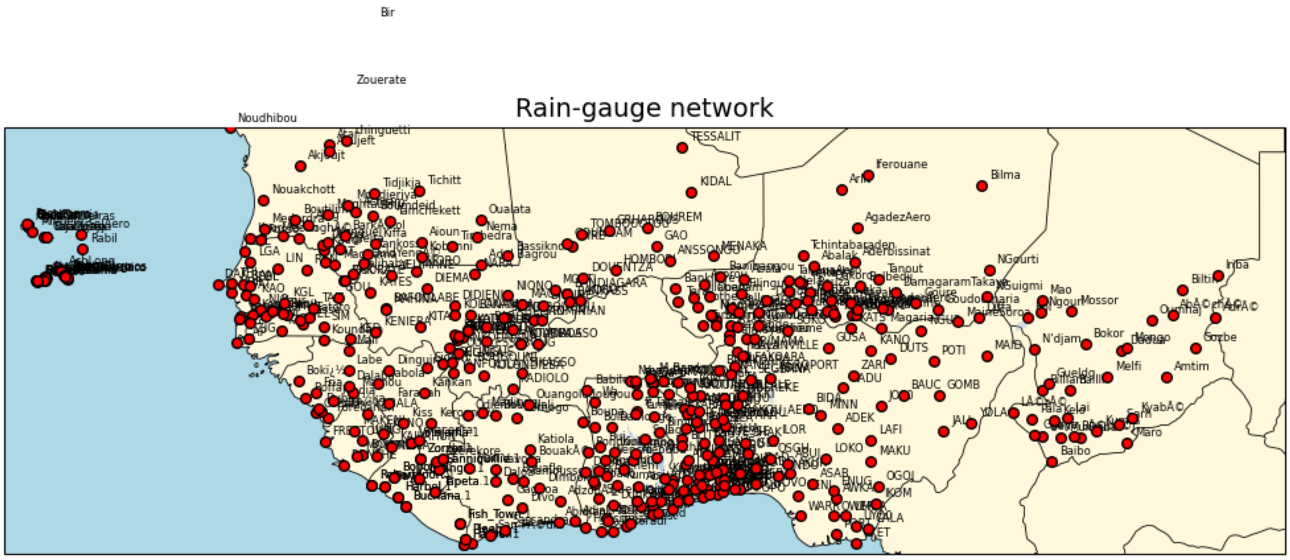
Process Obs in combining with Seasonal cumulative Ground-Observation


Rain-gauge network

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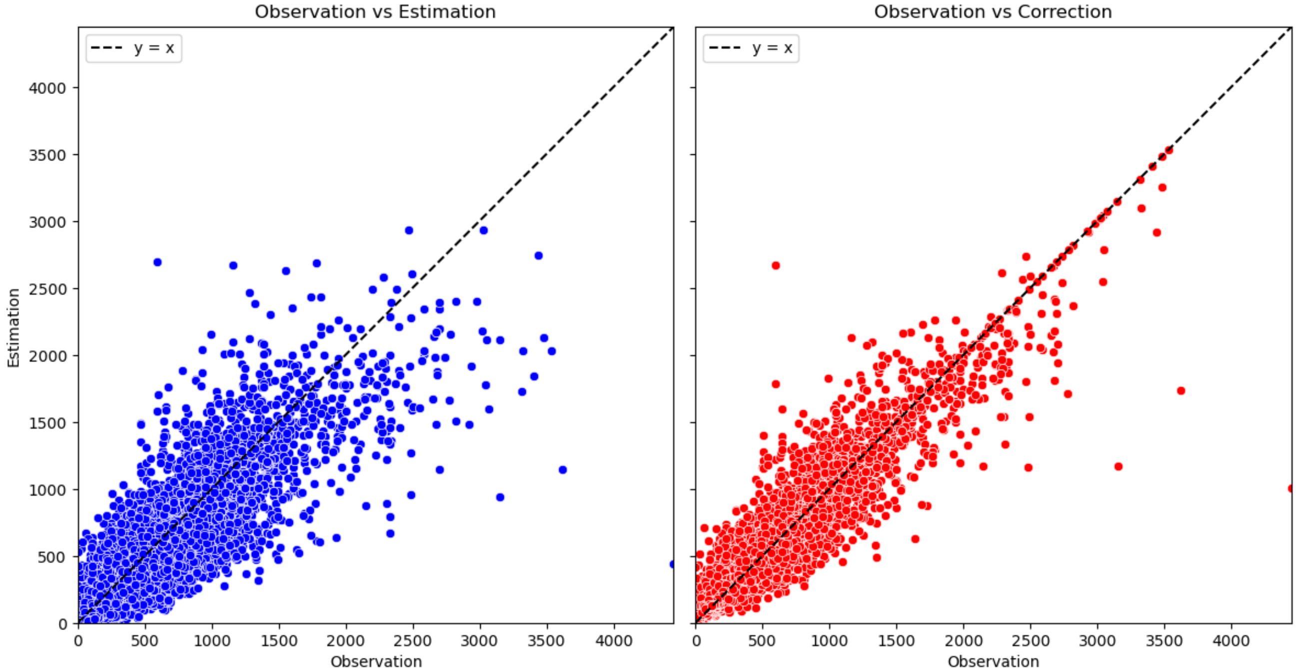

1st - Approach: Choice of Best Dynamical models and weighted Average

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Validation of GCM

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Pearson Correlation
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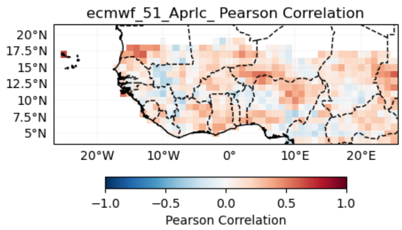
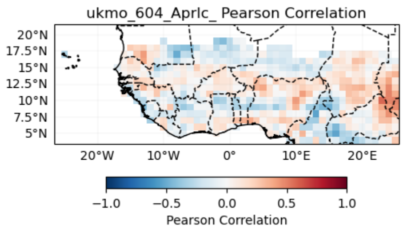
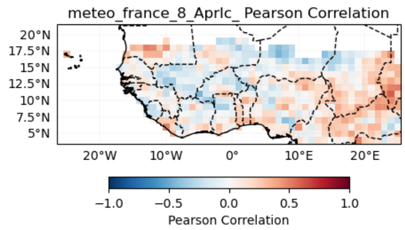
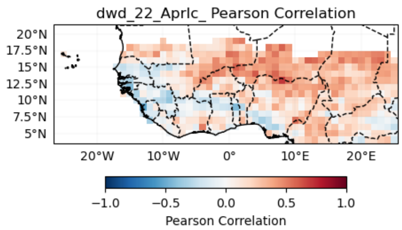
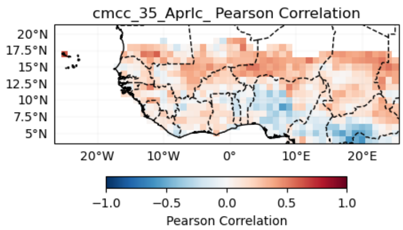
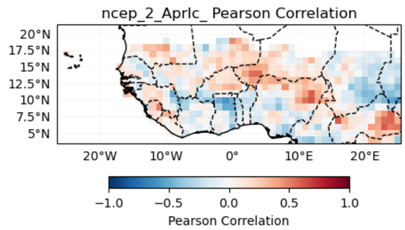


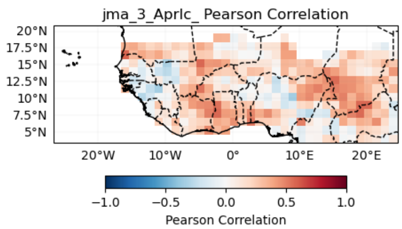


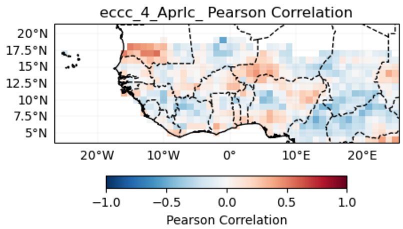


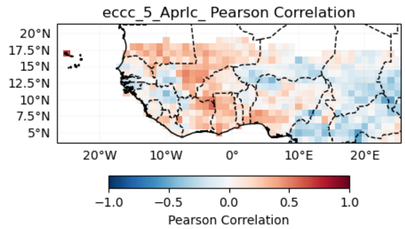


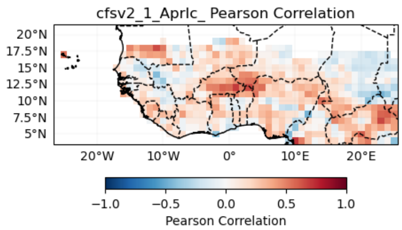


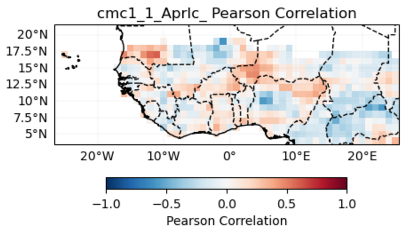


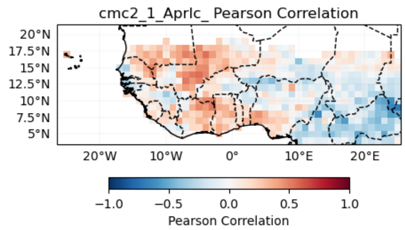


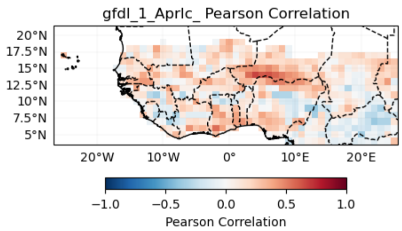


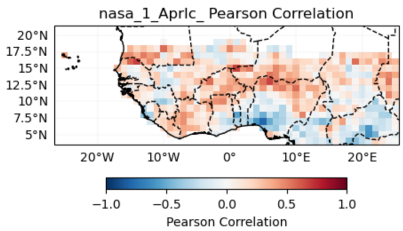


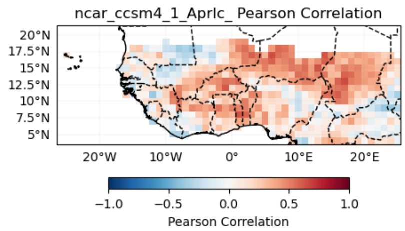


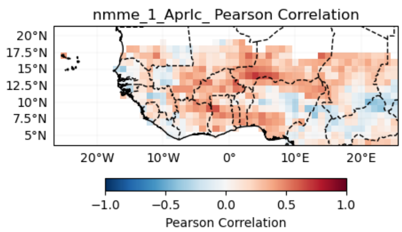




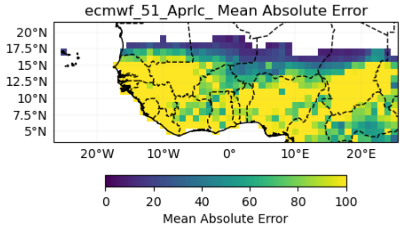


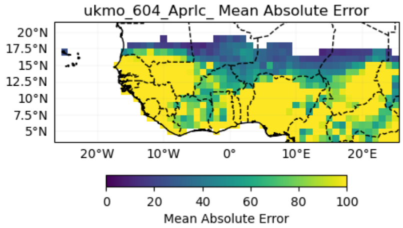


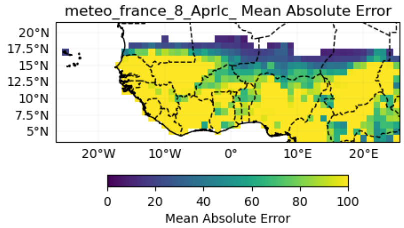


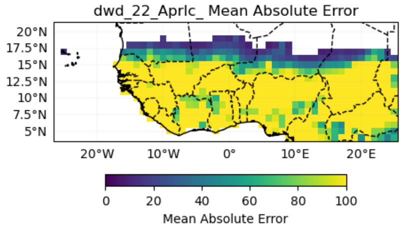


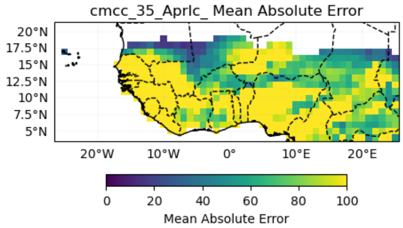
Mean Absolute Error

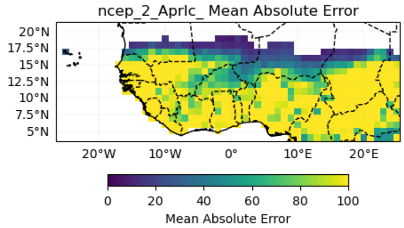


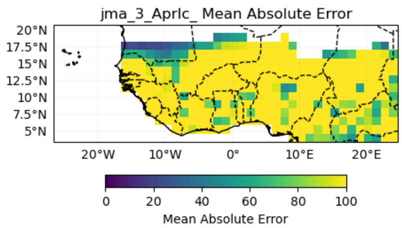


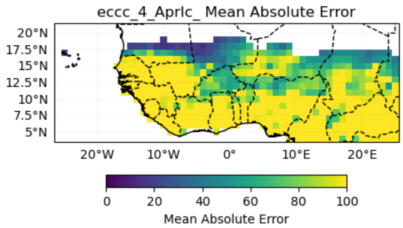


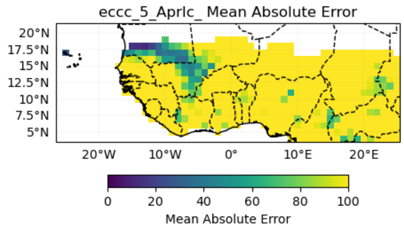


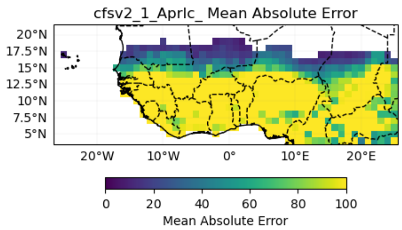


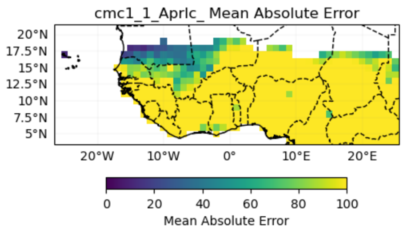


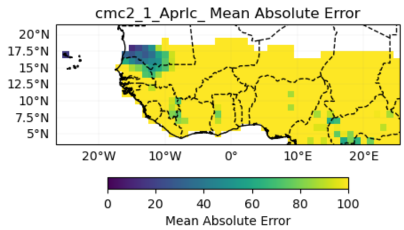


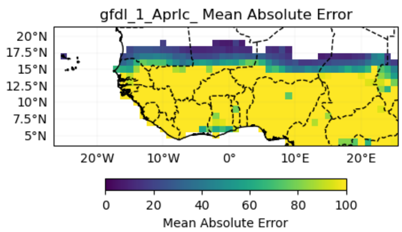


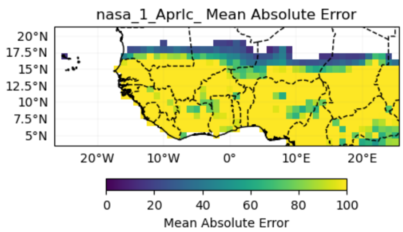


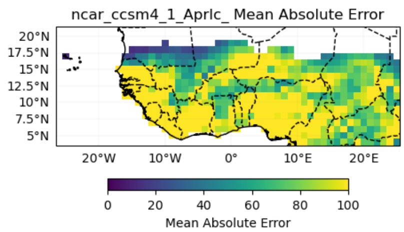


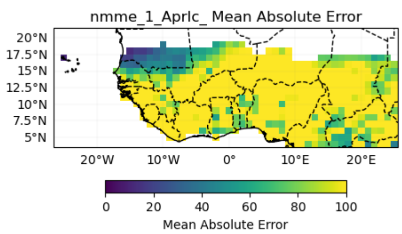












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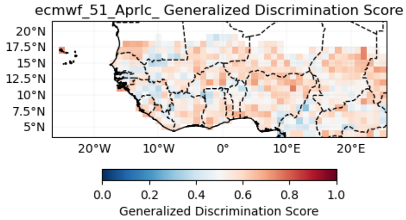

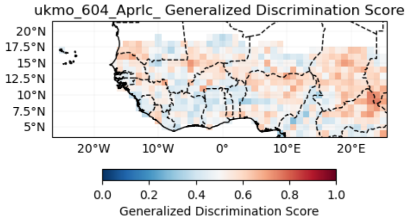

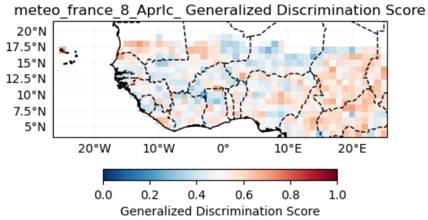

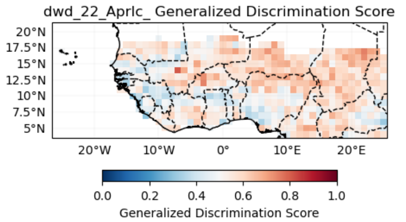

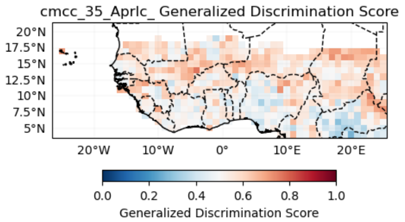

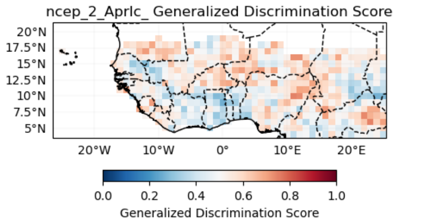

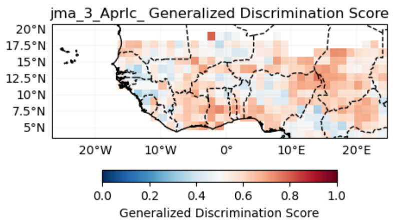

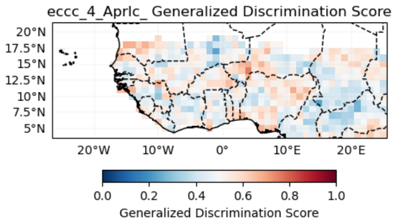

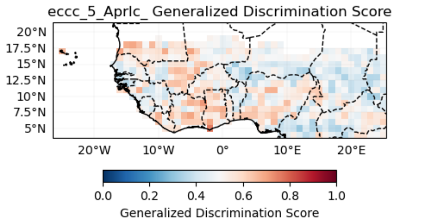

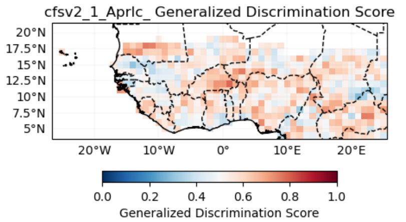

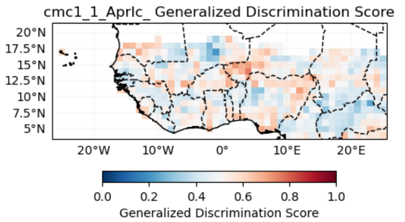

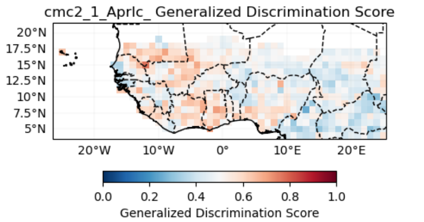

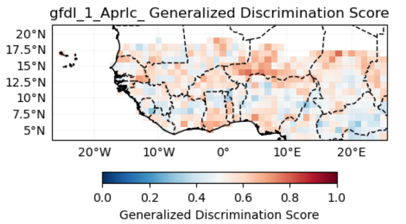

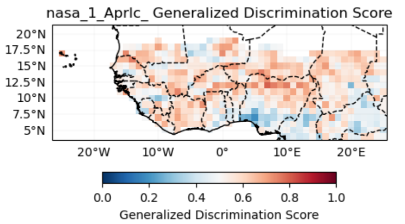

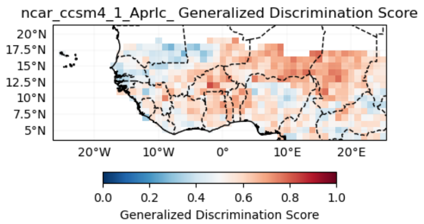

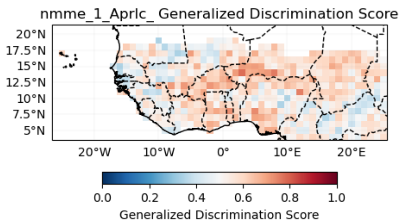

Choice of best GCM models

```
['ECMWF_51.PRCP', 'NMME_1.PRCP', 'DWD_22.PRCP', 'CFSV2_1.PRCP']
```

MME for best GCM models

Cross-validation

```
Cross-validation ongoing
100%|████████████████████████████████████████| 24/24 [08:15<00:00, 20.63s/it]
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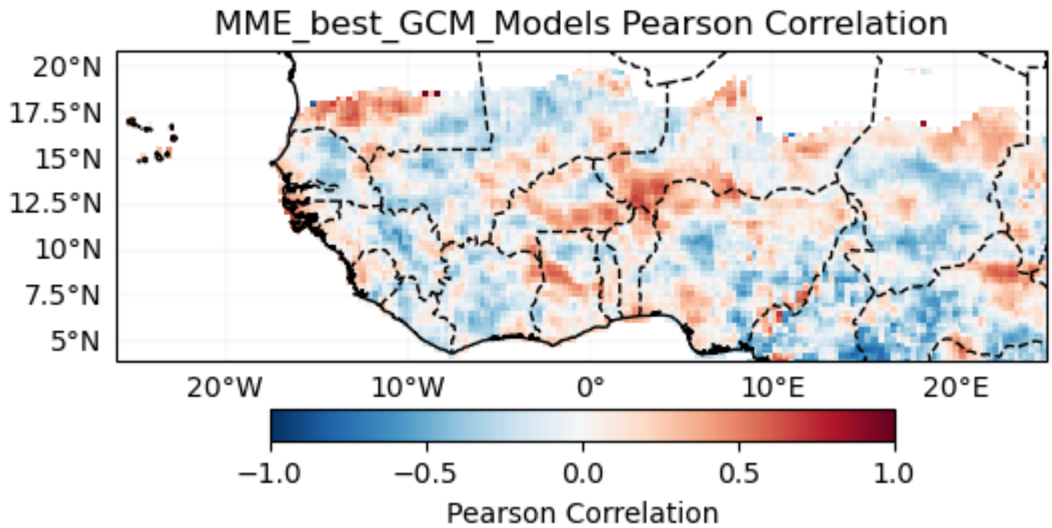
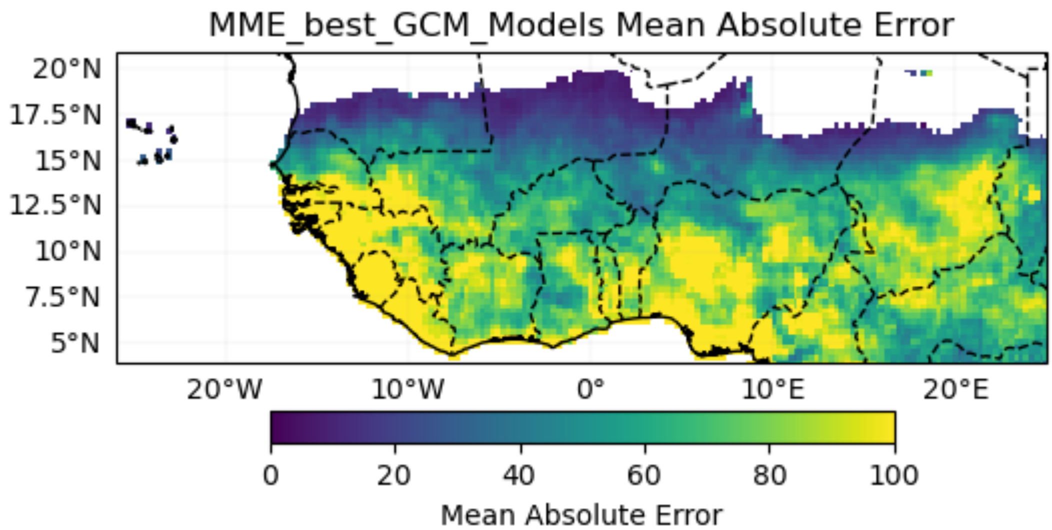
Verification





Probabilistic

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Generalized Discrimination Score
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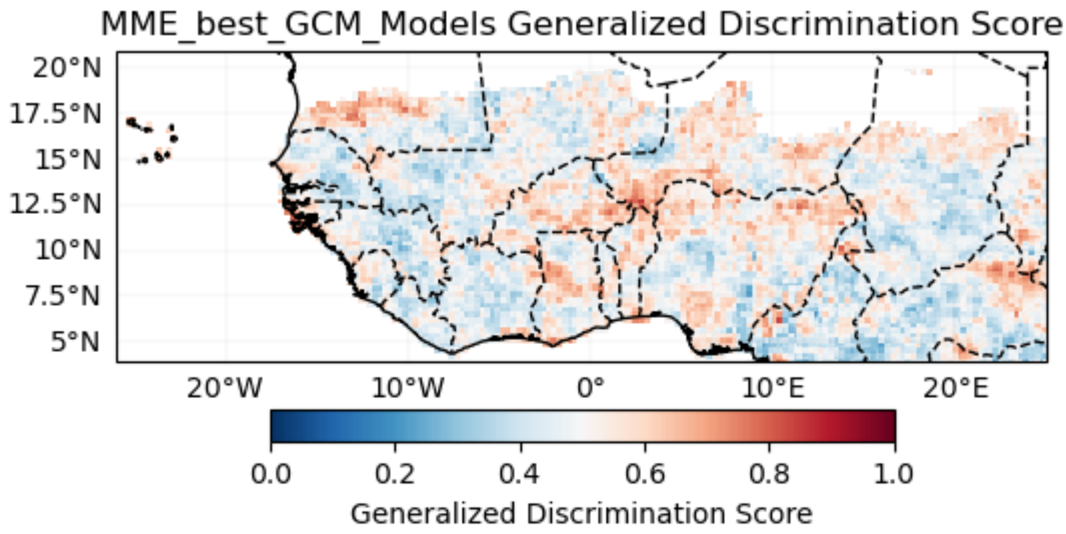
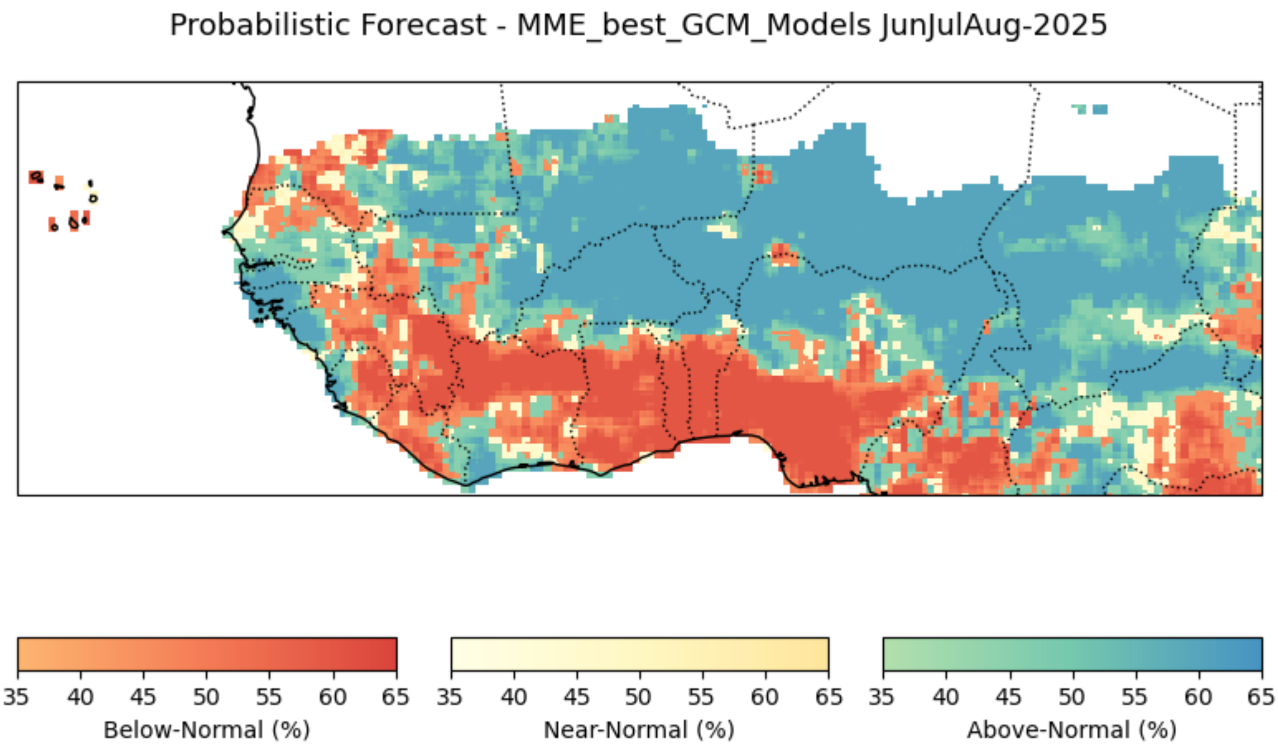


Forecast

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8508
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2nd - Approach: Statistico-Dynamical

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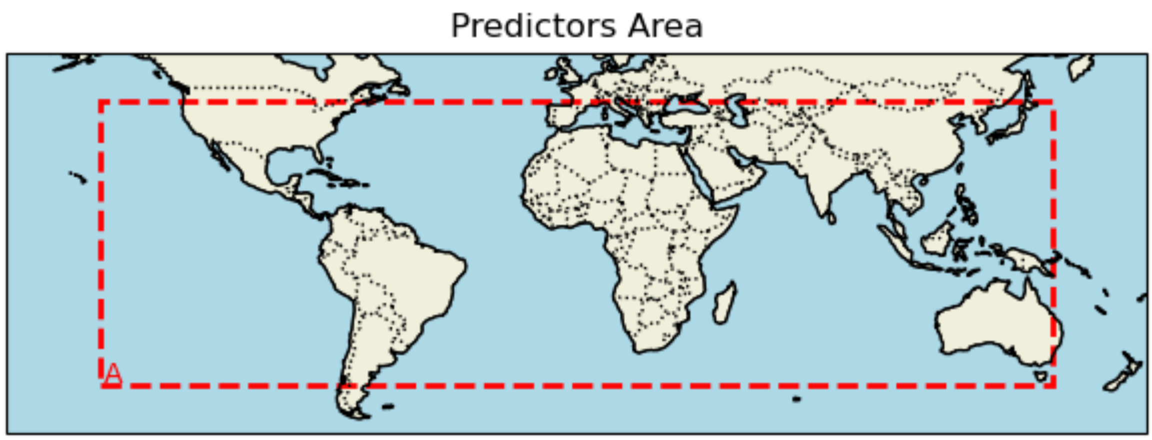
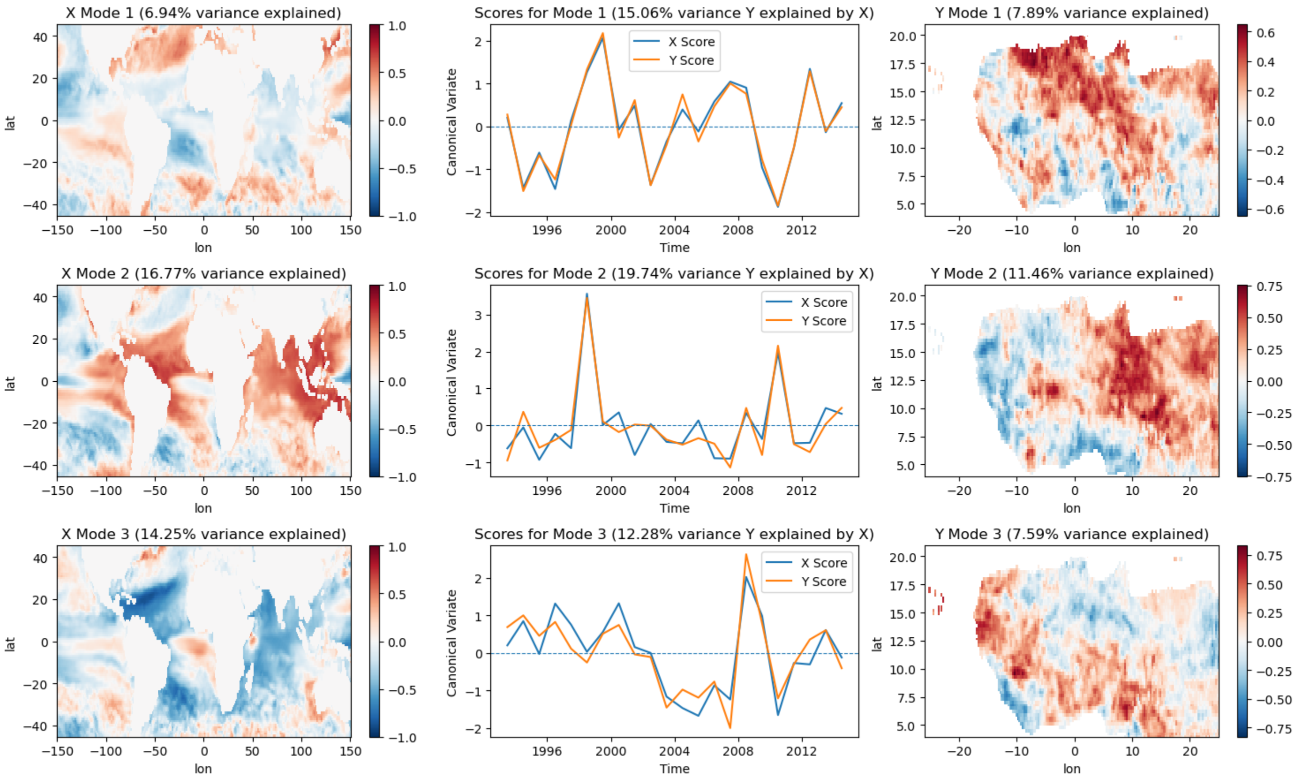
Initialize the WAS_CCA class (default)

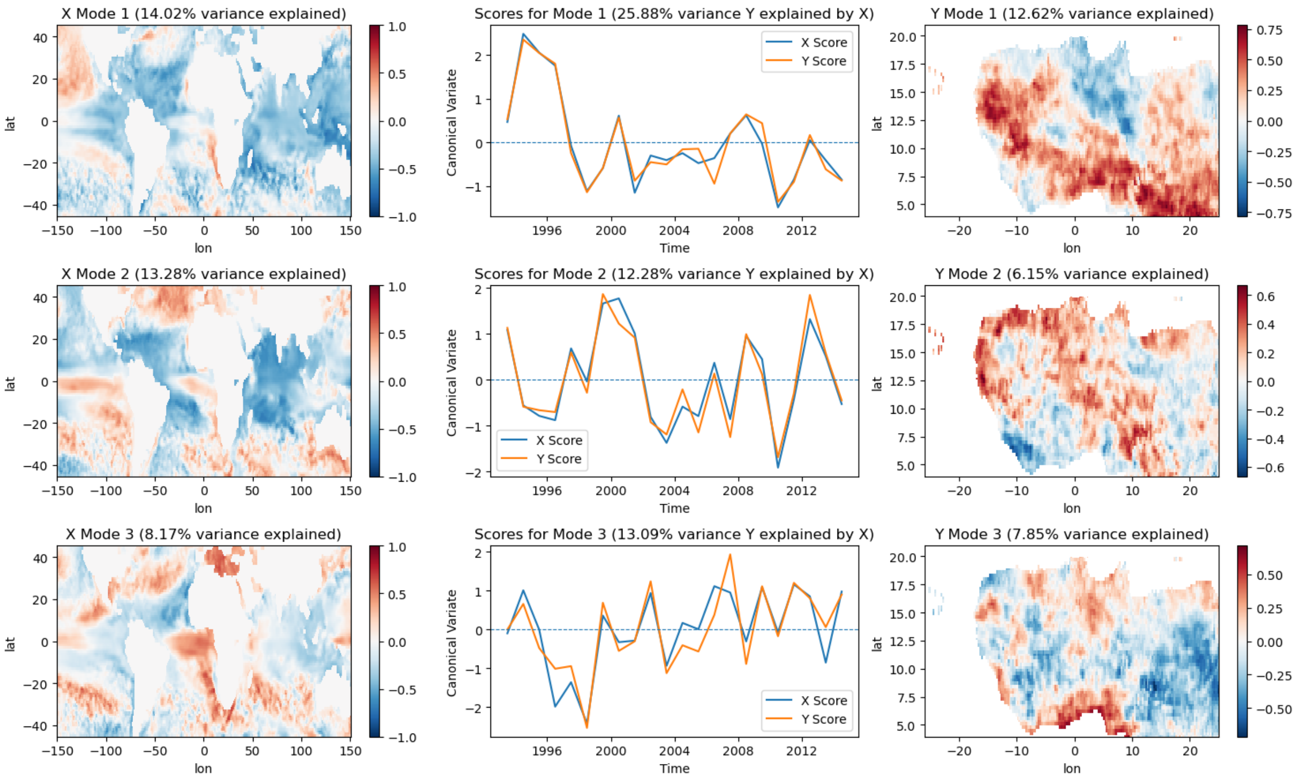


Modes of CCA for each model

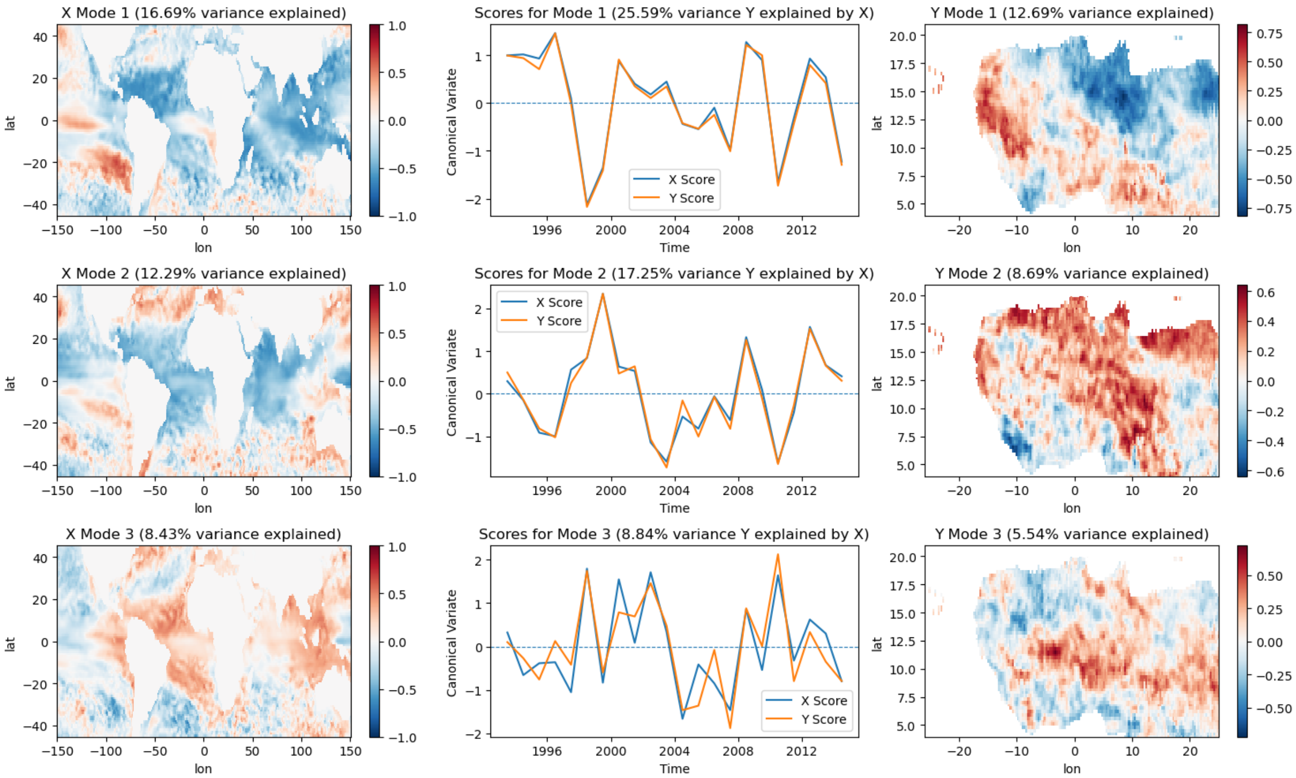
```
CCA Modes for model ECMWF_51.SST
```



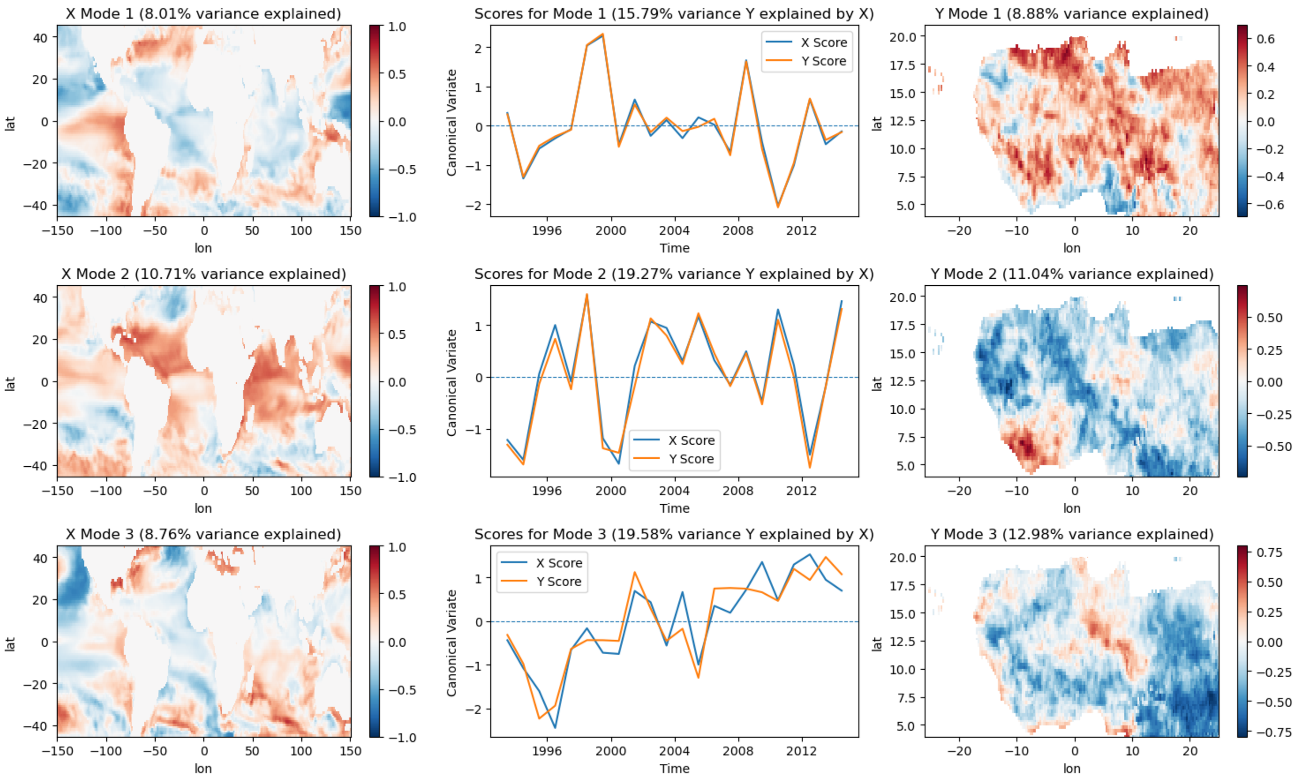
CCA Modes for model UKMO_604.SST



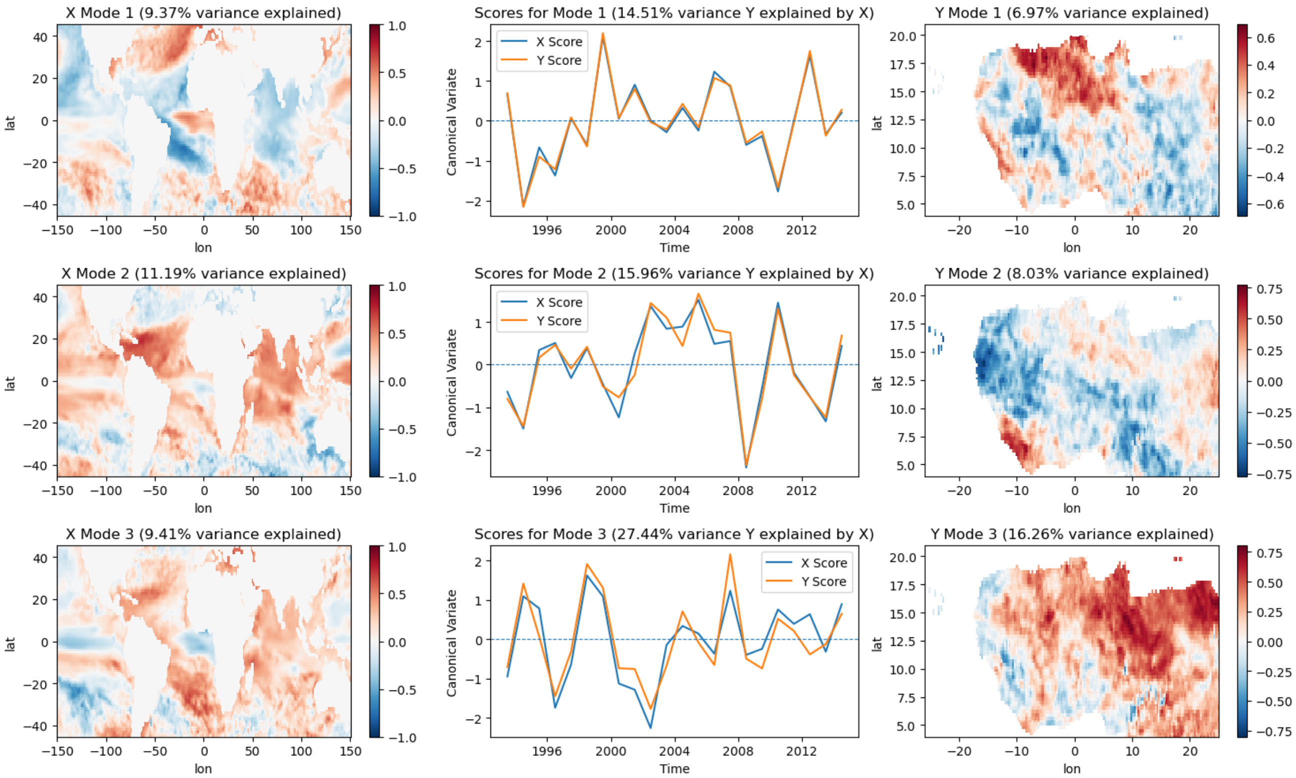
CCA Modes for model METEOFRANCE_8.SST



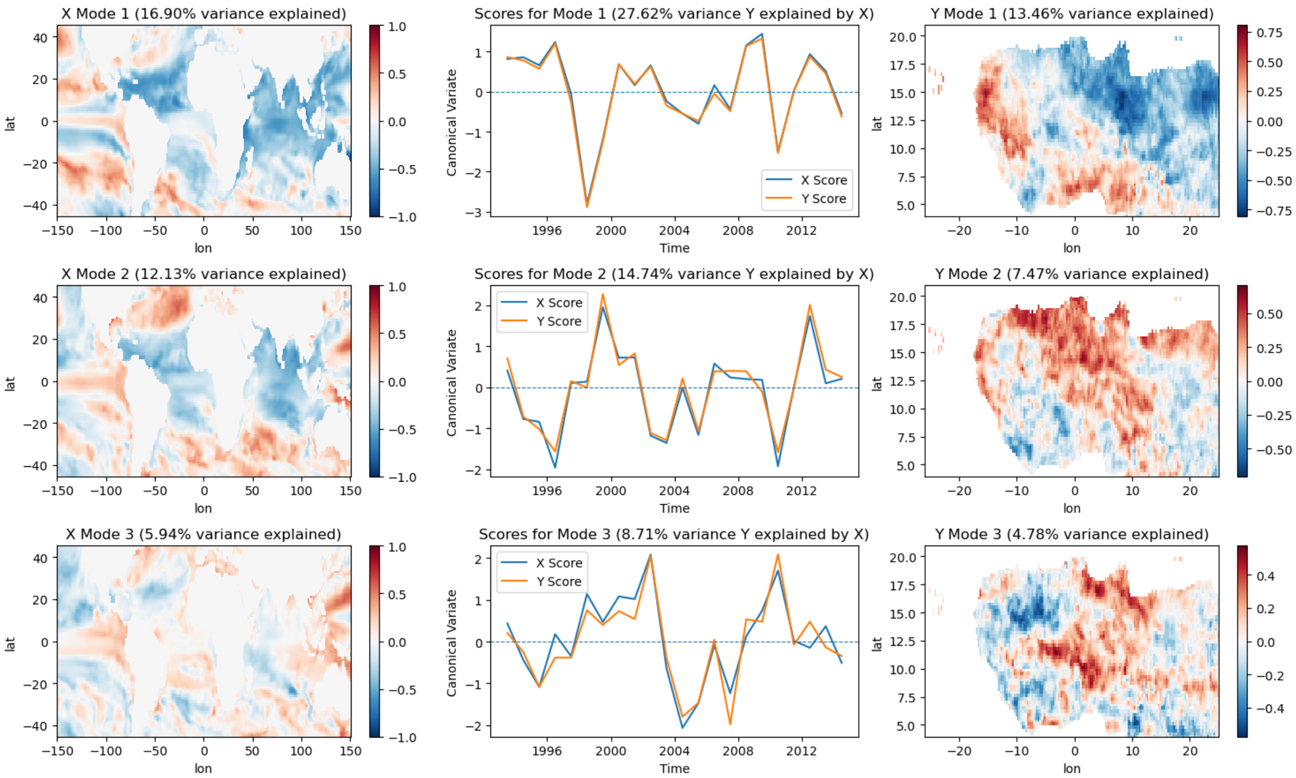
CCA Modes for model DWD_22.SST



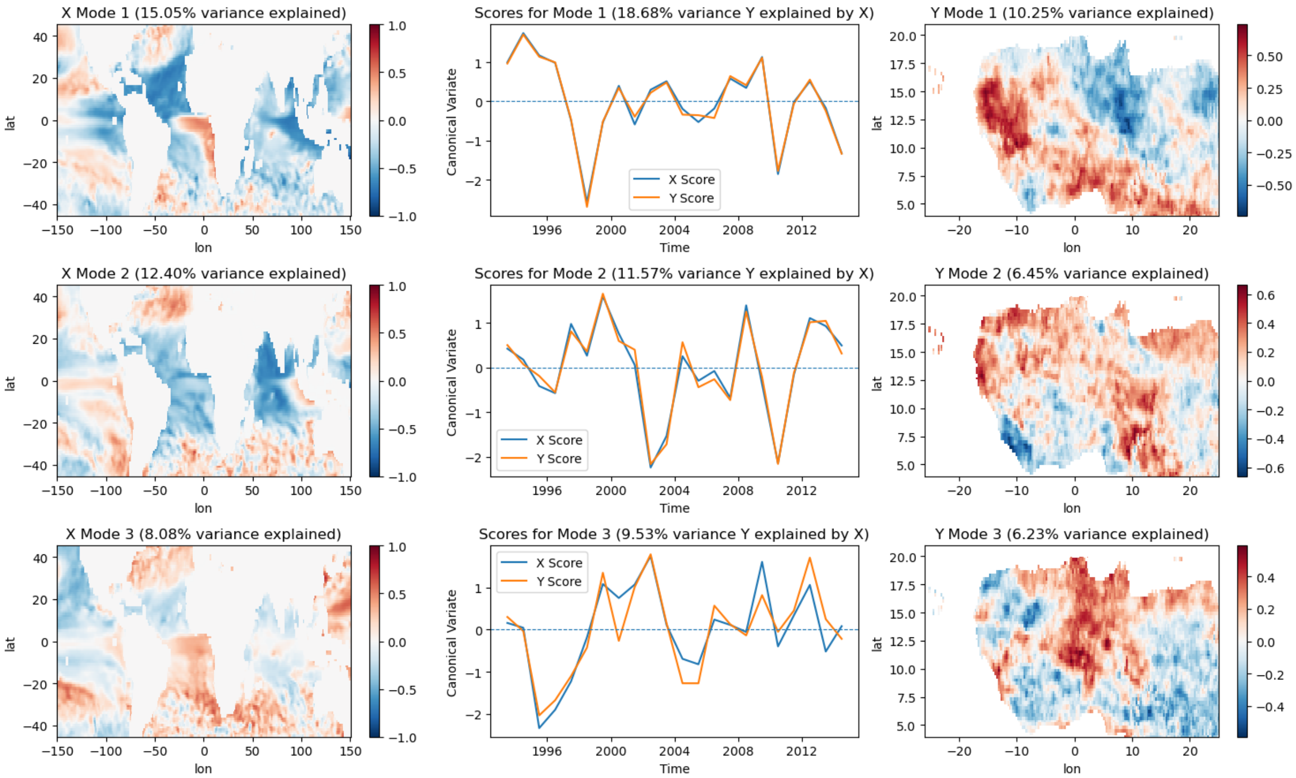
CCA Modes for model CMCC_35.SST



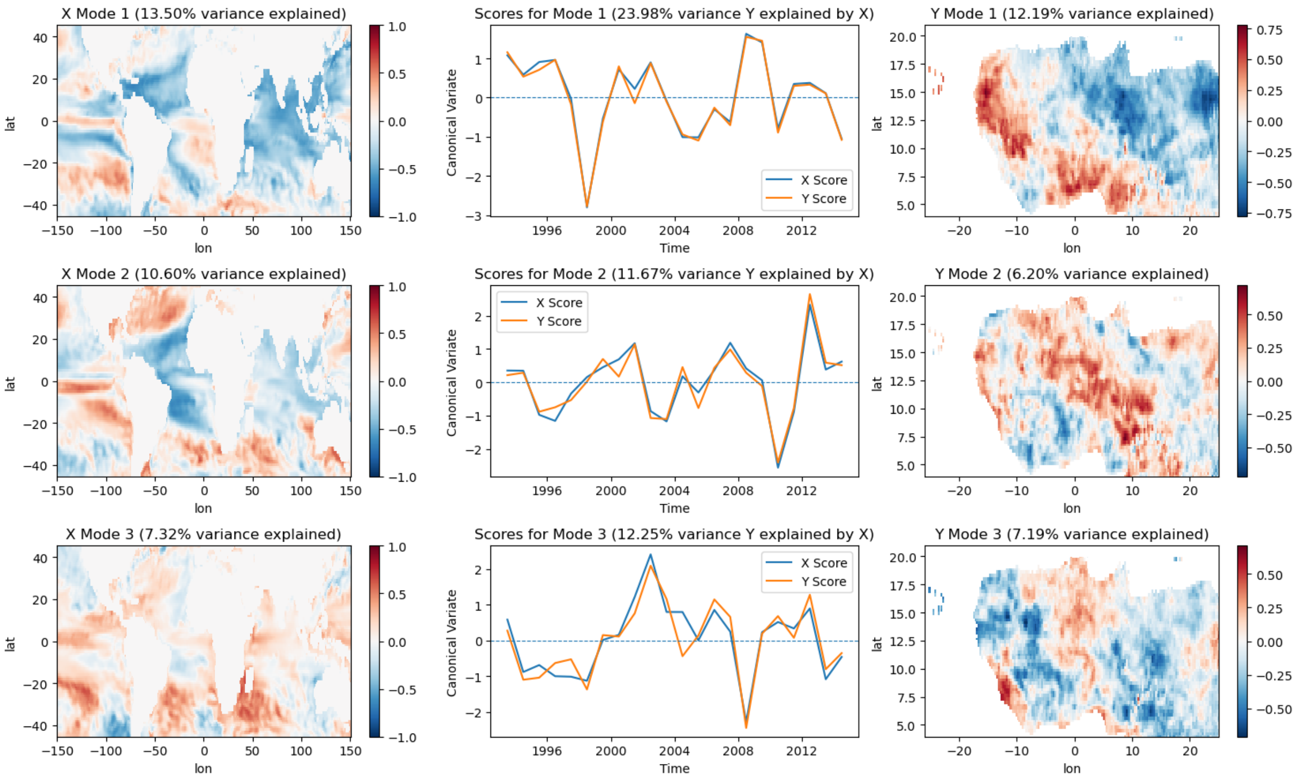
CCA Modes for model NCEP_2.SST



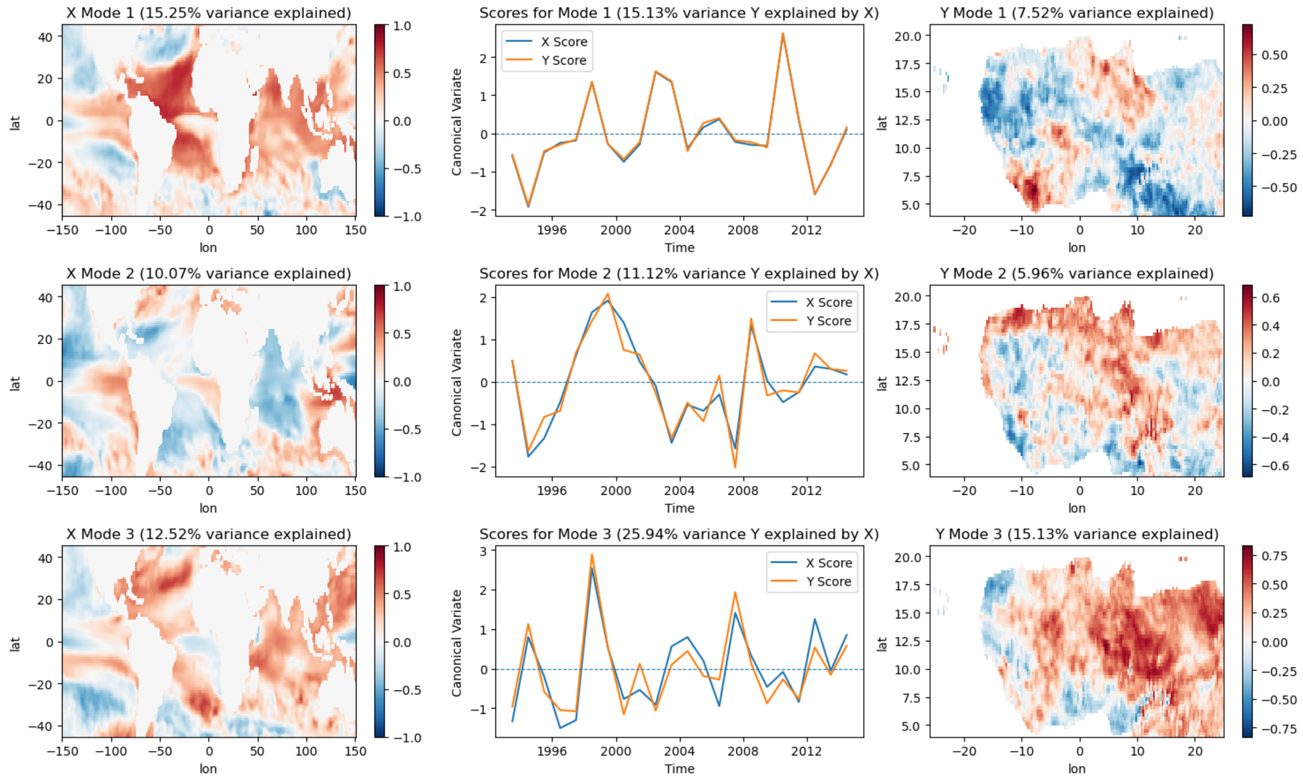
CCA Modes for model JMA_3.SST



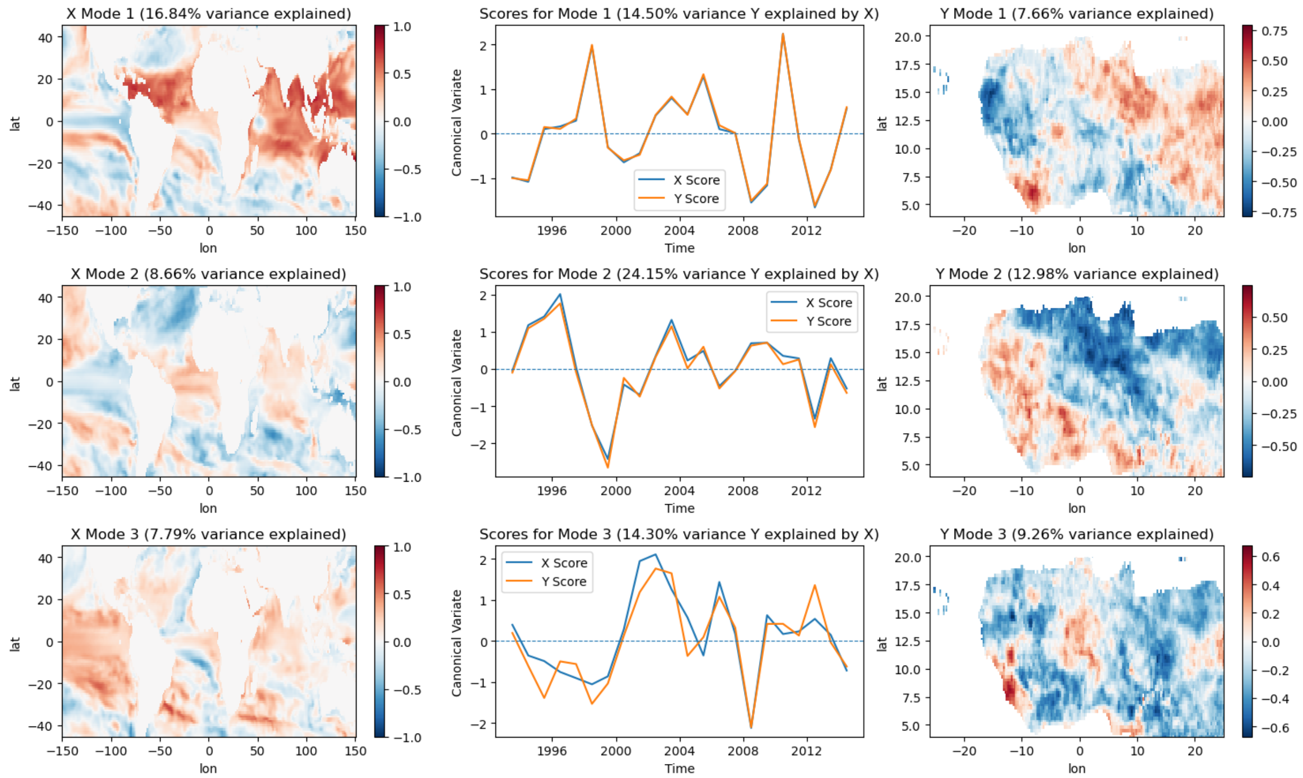
CCA Modes for model ECCC_4.SST



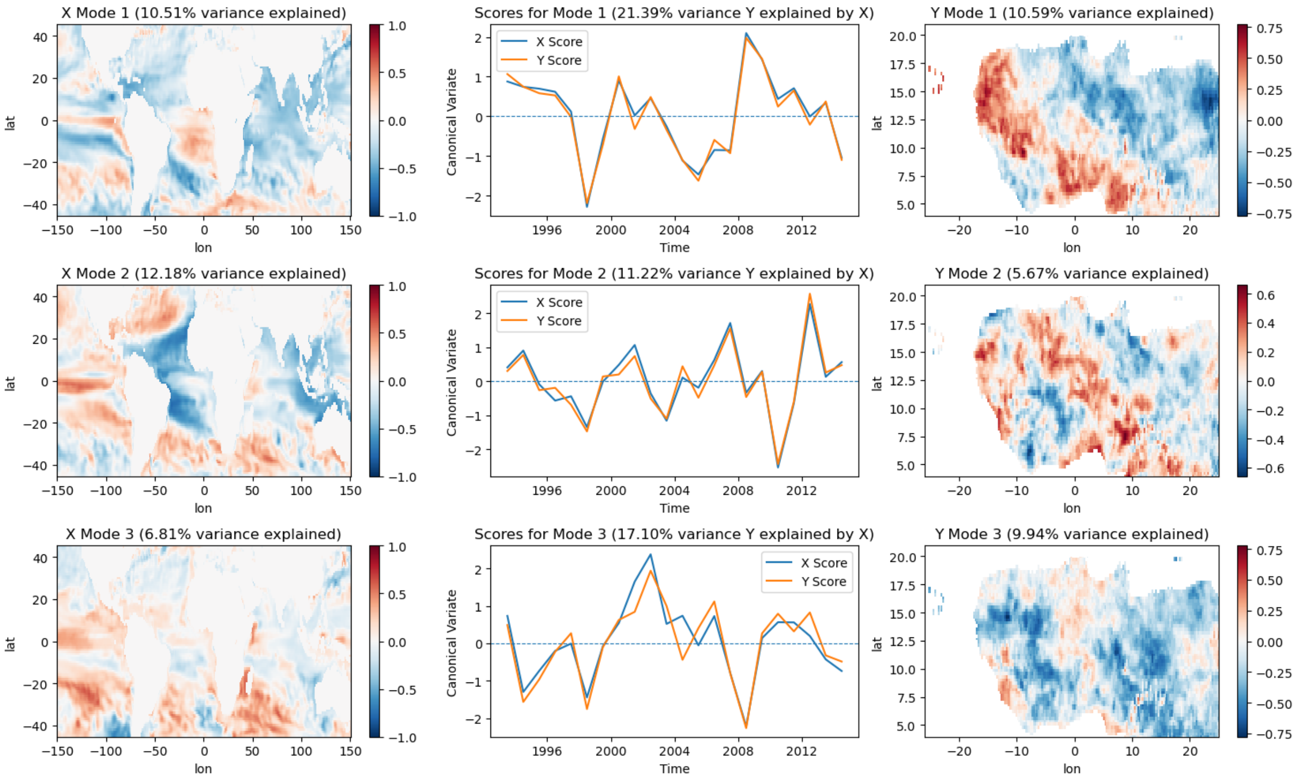
CCA Modes for model ECCC_5.SST



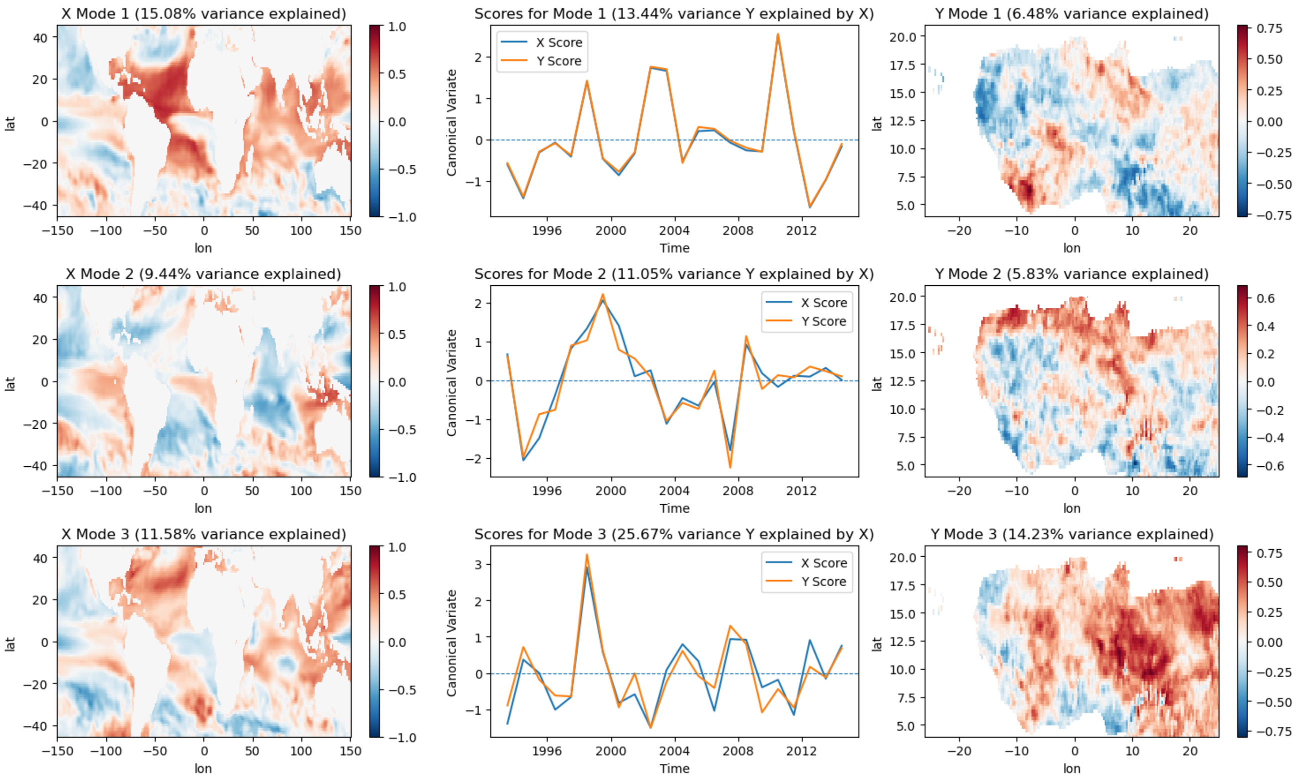
CCA Modes for model CFSV2_1.SST



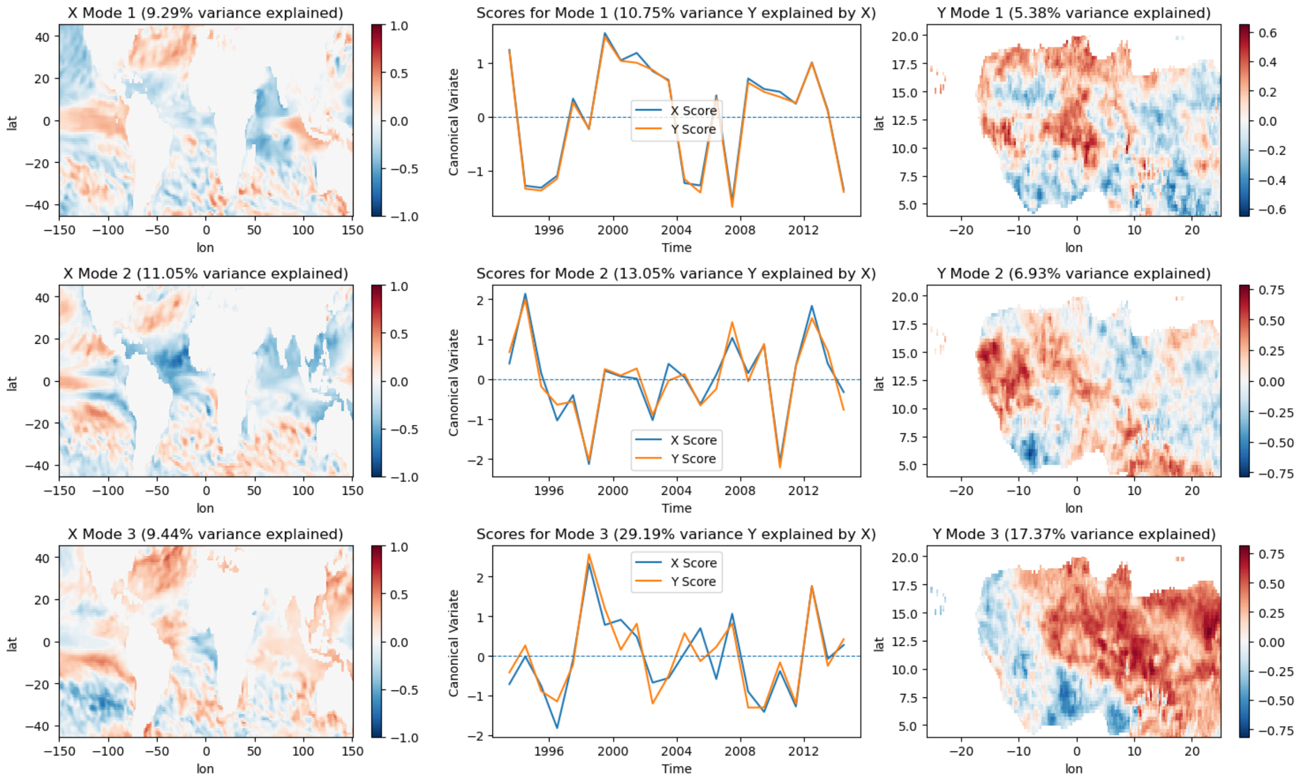
CCA Modes for model CMC1_1.SST



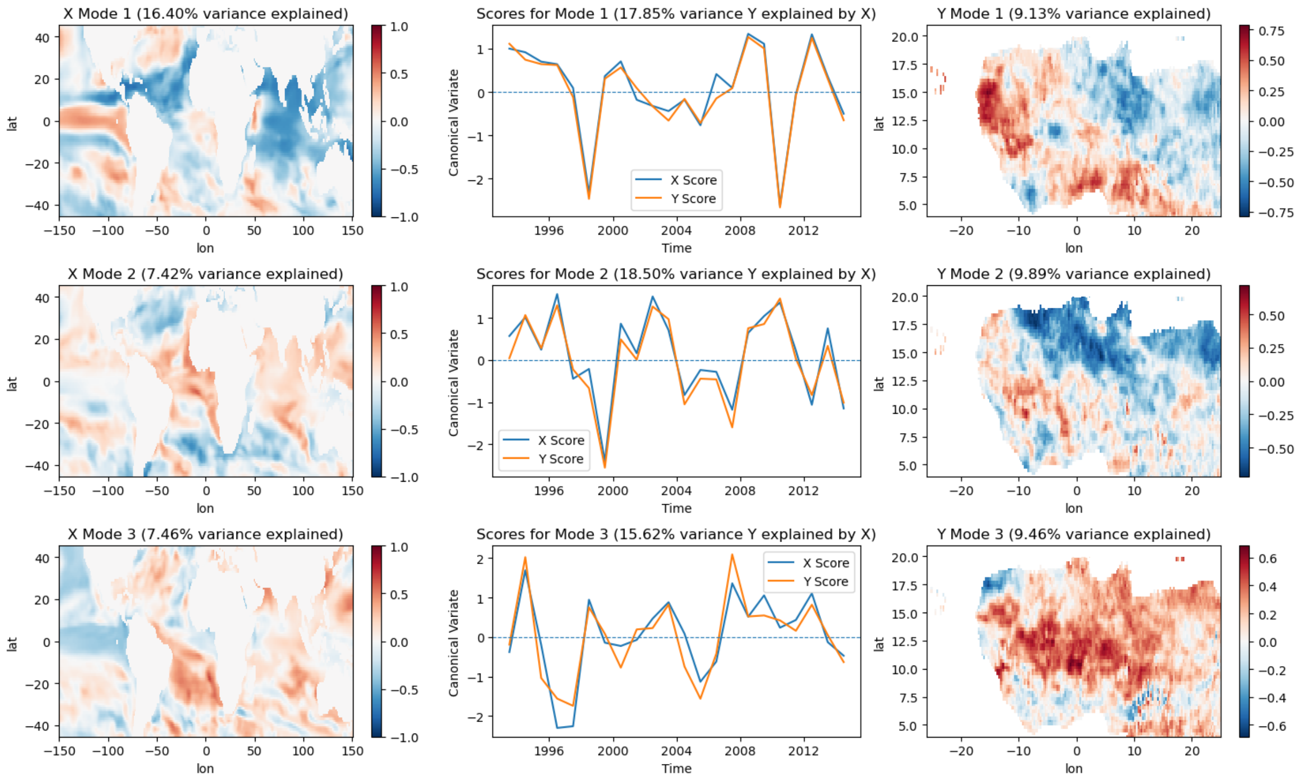
CCA Modes for model CMC2_1.SST



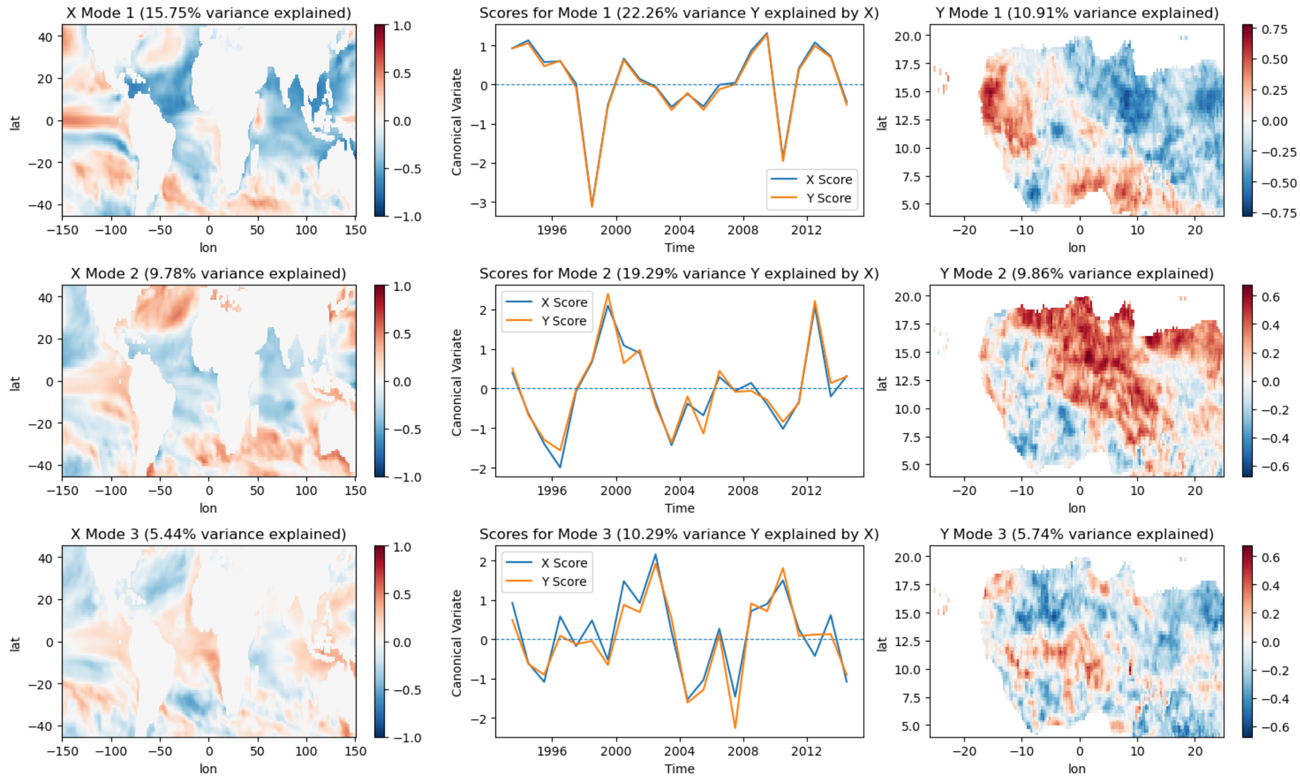
CCA Modes for model GFDL_1.SST



CCA Modes for model NCAR_CCSM4_1.SST



CCA Modes for model NMME_1.SST



Cross validation

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Cross-validation ongoing
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UKMO_604
Cross-validation ongoing
100%|████████████████████████████████████████| 24/24 [00:09<00:00,  2.47it/s]
METEOFRANCE_8
Cross-validation ongoing
100%|████████████████████████████████████████| 24/24 [00:09<00:00,  2.45it/s]
DWD_22
Cross-validation ongoing
100%|████████████████████████████████████████| 24/24 [00:10<00:00,  2.36it/s]
CMCC_35
Cross-validation ongoing
100%|████████████████████████████████████████| 24/24 [00:09<00:00,  2.45it/s]
NCEP_2
Cross-validation ongoing
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JMA_3
Cross-validation ongoing
100%|████████████████████████████████████████| 24/24 [00:10<00:00,  2.35it/s]
ECCC_4
Cross-validation ongoing
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ECCC_5
Cross-validation ongoing
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CFSV2_1
Cross-validation ongoing
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CMC1_1
Cross-validation ongoing
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CMC2_1
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GFDL_1
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NCAR_CCSM4_1
Cross-validation ongoing
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NMME_1
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Verification (individual)

Deterministic

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Pearson Correlation
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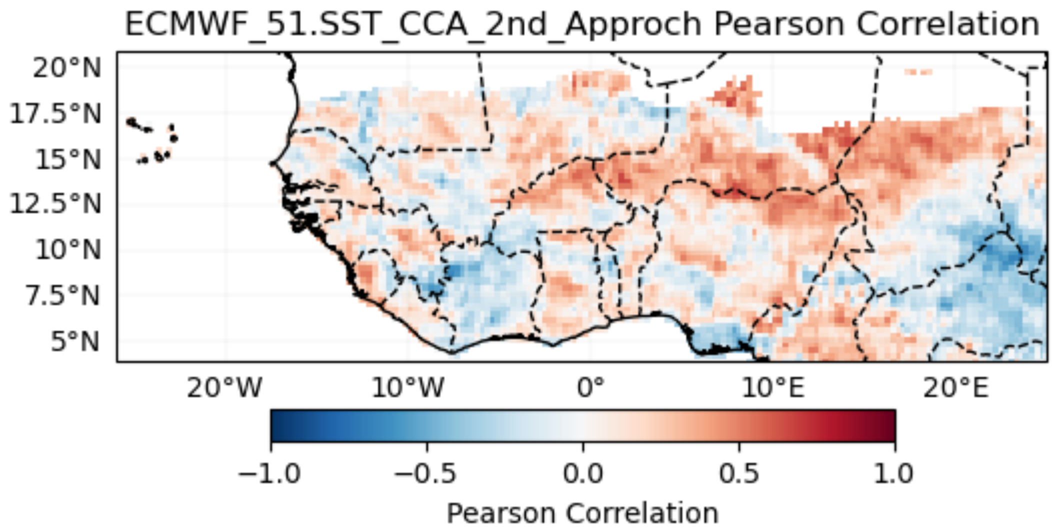

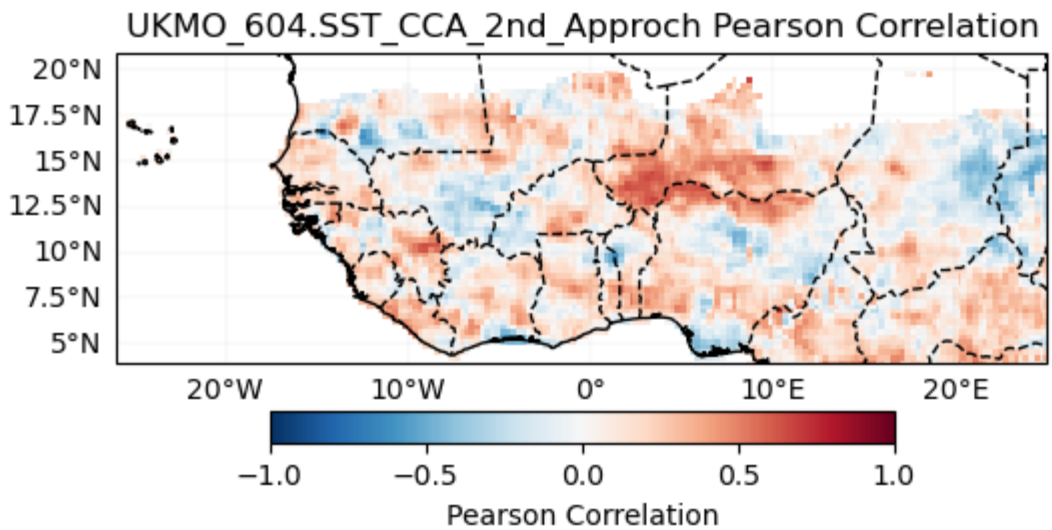

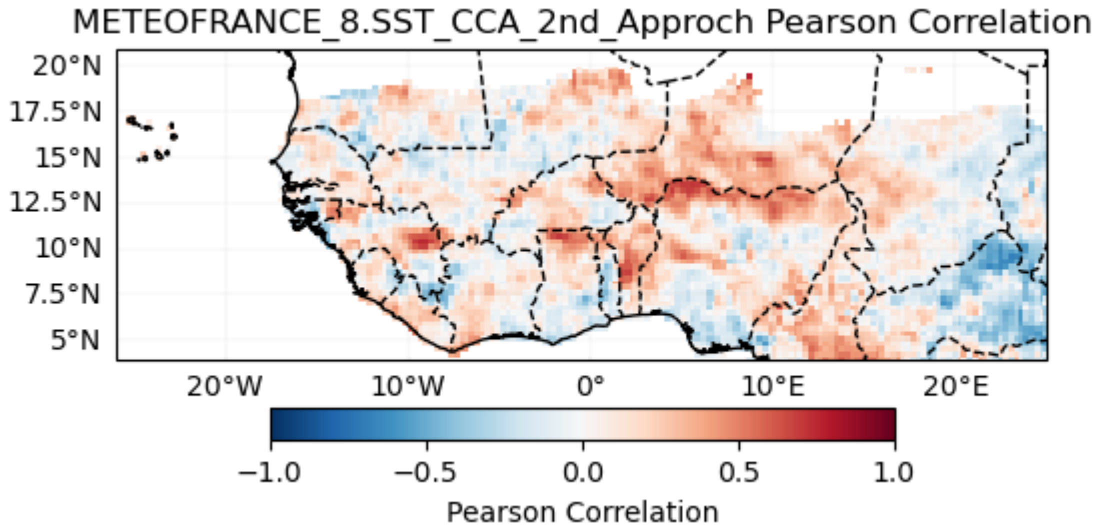
METEOFRANCE_8.SST_CCA_2nd_Approch Pearson Correlation
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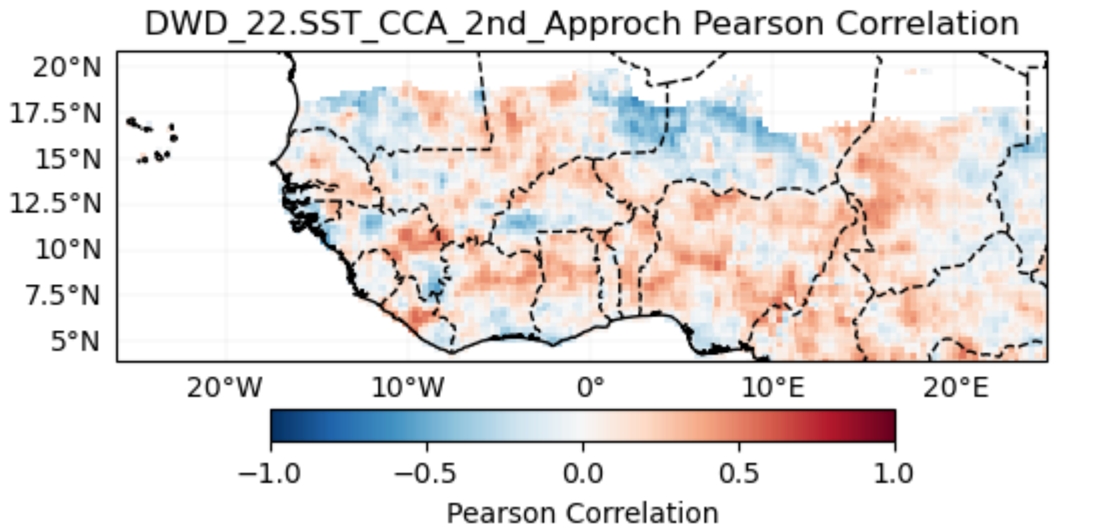
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
−1.0
−0.5
0.0
0.5
1.0
Pearson Correlation


DWD_22.SST_CCA_2nd_Approch Pearson Correlation
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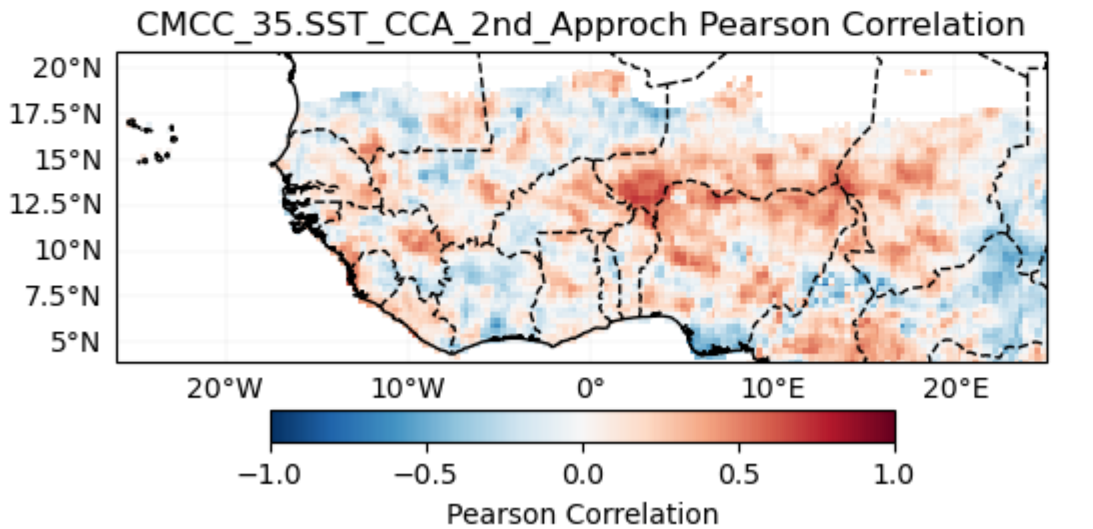
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
−1.0
−0.5
0.0
0.5
1.0
Pearson Correlation


CMCC_35.SST_CCA_2nd_Approch Pearson Correlation
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10°W
0°
10°E
20°E
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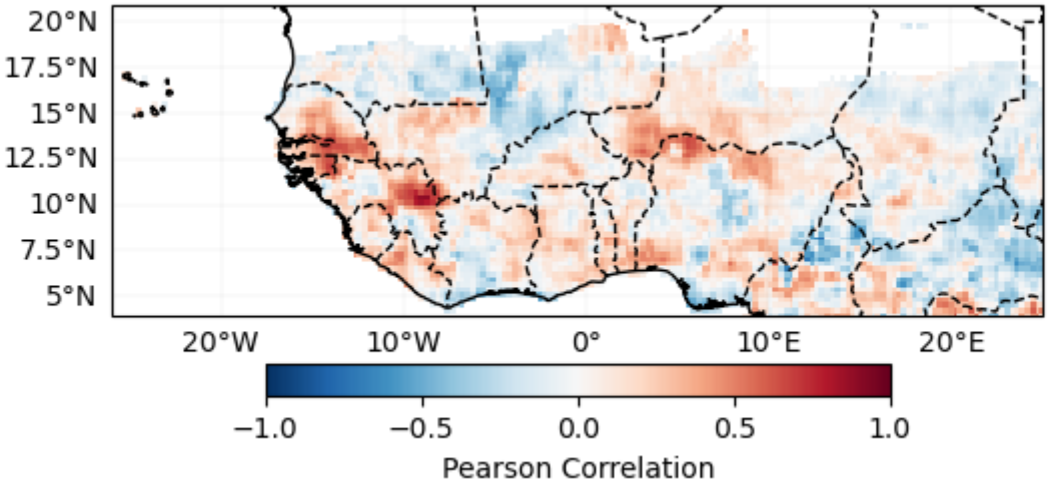
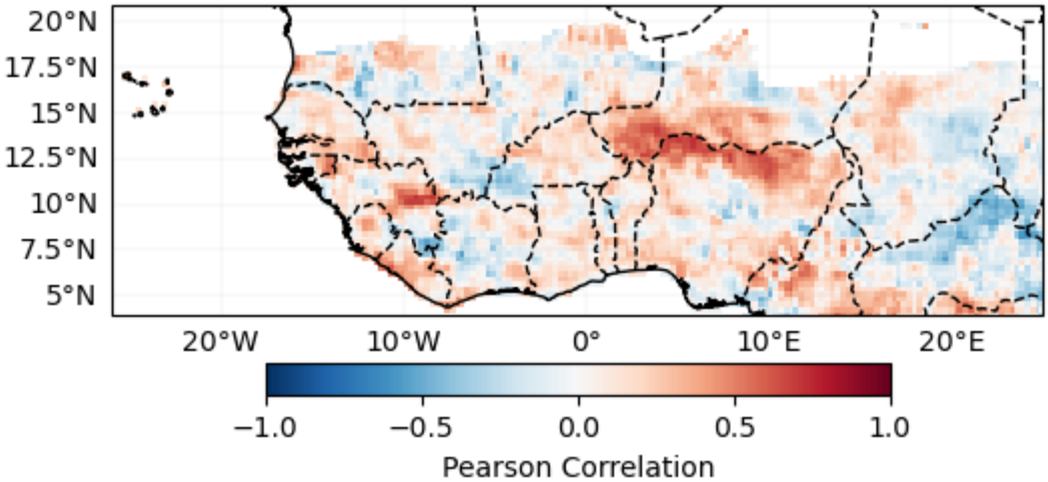
Pearson Correlation

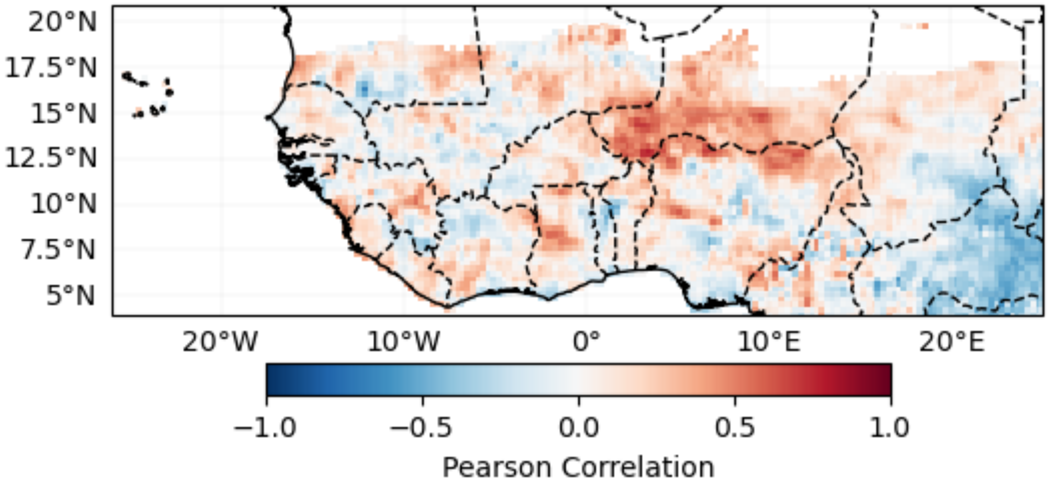
NCEP_2.SST_CCA_2nd_Approch Pearson Correlation

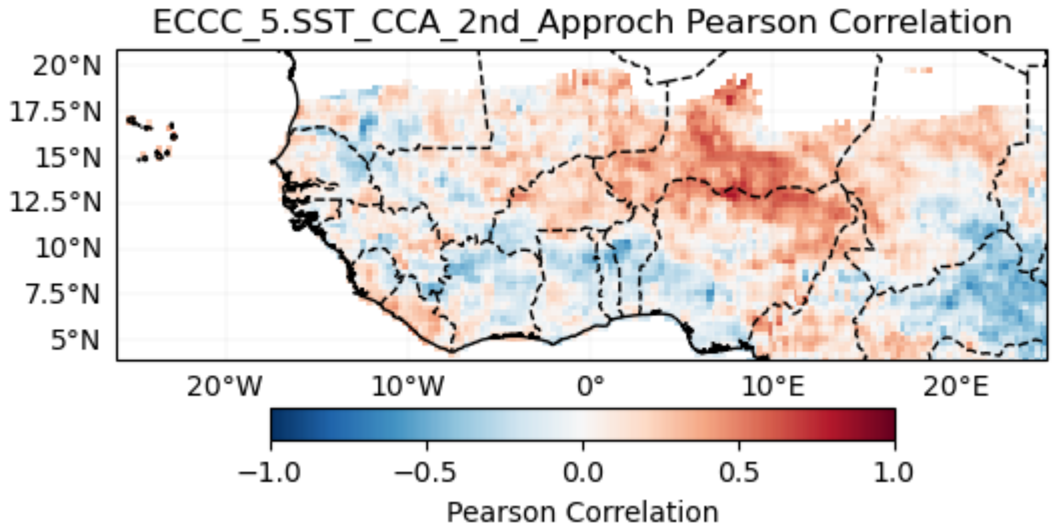


JMA_3.SST_CCA_2nd_Approch Pearson Correlation



ECCC_4.SST_CCA_2nd_Approch Pearson Correlation

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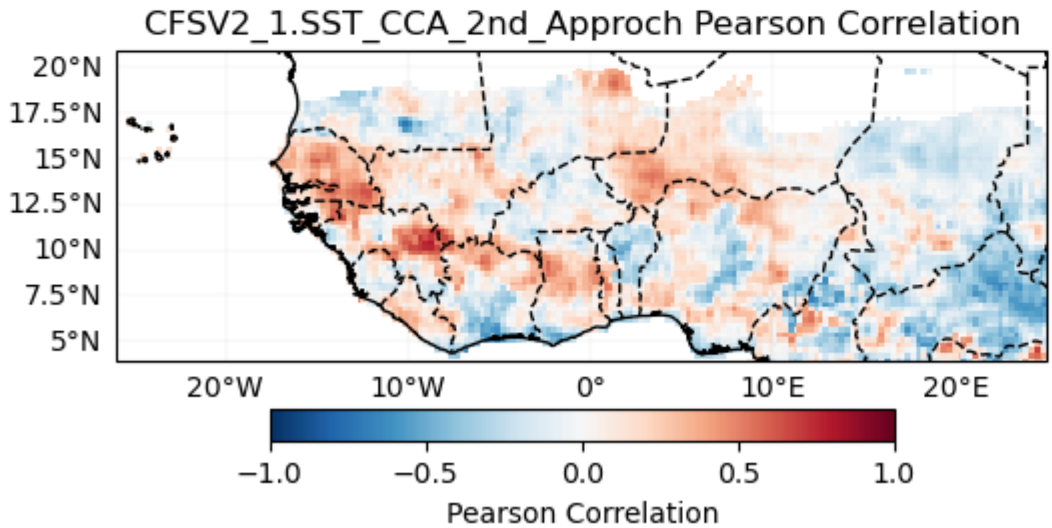
Pearson Correlation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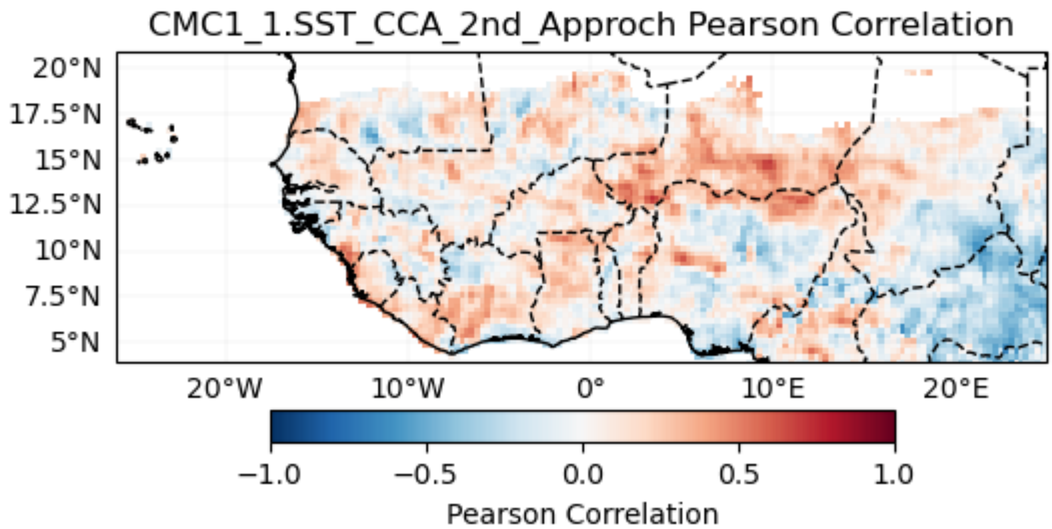
Pearson Correlation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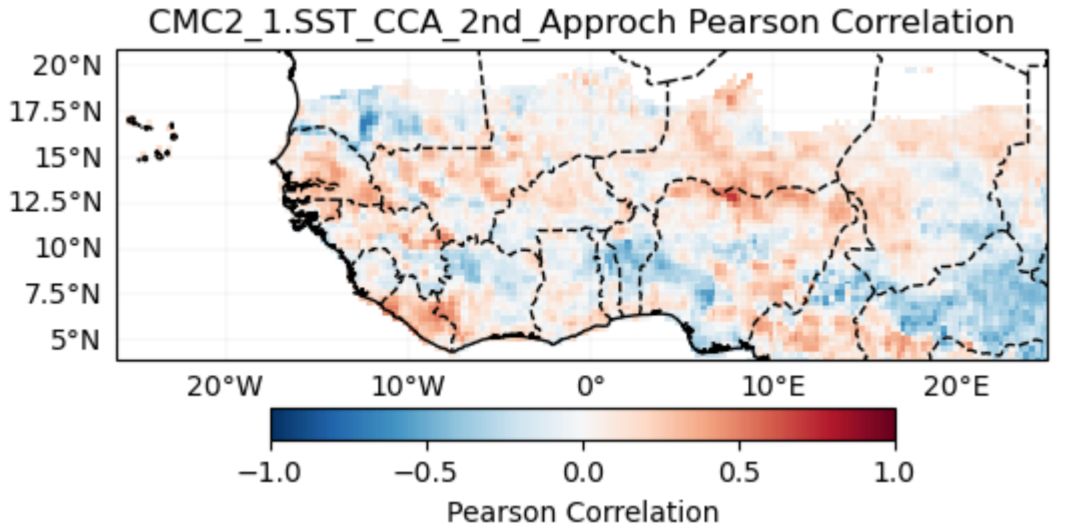
Pearson Correlation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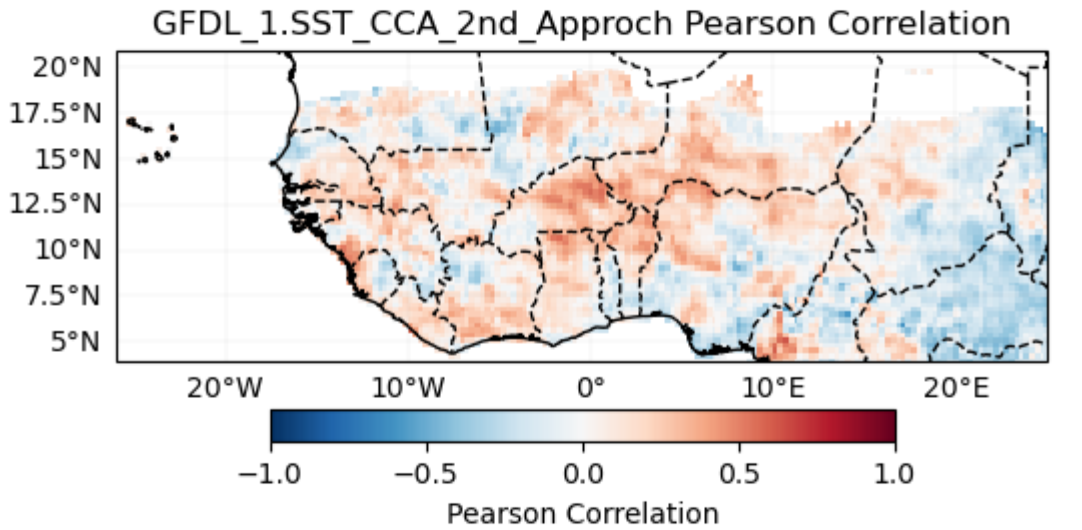
Pearson Correlation


GFDL_1.SST_CCA_2nd_Approch Pearson Correlation
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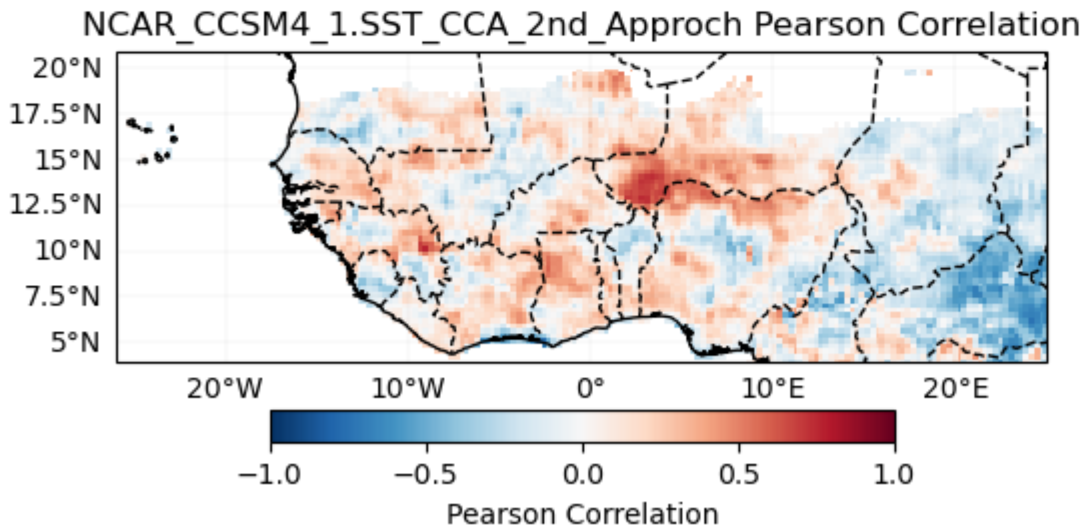
Pearson Correlation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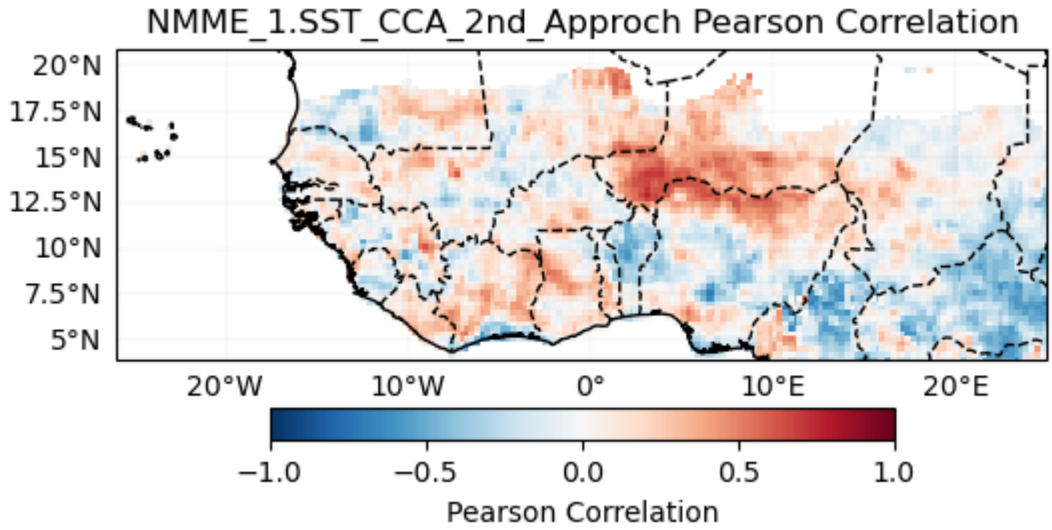
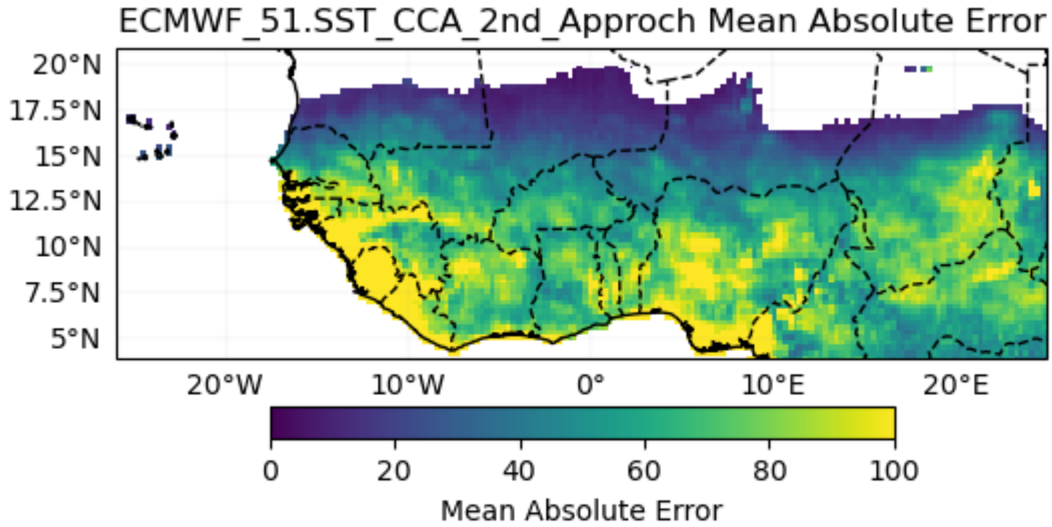
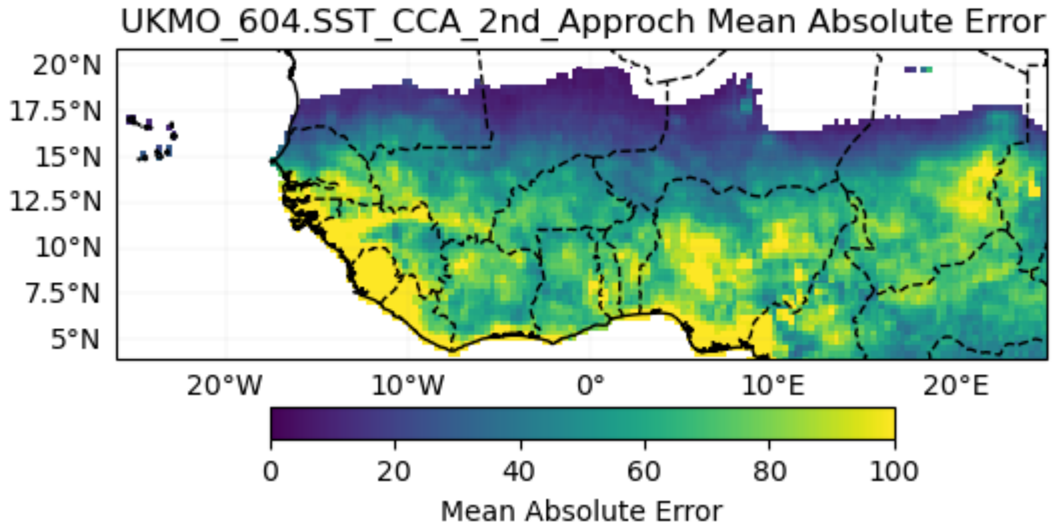
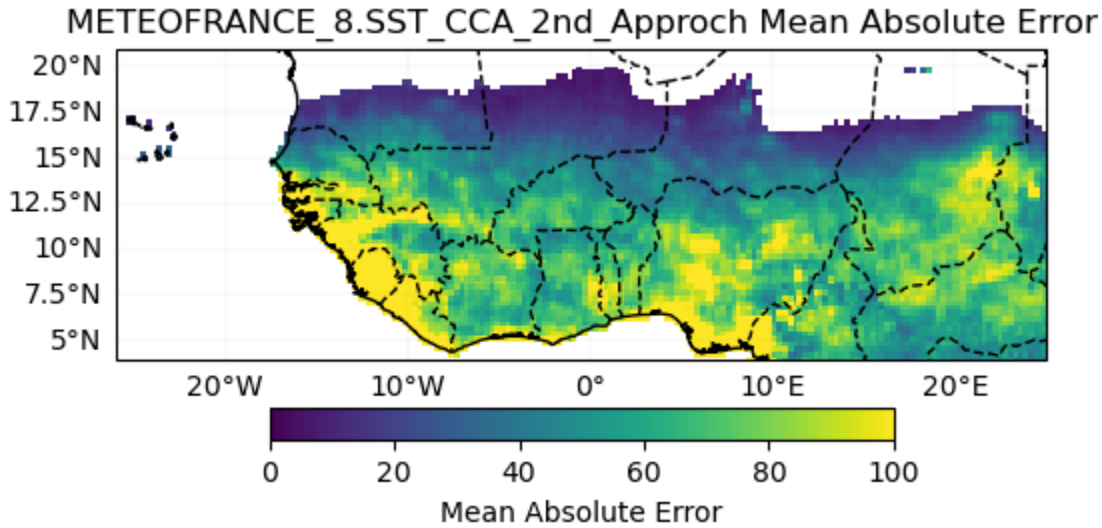
Pearson Correlation



Mean Absolute Error






METEOFRANCE_8.SST_CCA_2nd_Approch Mean Absolute Error
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15°N
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10°N
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10°W
0°
10°E
20°E
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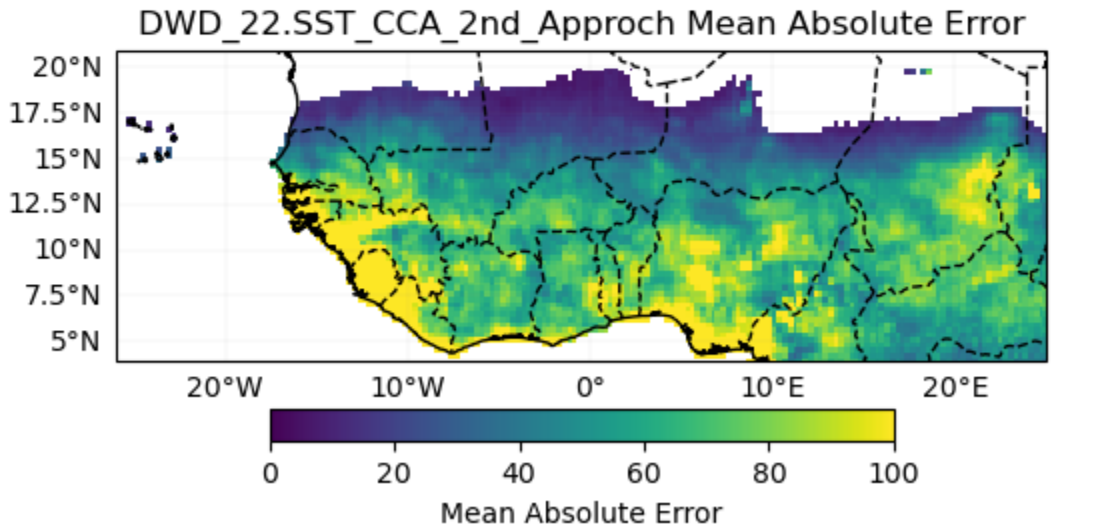
20
40
60
80
100
Mean Absolute Error


DWD_22.SST_CCA_2nd_Approch Mean Absolute Error
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10°E
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80
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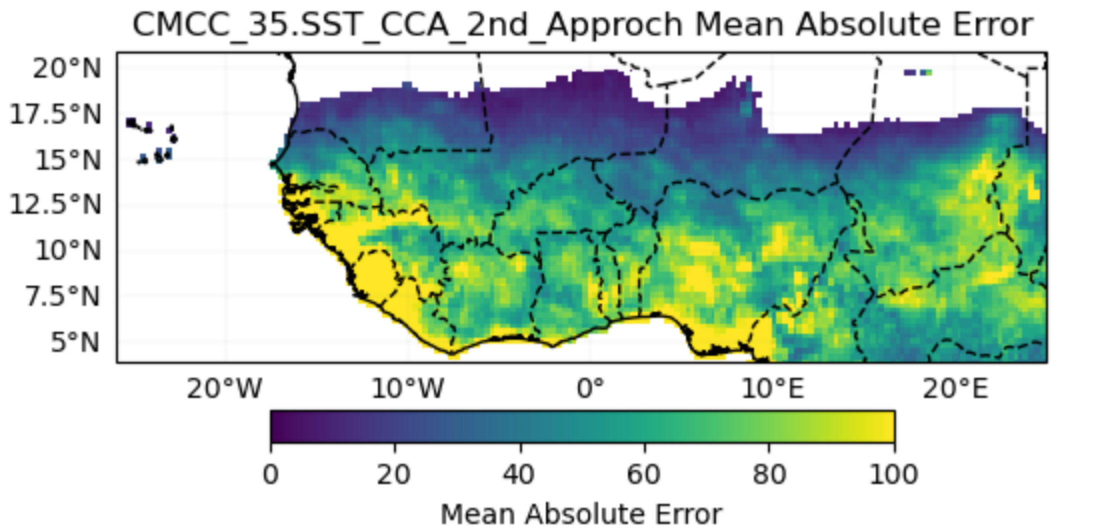
Mean Absolute Error


CMCC_35.SST_CCA_2nd_Approch Mean Absolute Error
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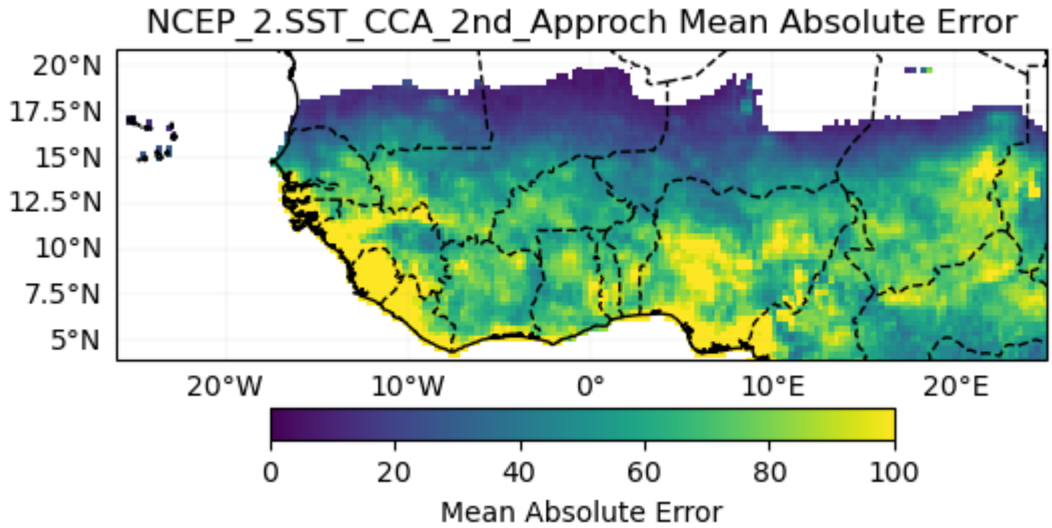
Mean Absolute Error


NCEP_2.SST_CCA_2nd_Approch Mean Absolute Error
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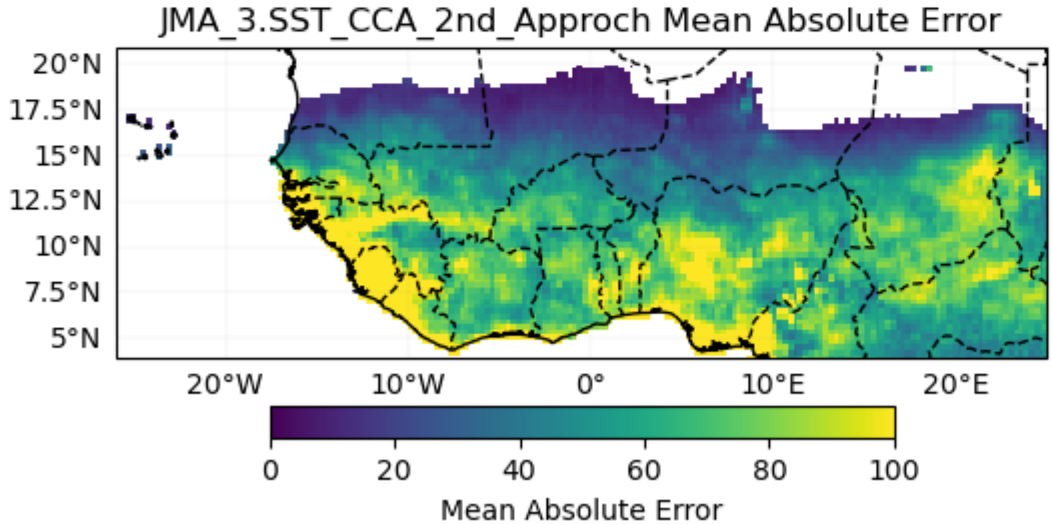
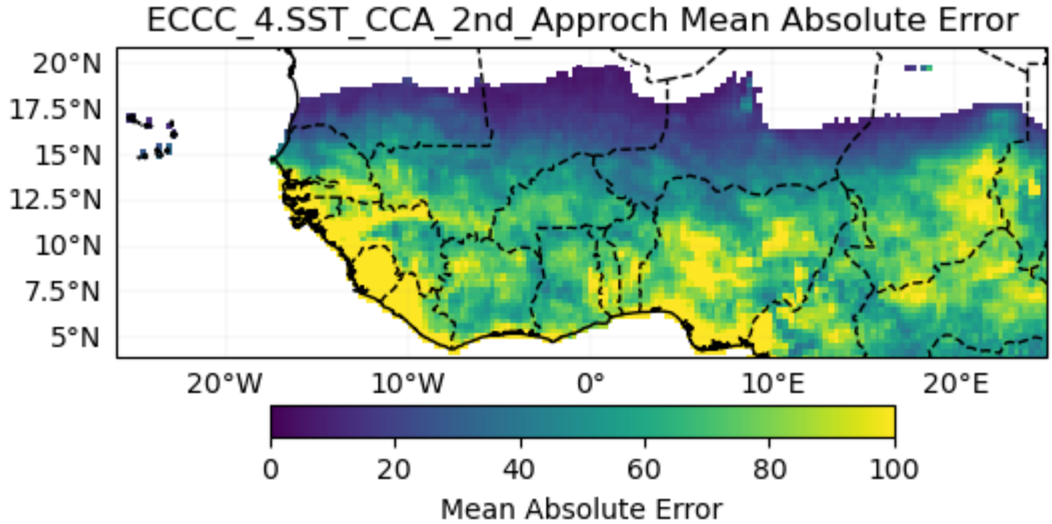
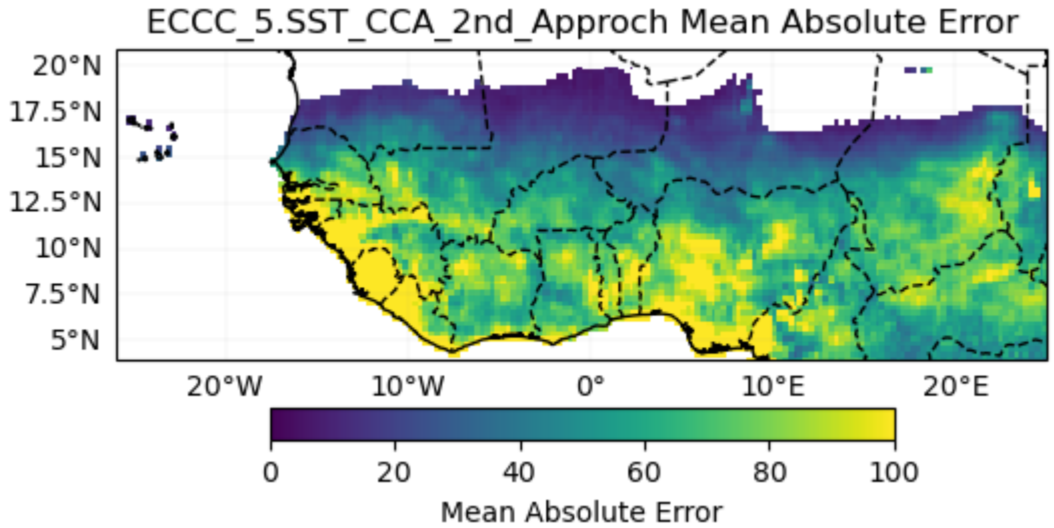
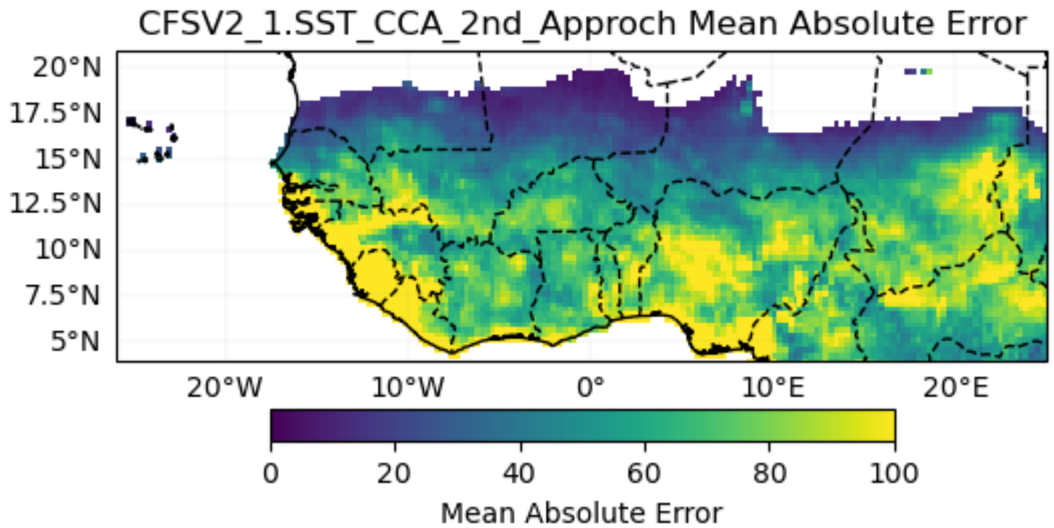
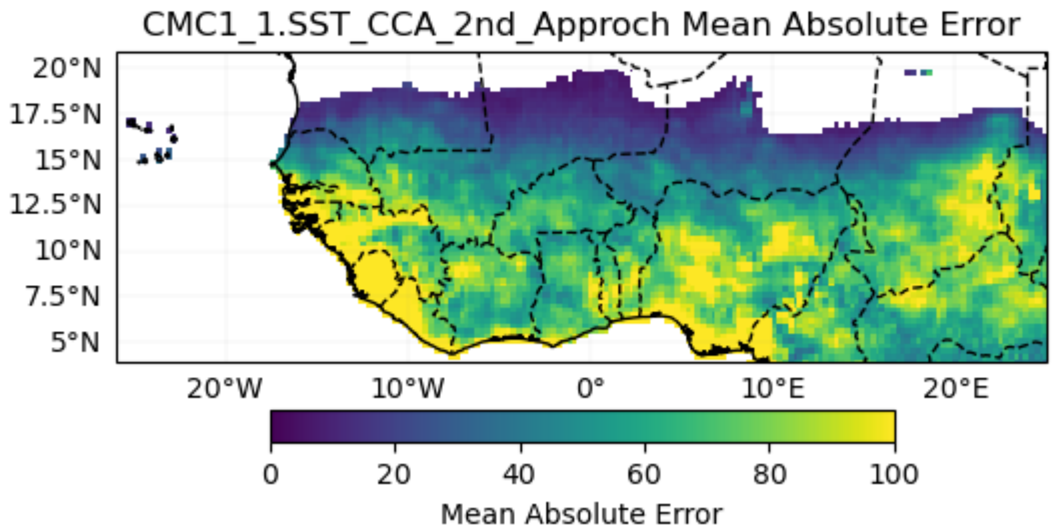
Mean Absolute Error


JMA_3.SST_CCA_2nd_Approch Mean Absolute Error
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0
20
40
60
80
100
Mean Absolute Error

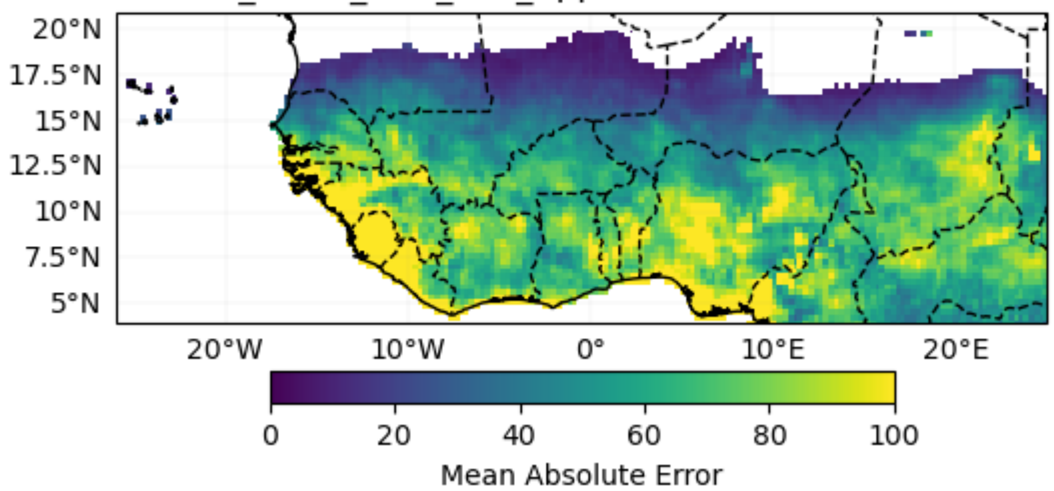

ECCC_4.SST_CCA_2nd_Approch Mean Absolute Error
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0
20
40
60
80
100
Mean Absolute Error


ECCC_5.SST_CCA_2nd_Approch Mean Absolute Error
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0
20
40
60
80
100
Mean Absolute Error

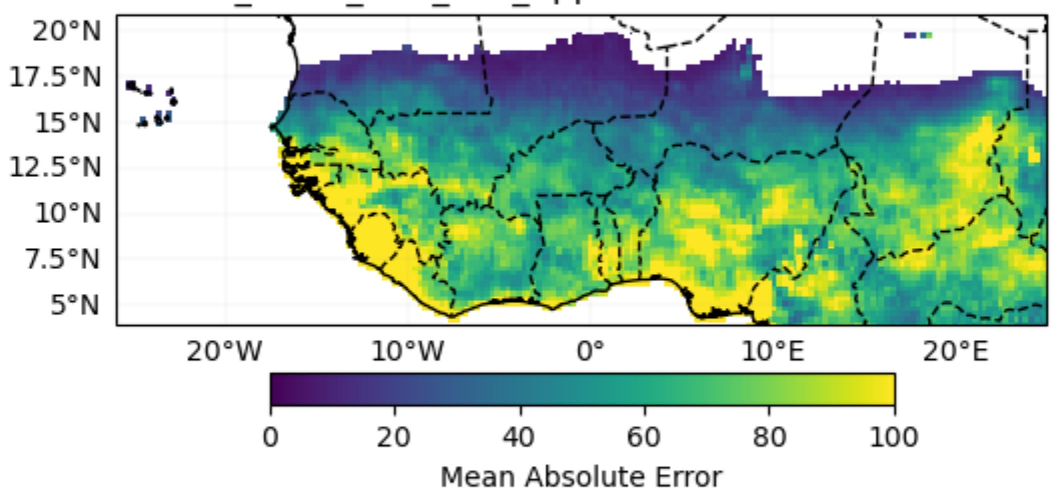

CFSV2_1.SST_CCA_2nd_Approch Mean Absolute Error
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0
20
40
60
80
100
Mean Absolute Error


CMC1_1.SST_CCA_2nd_Approch Mean Absolute Error
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0
20
40
60
80
100
Mean Absolute Error

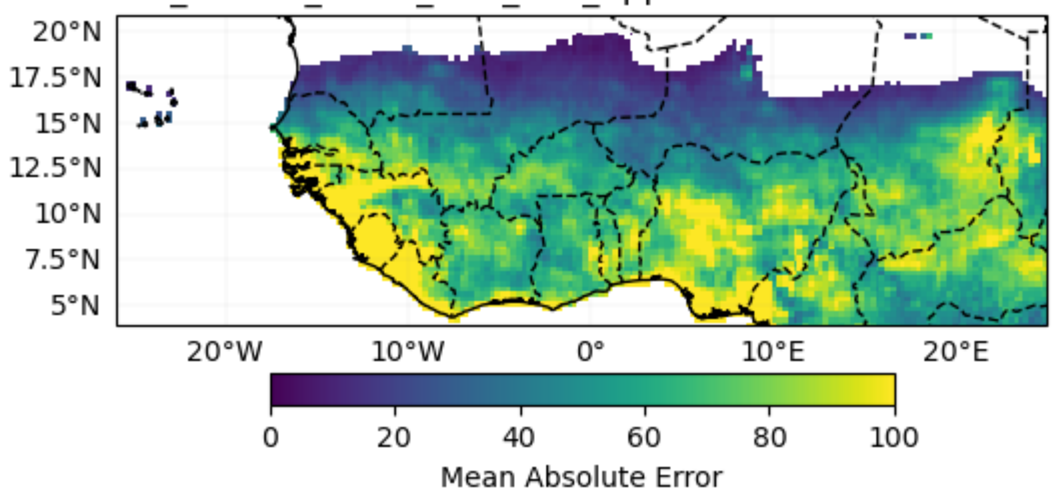
CMC2_1.SST_CCA_2nd_Approch Mean Absolute Error

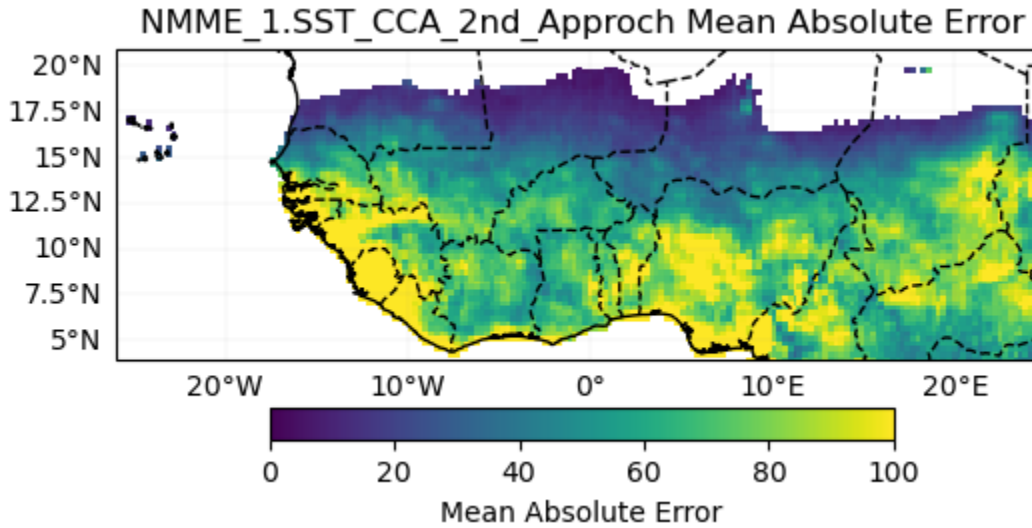


GFDL_1.SST_CCA_2nd_Approch Mean Absolute Error



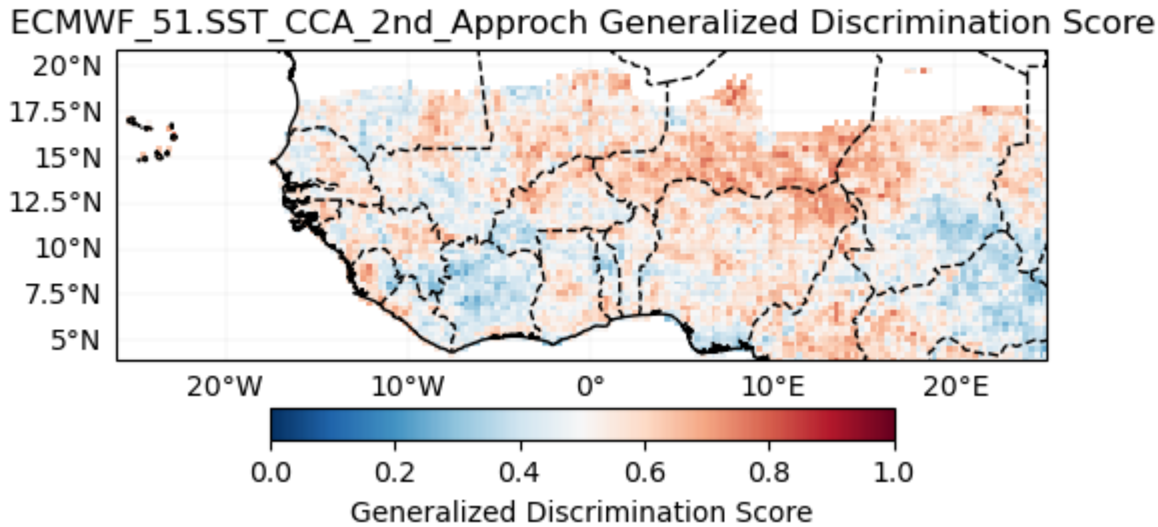
NCAR_CCSM4_1.SST_CCA_2nd_Approch Mean Absolute Error

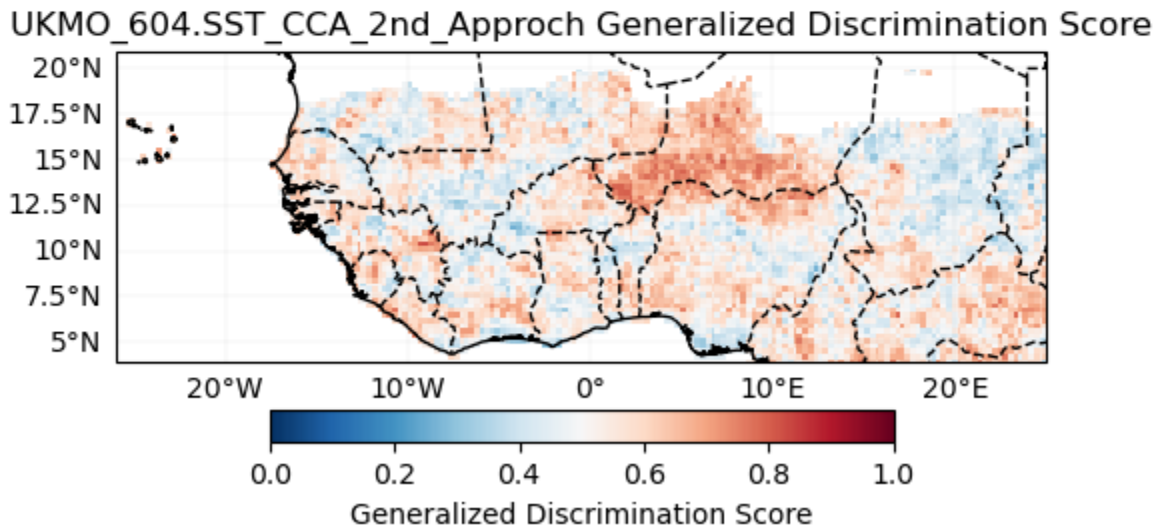


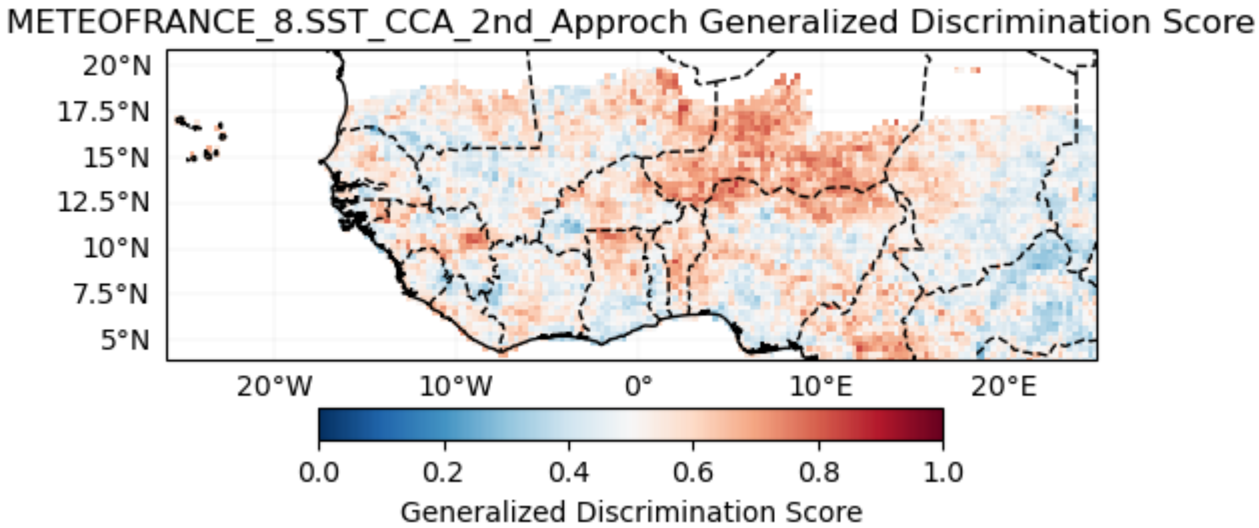


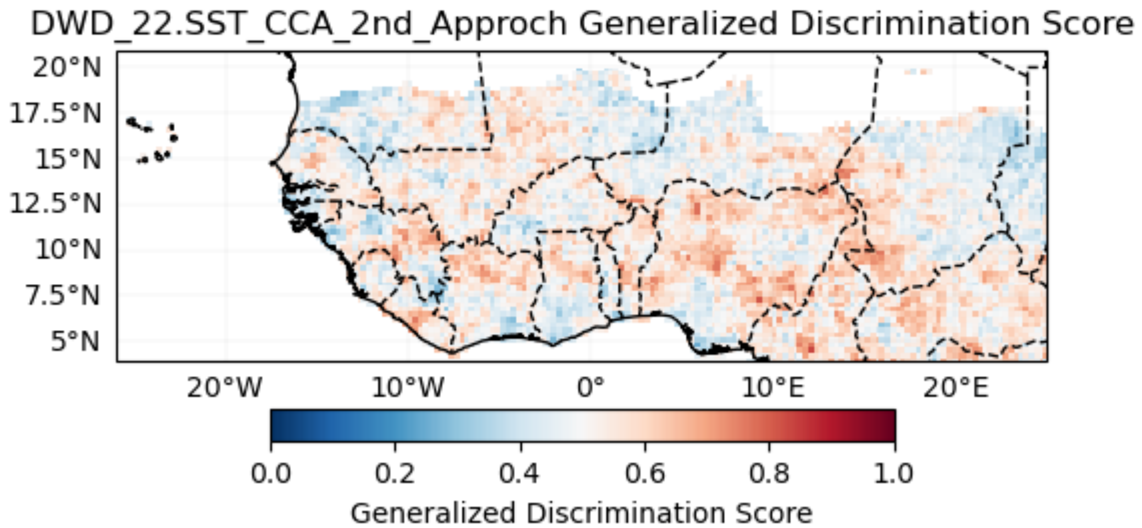
Probabilistic

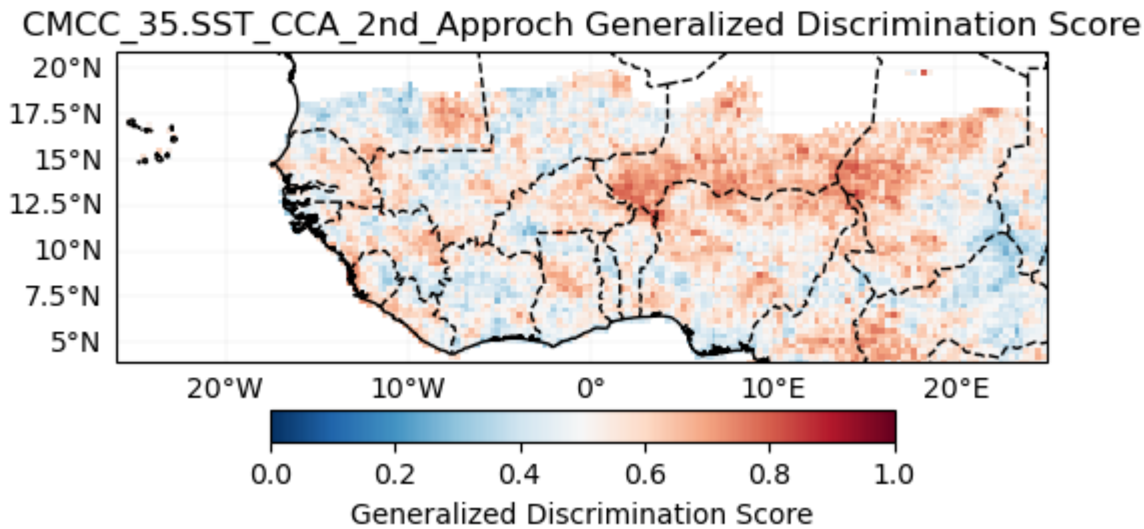
```
Generalized Discrimination Score
```

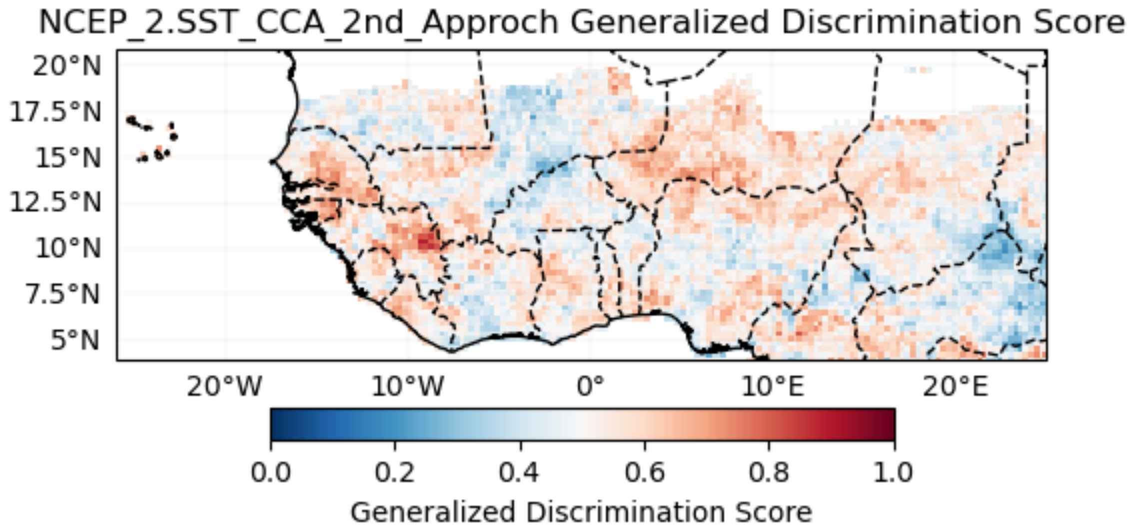


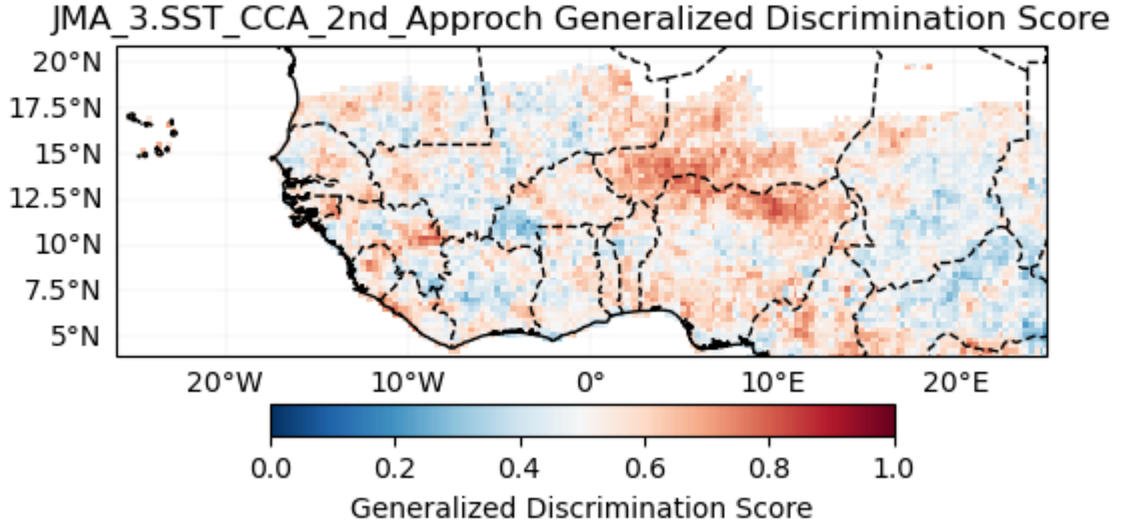


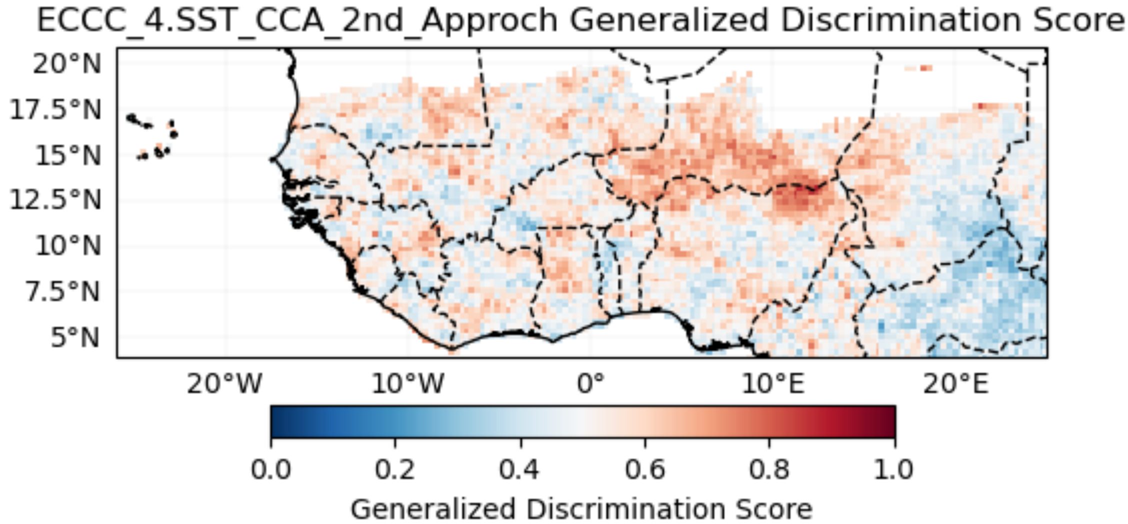

METEOFRANCE_8.SST_CCA_2nd_Approch Generalized Discrimination Score
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


DWD_22.SST_CCA_2nd_Approch Generalized Discrimination Score
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score

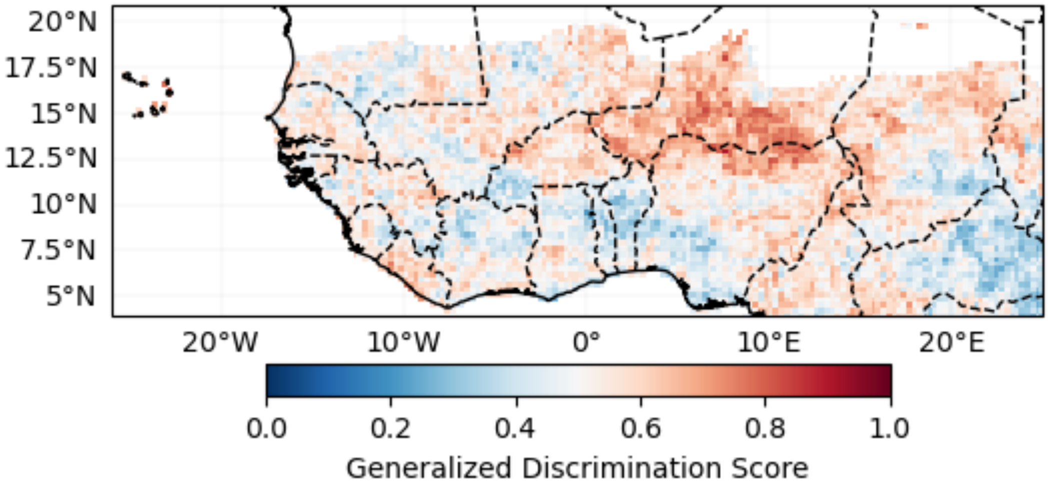

CMCC_35.SST_CCA_2nd_Approch Generalized Discrimination Score
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


NCEP_2.SST_CCA_2nd_Approch Generalized Discrimination Score
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score

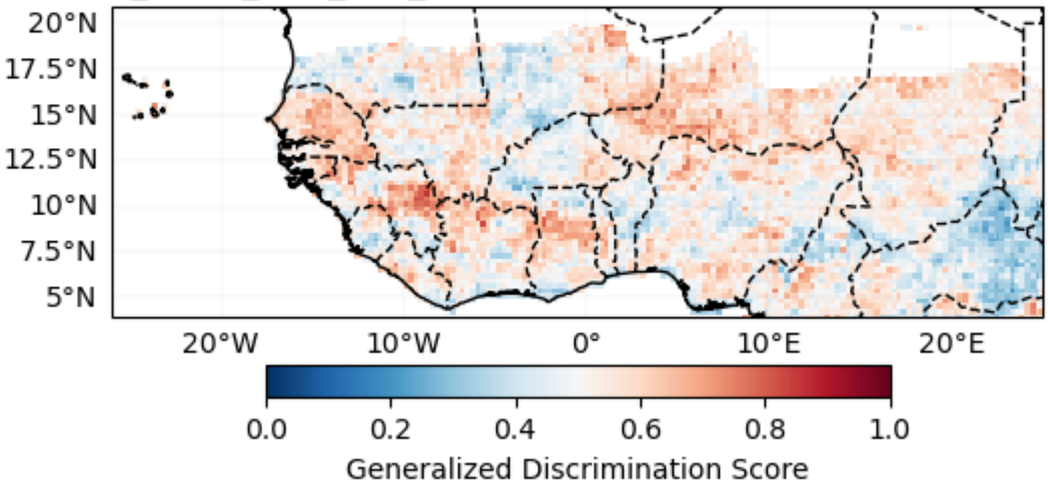

JMA_3.SST_CCA_2nd_Approch Generalized Discrimination Score
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


ECCC_4.SST_CCA_2nd_Approch Generalized Discrimination Score
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score

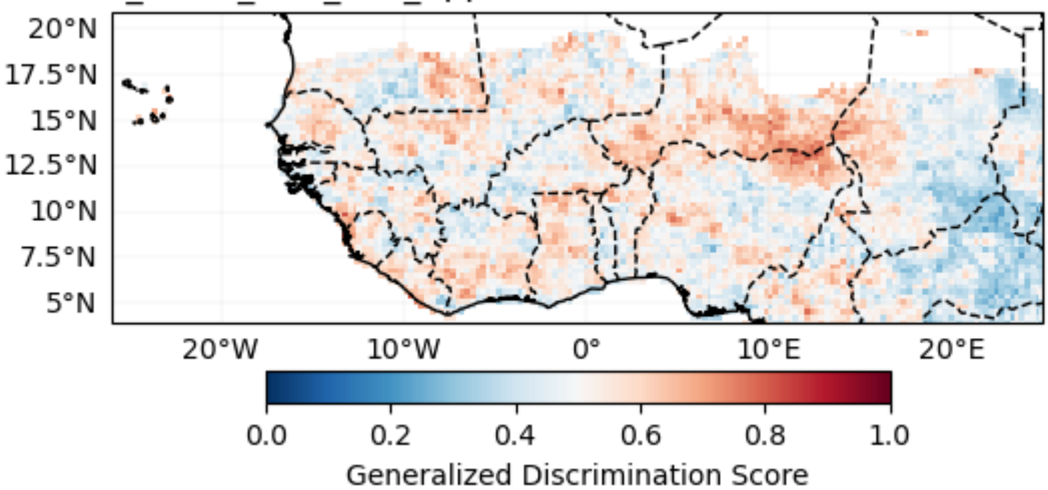
ECCC_5.SST_CCA_2nd_Approch Generalized Discrimination Score



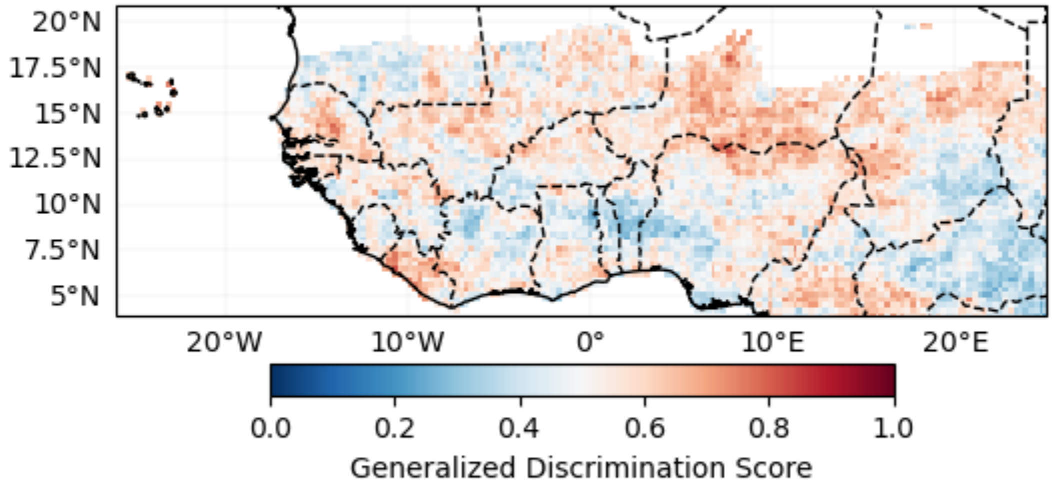
CFSV2_1.SST_CCA_2nd_Approch Generalized Discrimination Score



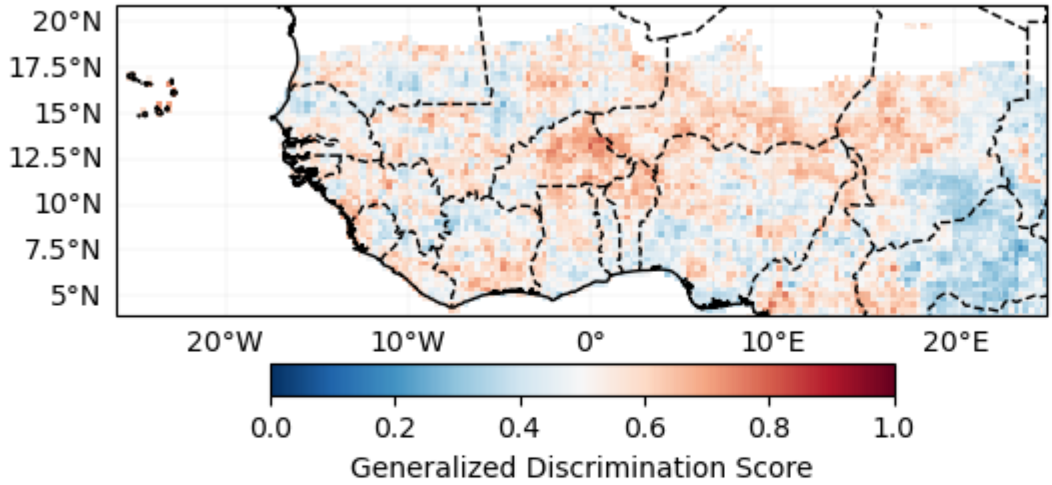
CMC1_1.SST_CCA_2nd_Approch Generalized Discrimination Score



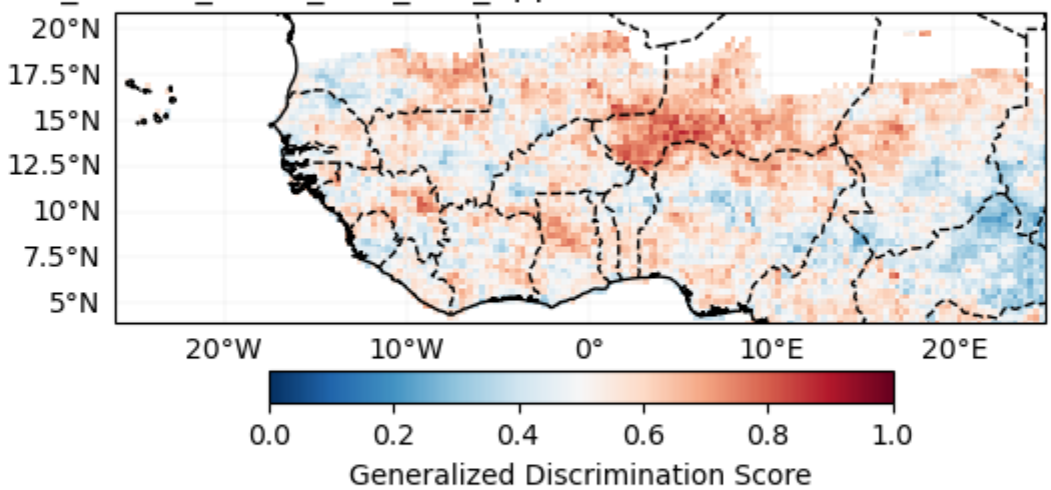
CMC2_1.SST_CCA_2nd_Approch Generalized Discrimination Score

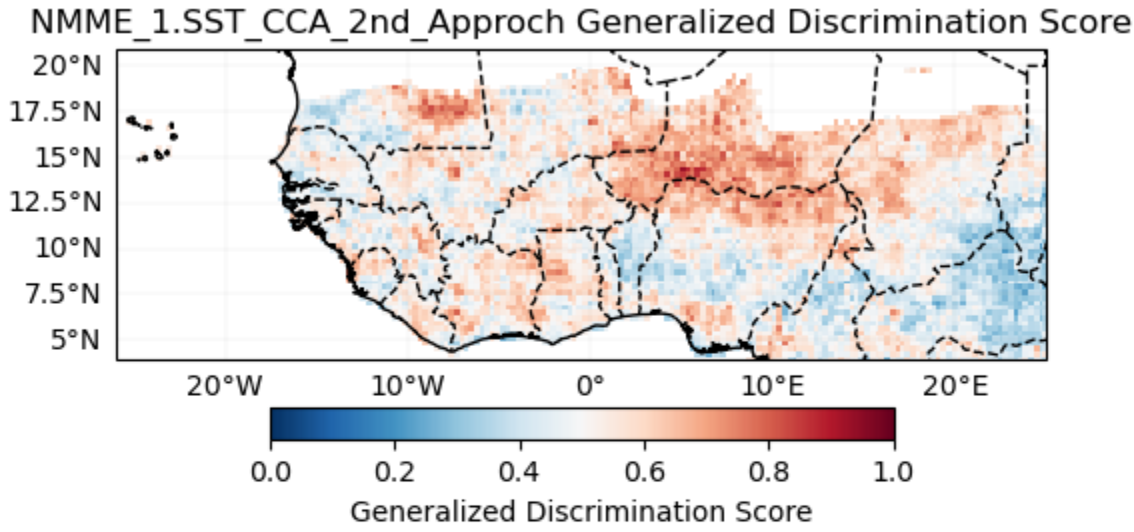


GFDL_1.SST_CCA_2nd_Approch Generalized Discrimination Score



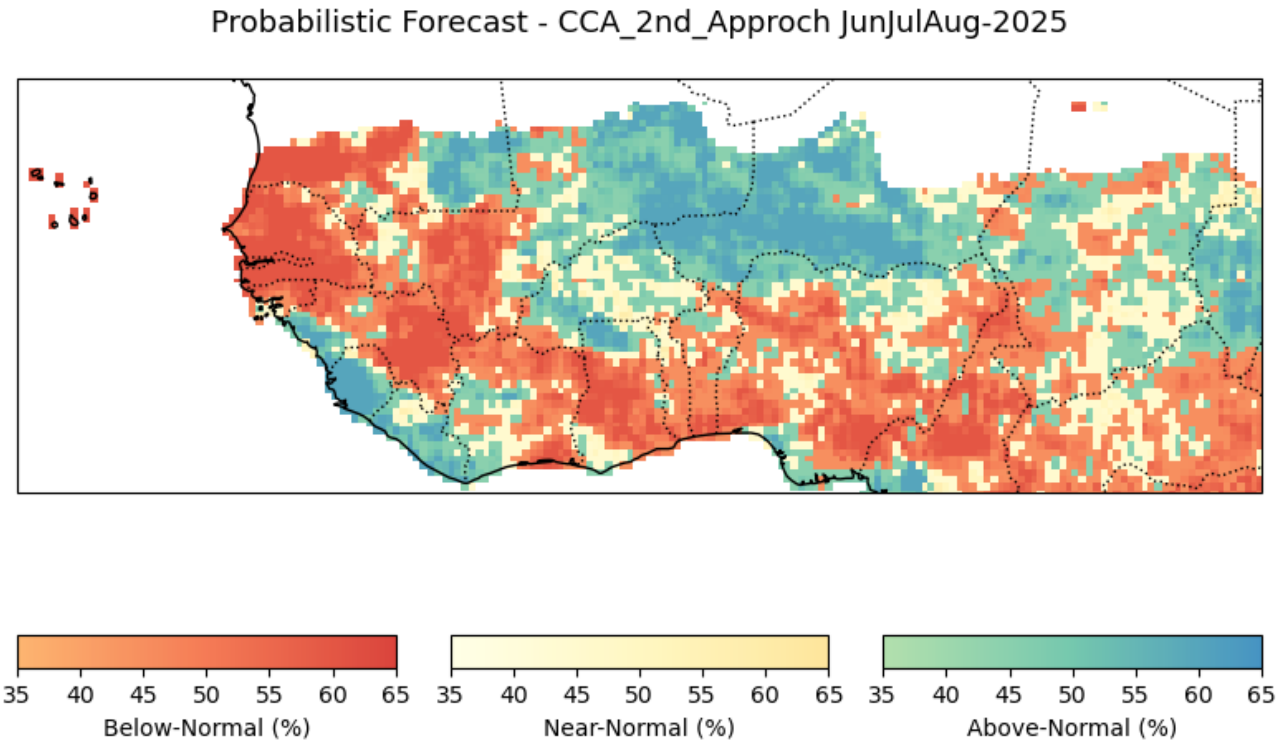
NCAR_CCSM4_1.SST_CCA_2nd_Approch Generalized Discrimination Score





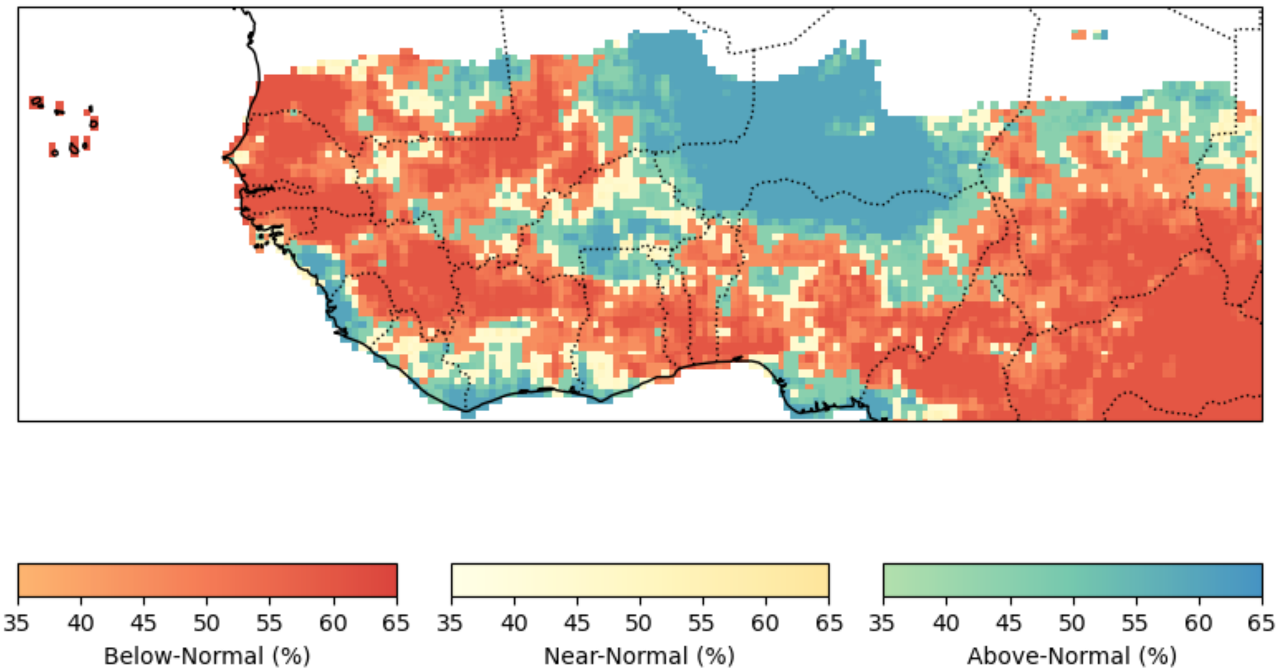
Forecast (individual)

ECMWF_51



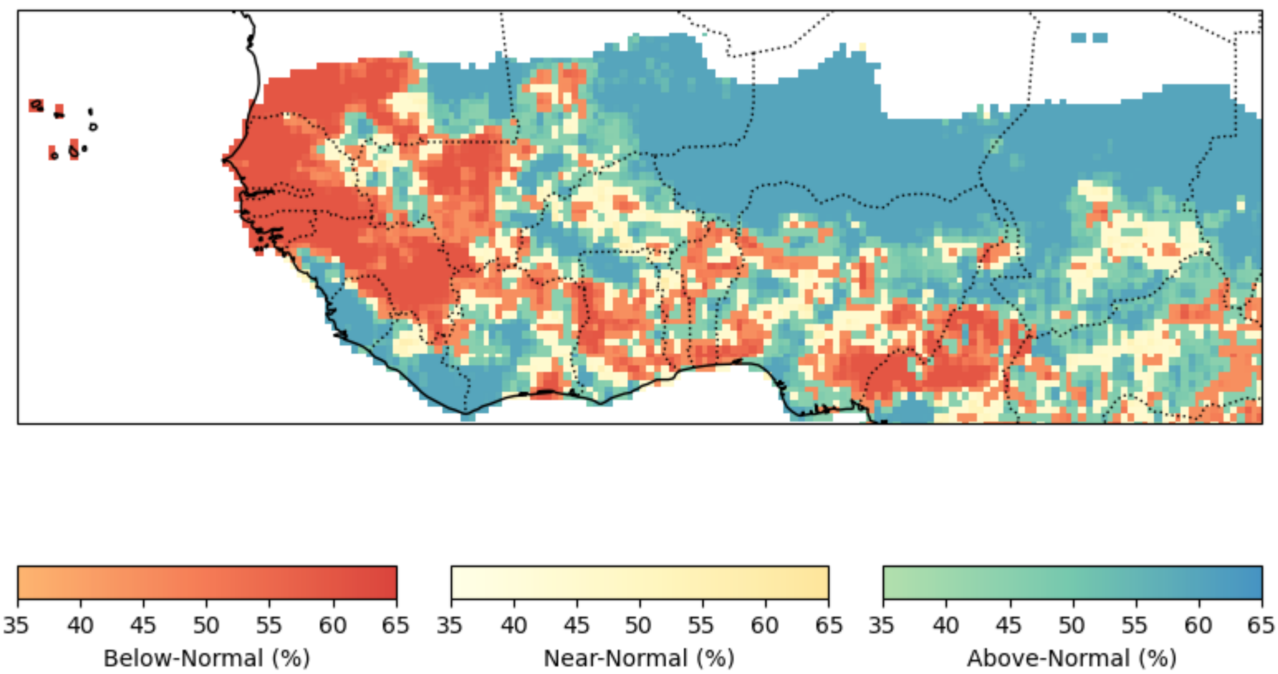
UKMO_604

Probabilistic Forecast - CCA_2nd_Approch JunJulAug-2025

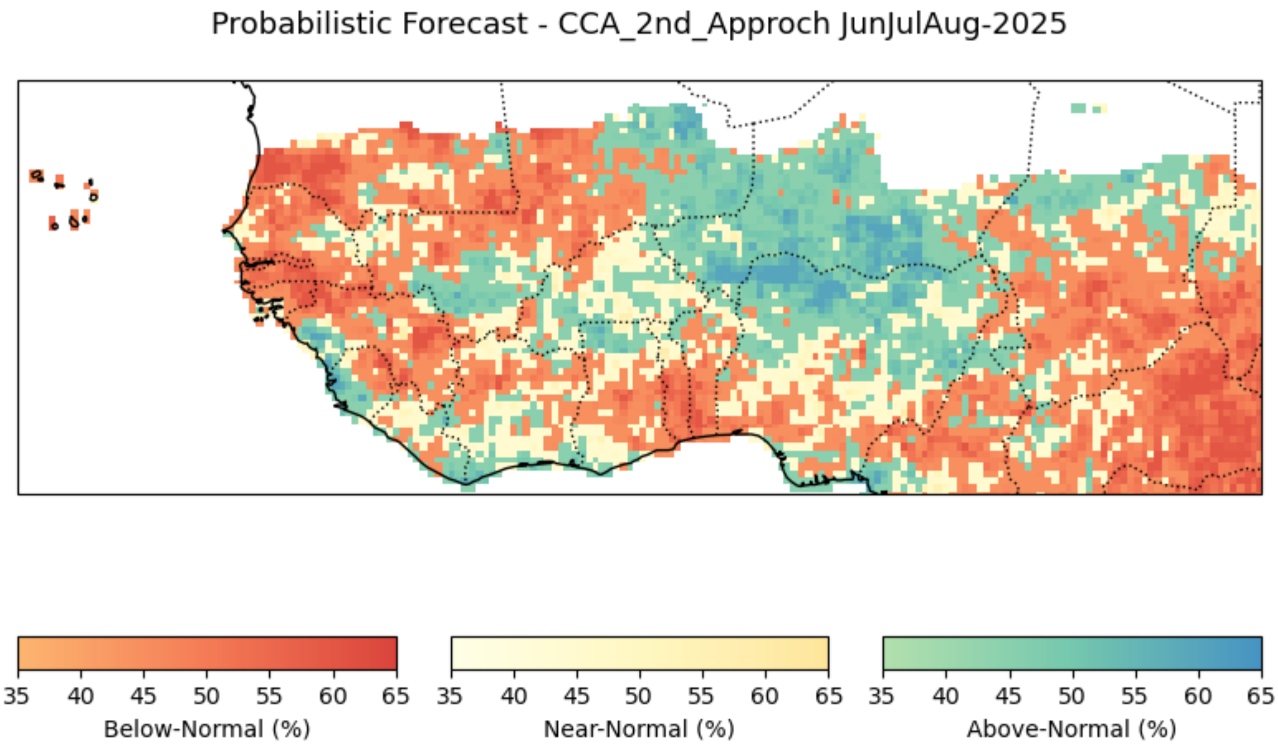


METEOFRANCE_8

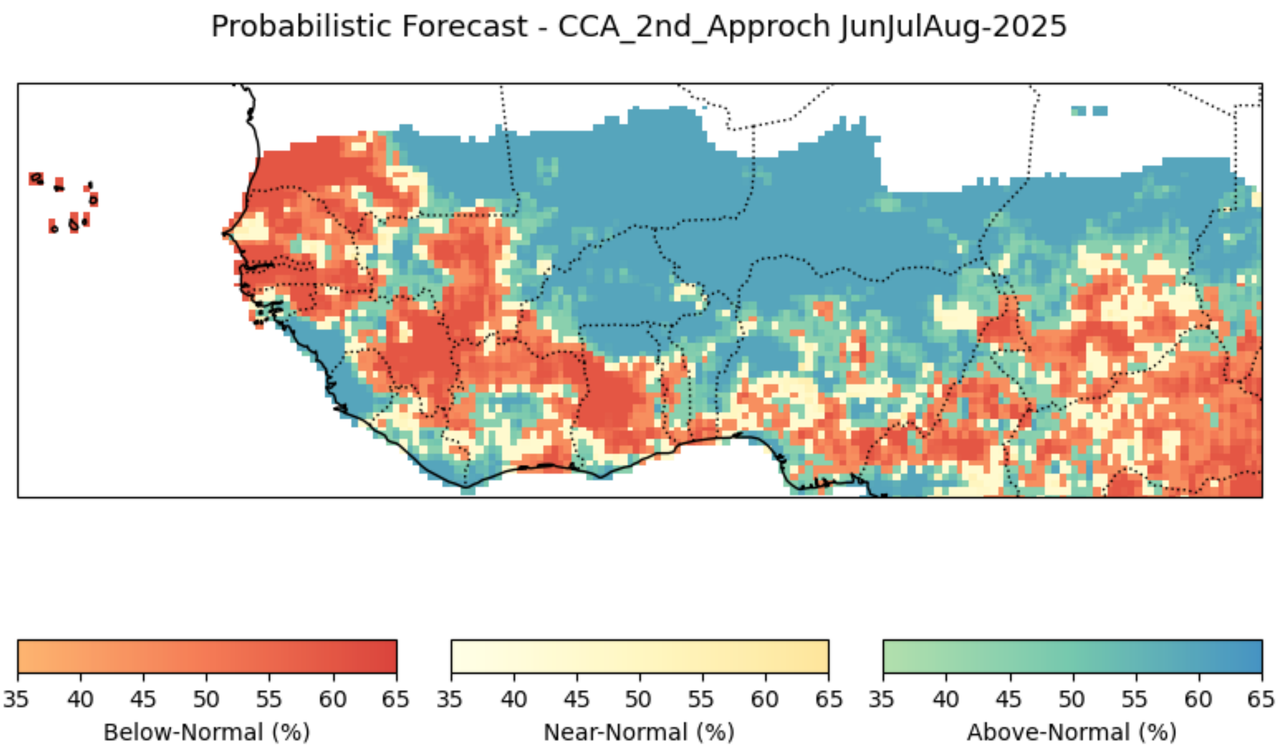
Probabilistic Forecast - CCA_2nd_Approch JunJulAug-2025



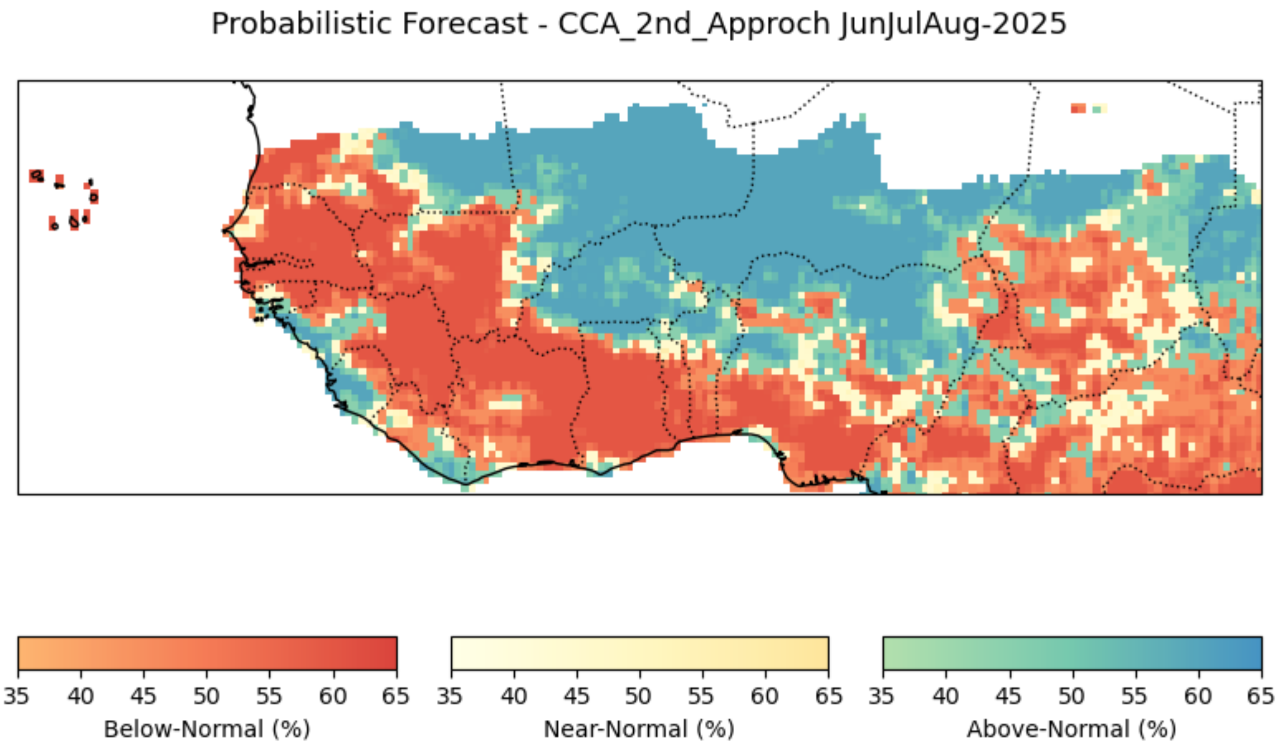
DWD_22



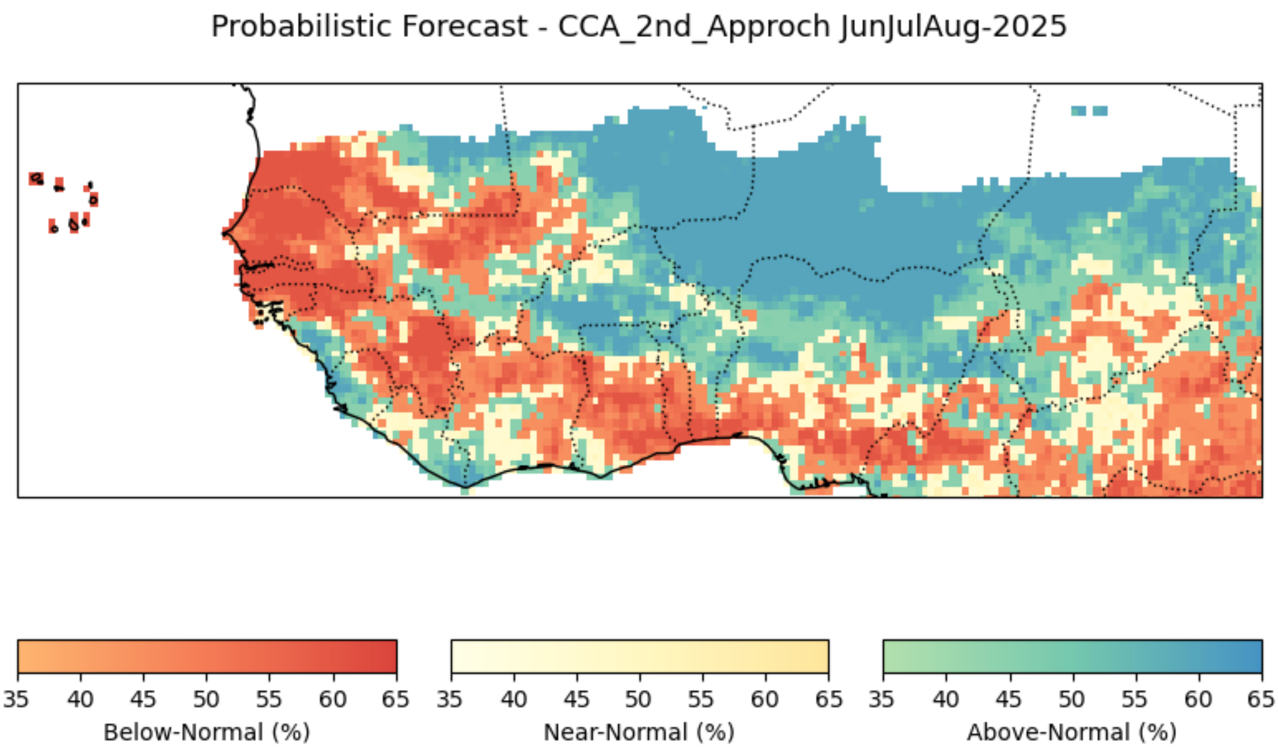
CMCC_35



NCEP_2

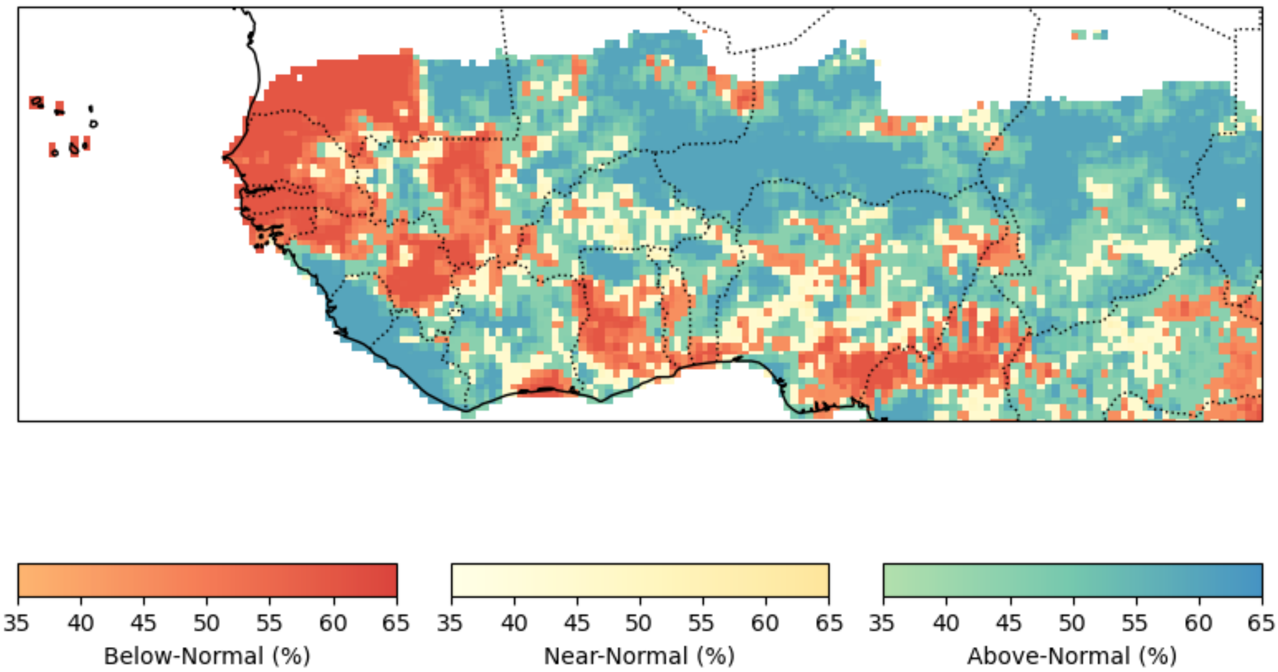

Probabilistic Forecast - CCA_2nd_Approch JunJulAug-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

JMA_3


Probabilistic Forecast - CCA_2nd_Approch JunJulAug-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

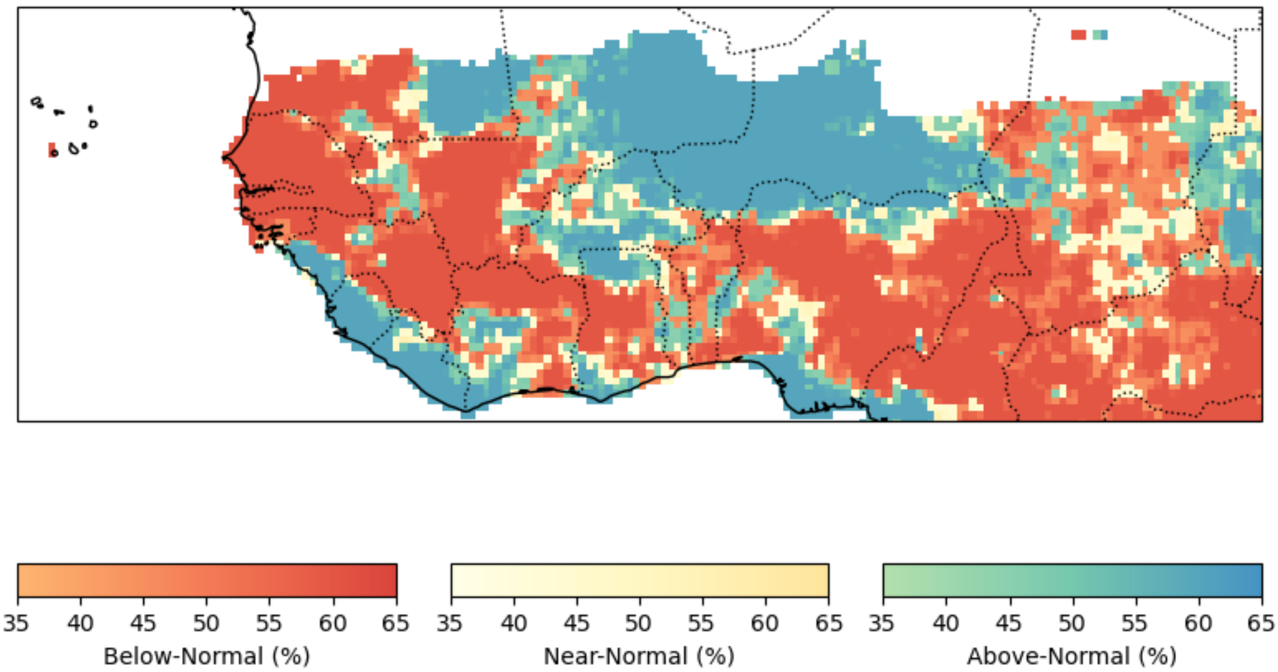
ECCC_4

Probabilistic Forecast - CCA_2nd_Approch JunJulAug-2025

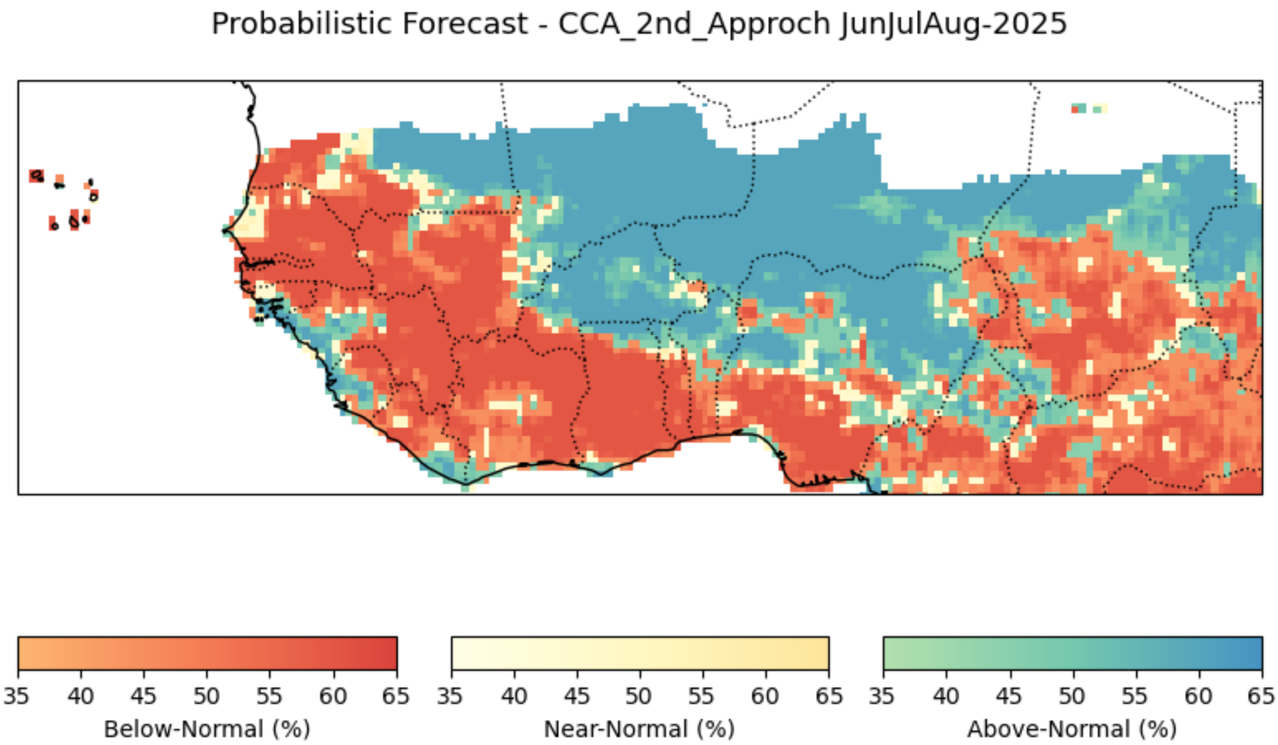


ECCC_5

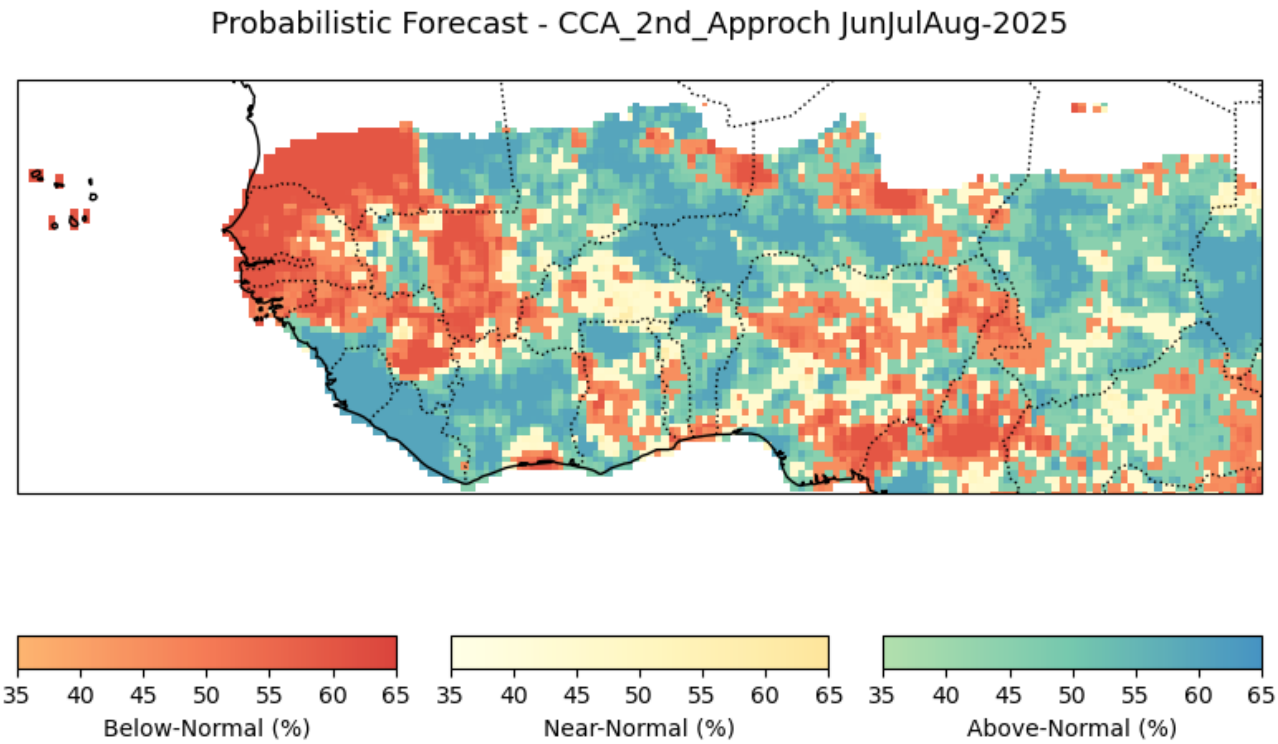
Probabilistic Forecast - CCA_2nd_Approch JunJulAug-2025



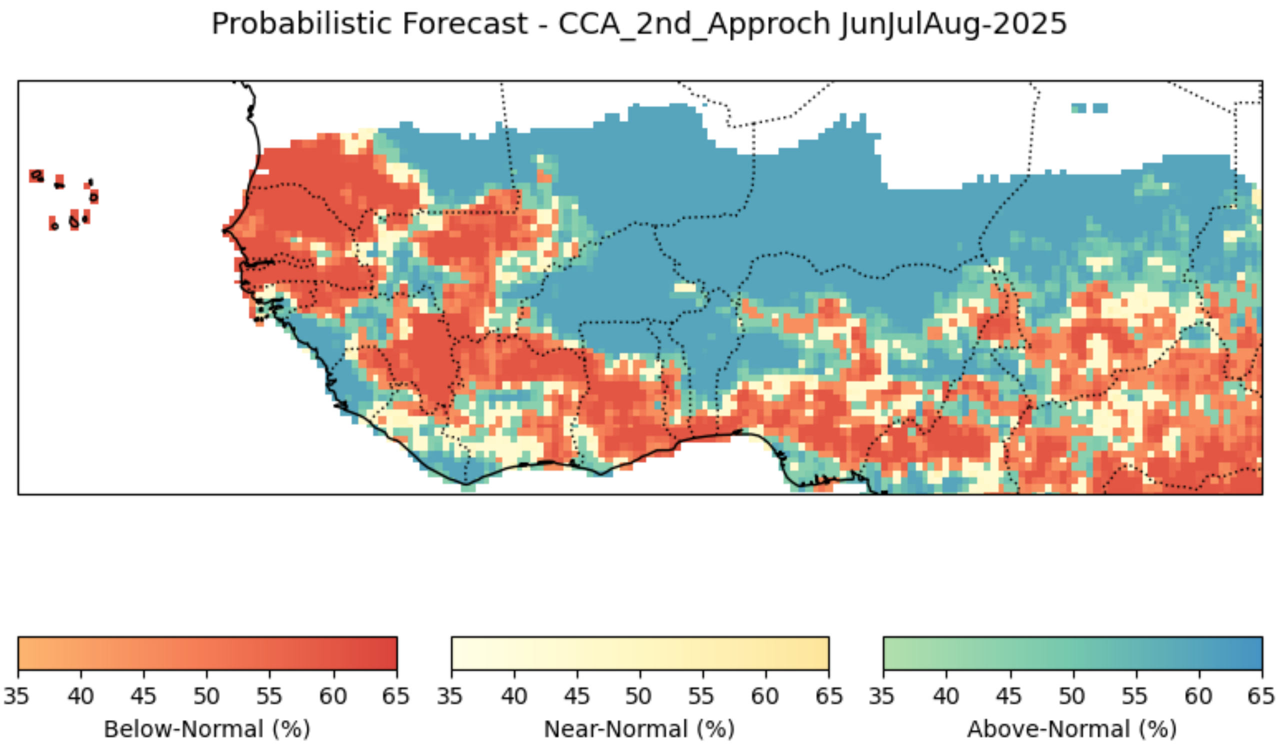
CFSV2_1


Probabilistic Forecast - CCA_2nd_Approch JunJulAug-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

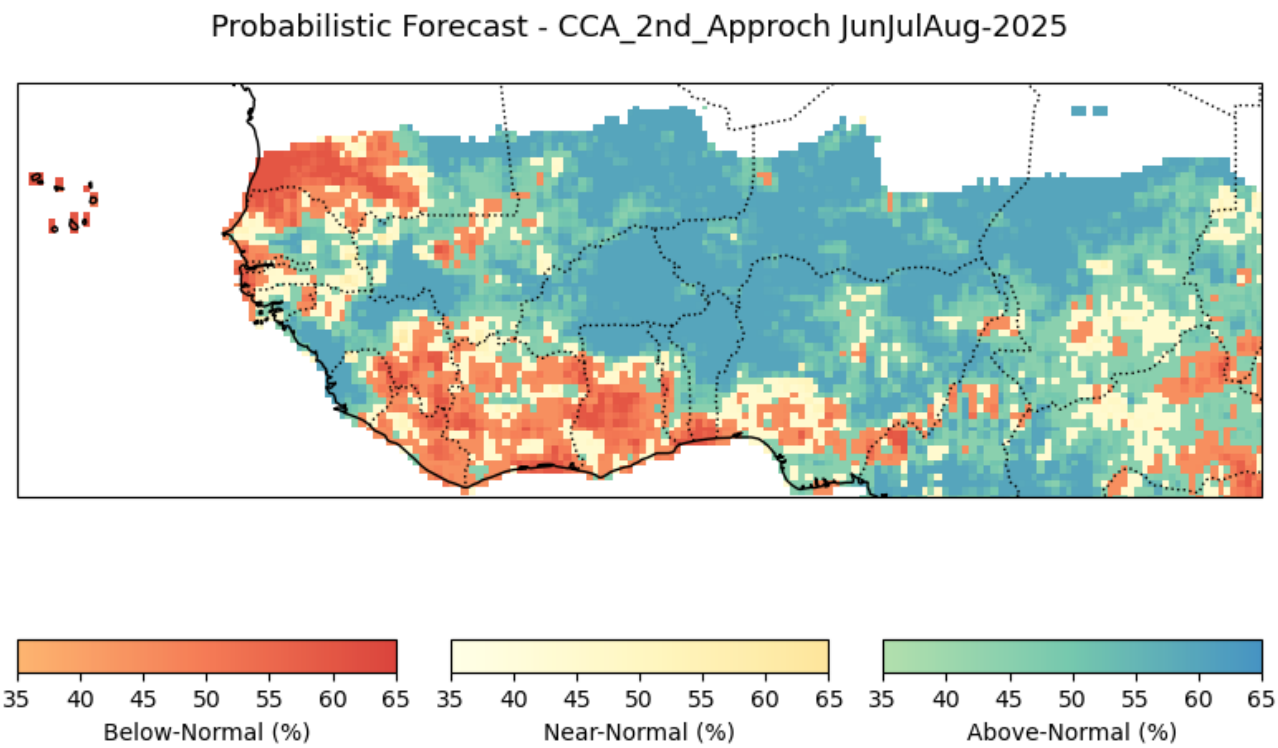
CMC1_1


Probabilistic Forecast - CCA_2nd_Approch JunJulAug-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

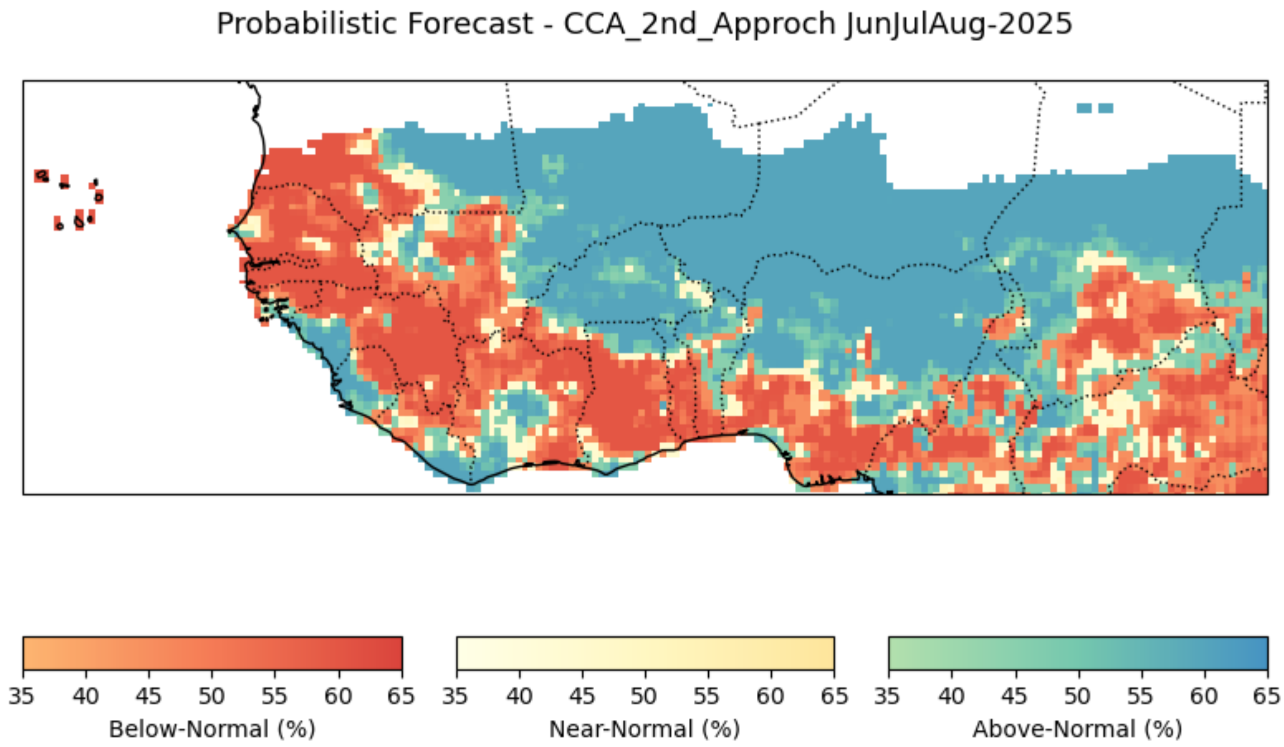
CMC2_1



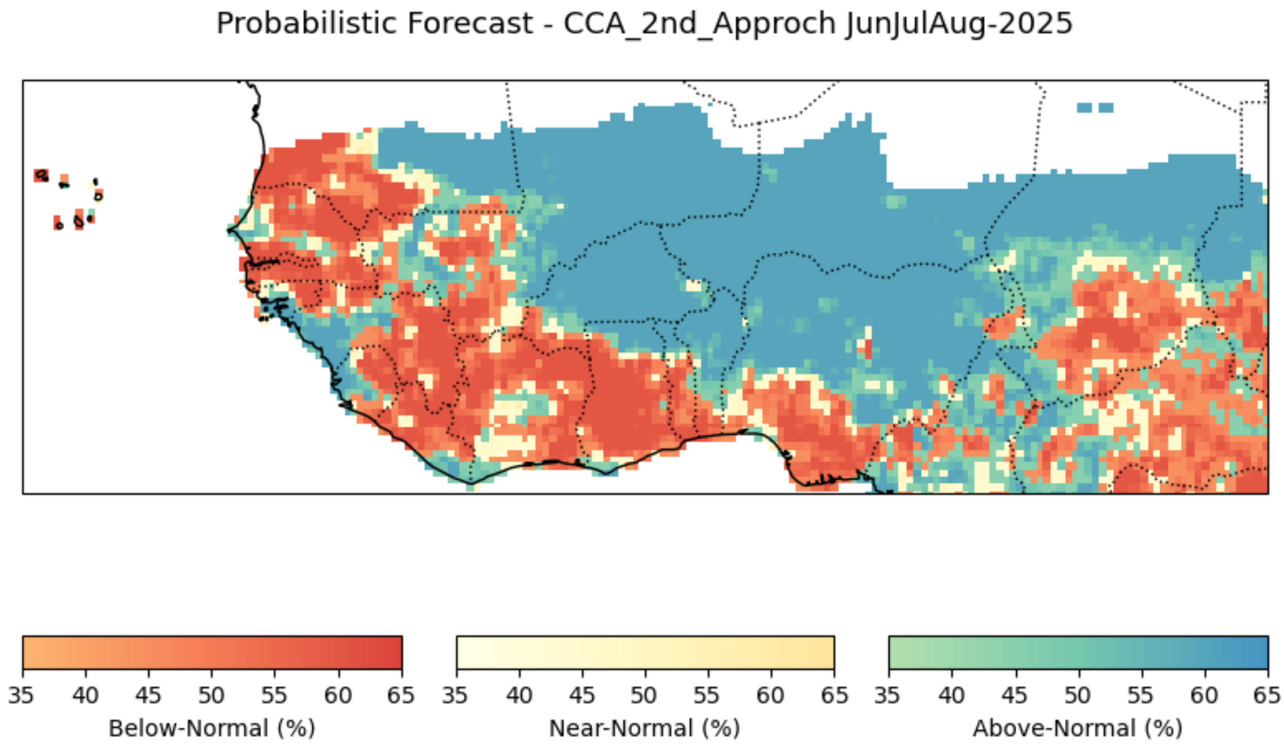
GFDL_1



NCAR_CCSM4_1




```
NMME_1
```

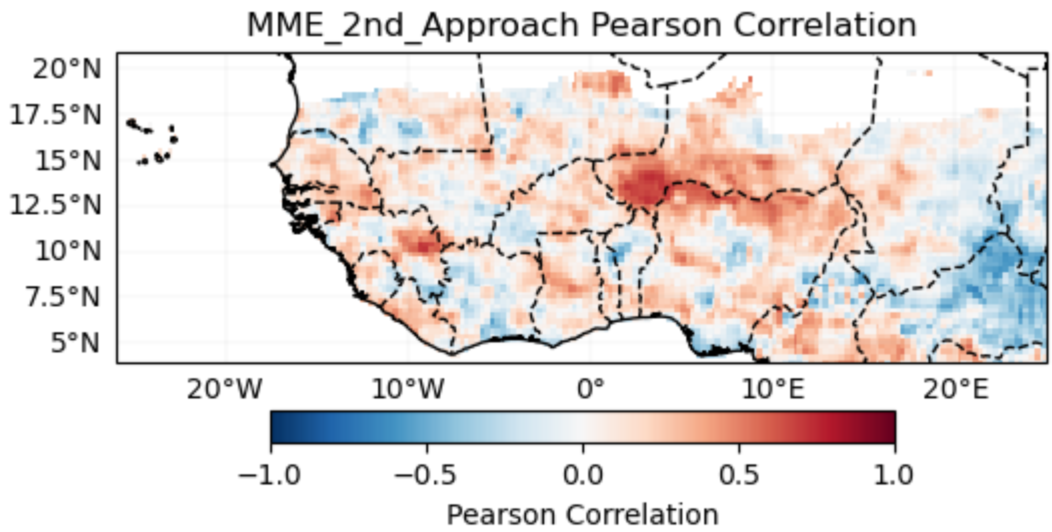


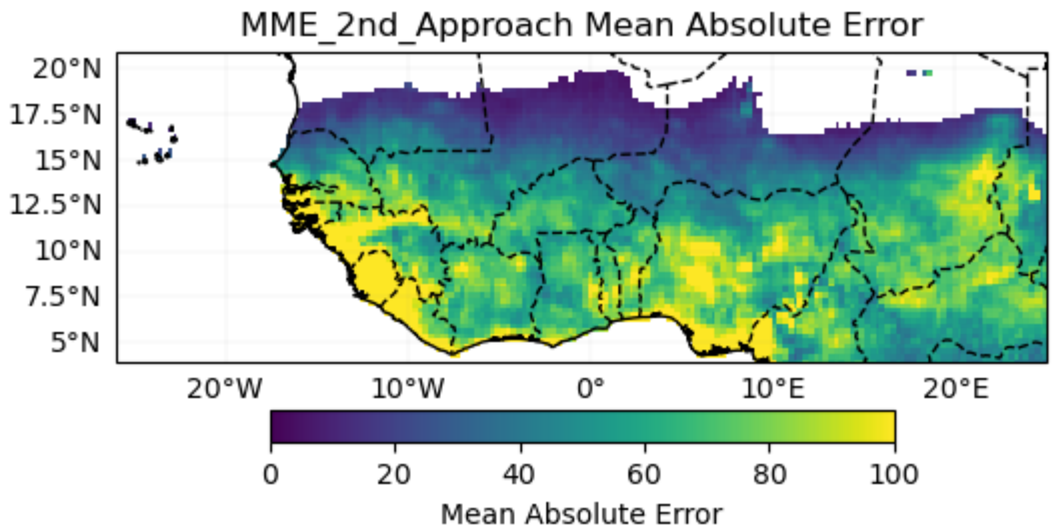
MME 2nd-Approach: Statistico-Dynamical

```
['CMCC_35.SST',
 'UKMO_604.SST',
 'ECCC_5.SST',
 'NCAR_CCSM4_1.SST',
 'ECMWF_51.SST',
 'CFSV2_1.SST',
 'JMA_3.SST']
```

Verification

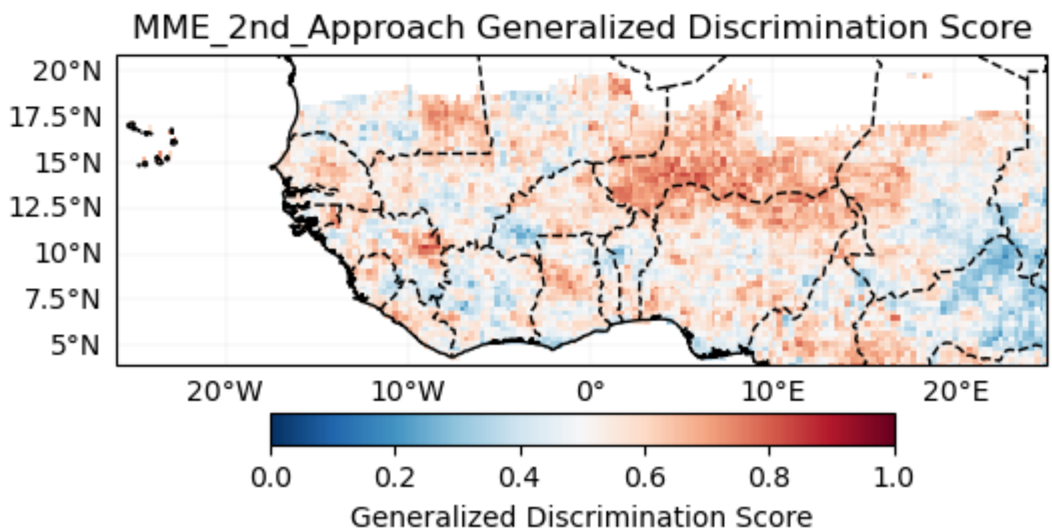
Deterministic



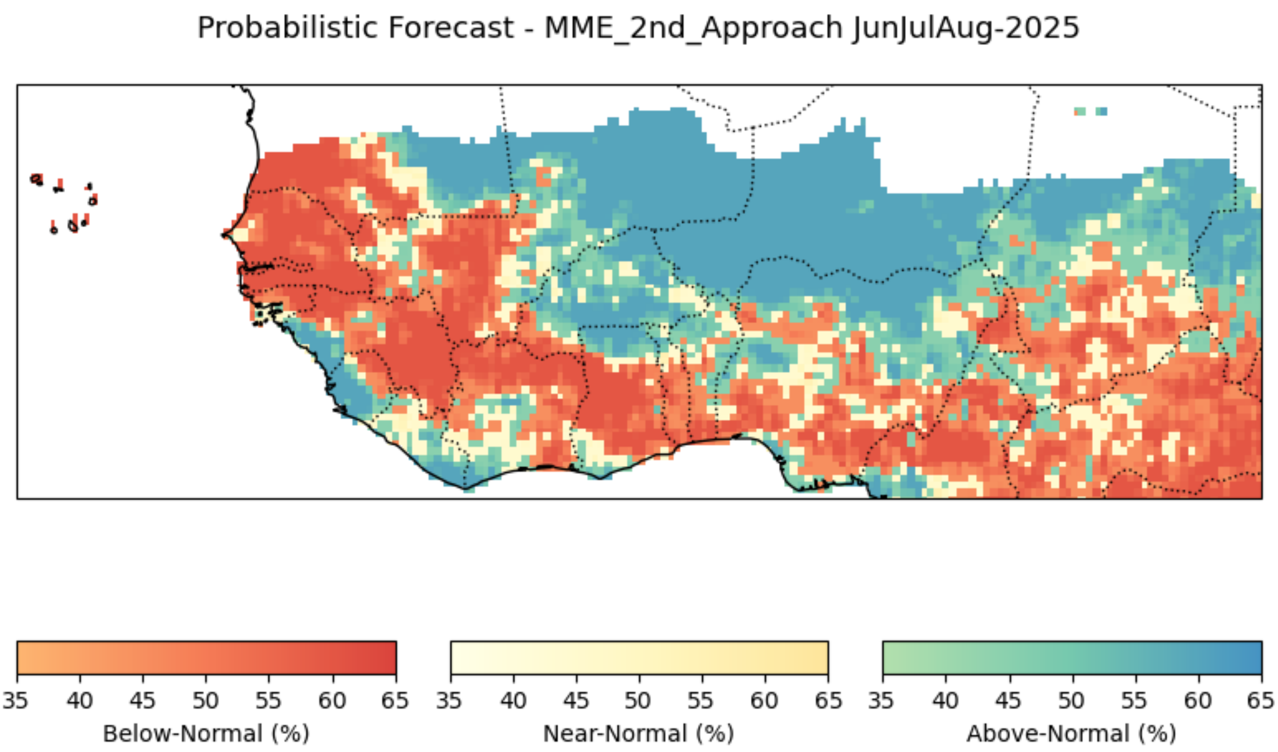


Probabilistic

```
Generalized Discrimination Score
```



Forecast



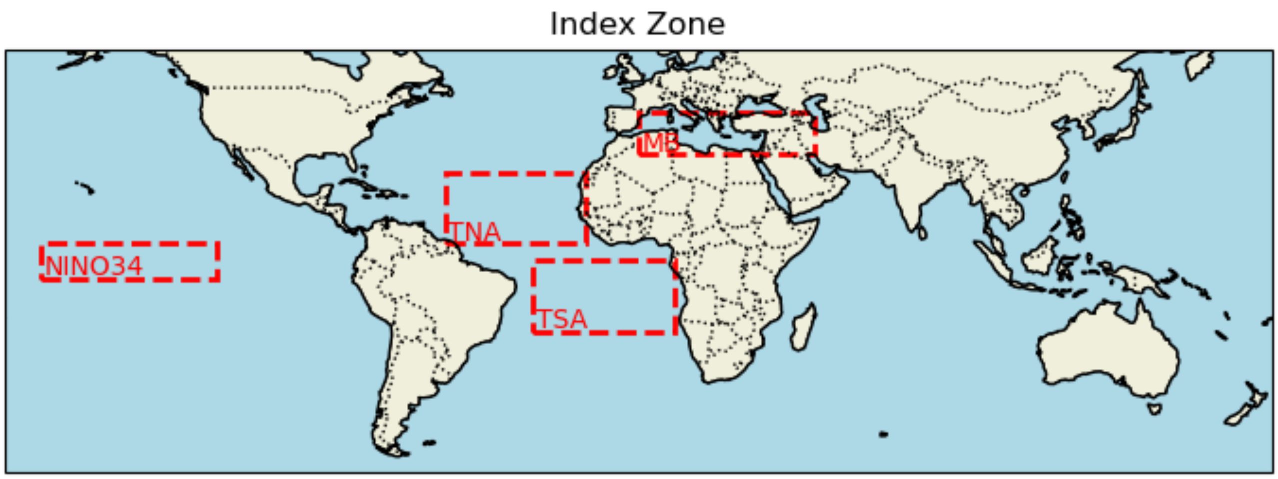

```
144300
```

3rd - Approach: Observational-based seasonal forecast

```
JJA_2025/reanalysis_data/NOAA_SST_1991_2025_JanFebMar.nc exists. Skipping.
```

Process SST index

```
['NINO34', 'NINO12', 'NINO3', 'NINO4', 'NINO_Global', 'TNA', 'TSA', 'NAT', 'SAT',
'TASI', 'WTIO', 'SETIO', 'DMI', 'MB']
```




```
NOAA SST
```

Calculate variance inflation factor to see multicolinearity predictor

```
  feature       VIF
0  NINO34  1.268913
1     TNA  1.698945
2     TSA  2.233701
3     DMI  1.298347
4      MB  1.718974
```

```
['NINO34', 'TNA', 'TSA', 'DMI', 'MB']
```

Initialize WAS_LinearRegression_Model

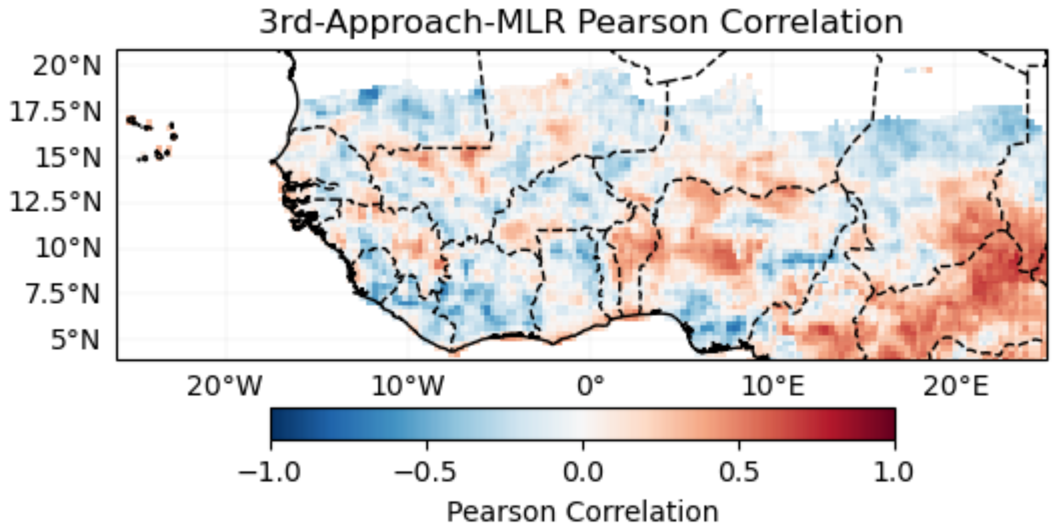
```
8
```

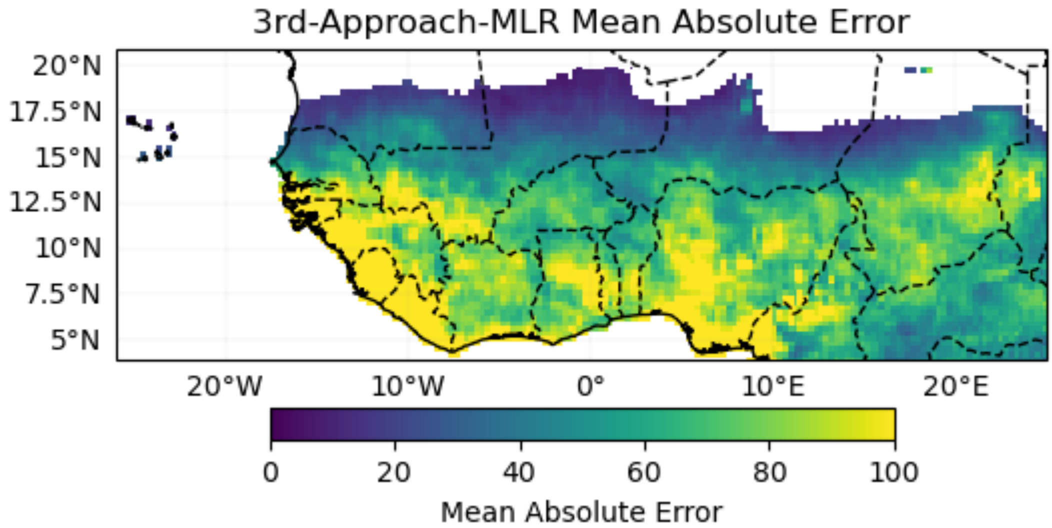
Cross validation

```
Cross-validation ongoing
100%|████████████████████████████████████████████| 34/34 [08:30<00:00, 15.01s/it]
```

Verification

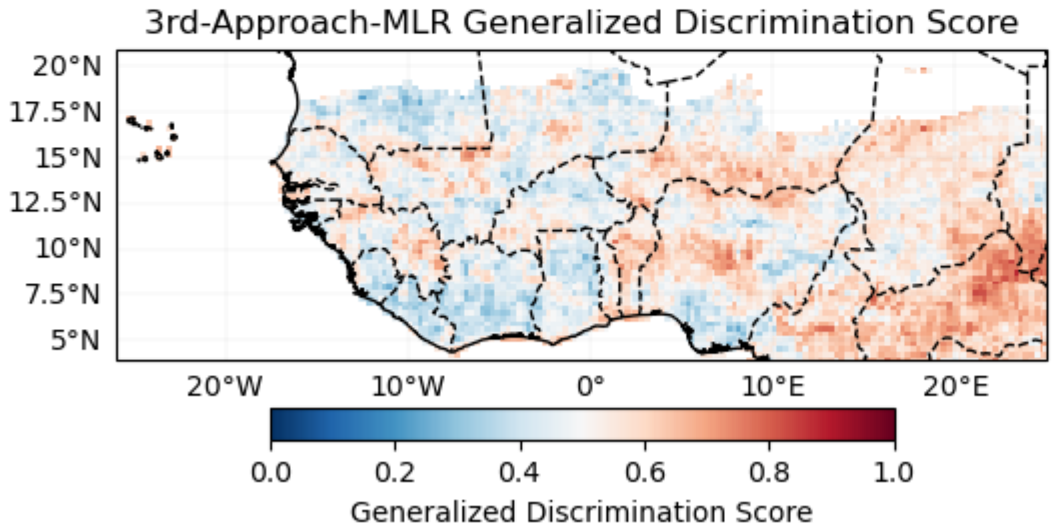
Deterministic



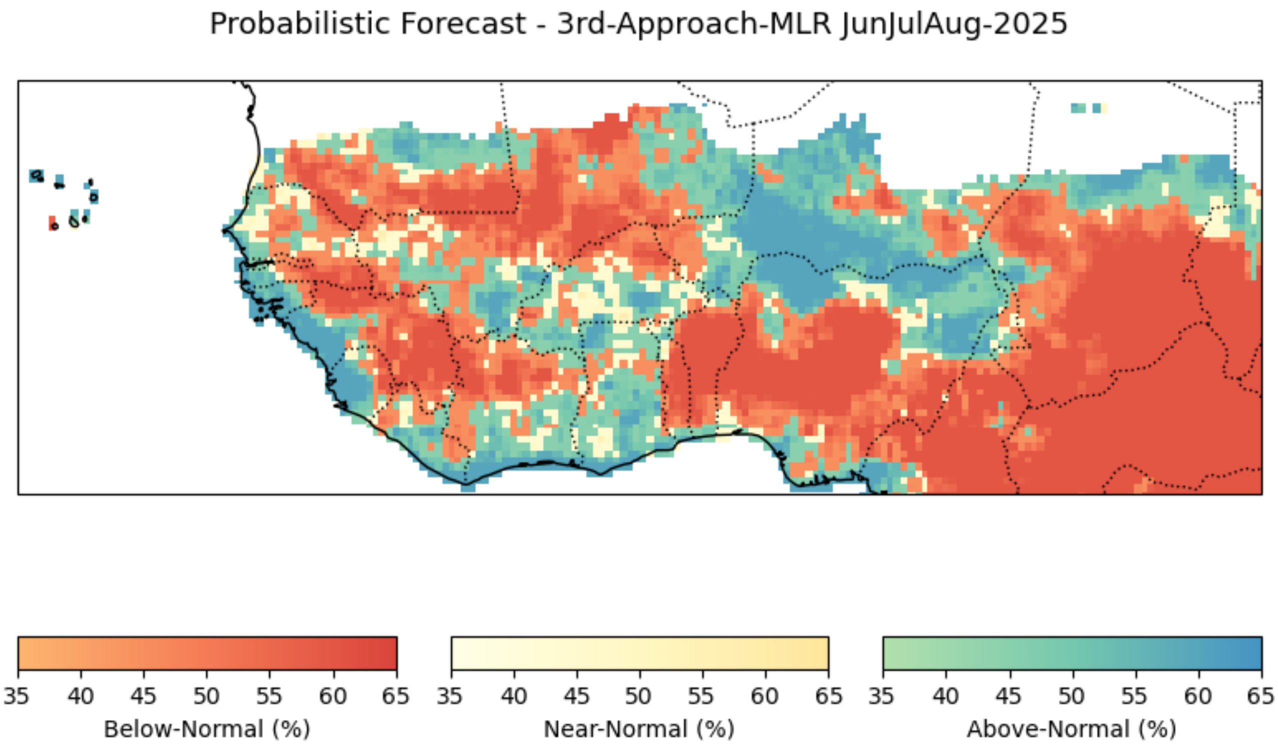


Probabilistic

```
Generalized Discrimination Score
```



Forecast



4th - Analog-based seasonal forecast

Initialize WAS_Analog class

Cross-validation

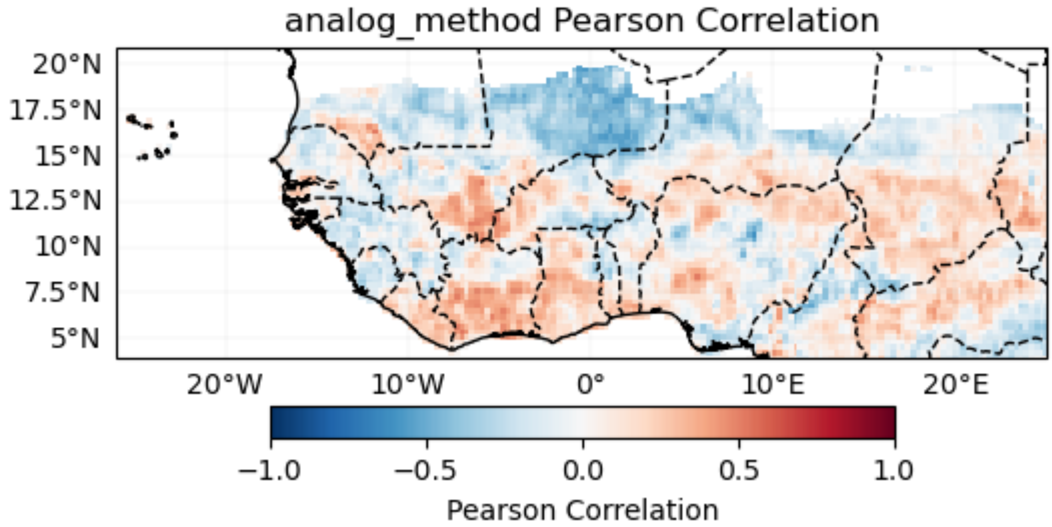
```
JJA_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc alrea
dy exists. Skipping download.
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization m
onth
JJA_2025/analog/forecast_ecmwf51_SST_MarIc_AprMayJunJulAug.nc
File 'JJA_2025/analog/forecast_ecmwf51_SST_MarIc_AprMayJunJulAug.nc' already exist
s. Skipping download.
All requested SST data has been processed.
Cross-validation ongoing
  3%|█                                 | 1/34 [00:00<00:06,  5.40it/s]
similar years for 1991 are [2002 1997 2023]
  6%|██                                | 2/34 [00:00<00:05,  5.45it/s]
similar years for 1992 are [2020 2019 2005]
  9%|███                               | 3/34 [00:00<00:05,  5.46it/s]
similar years for 1993 are [2020 1991 1996]
 12%|████                              | 4/34 [00:00<00:05,  5.47it/s]
similar years for 1994 are [2003 1991 2009]
 15%|█████                             | 5/34 [00:00<00:05,  5.49it/s]
similar years for 1995 are [2016 1998 2020]
 18%|██████                            | 6/34 [00:01<00:05,  5.47it/s]
similar years for 1996 are [1993 2018 2002]
 21%|███████                           | 7/34 [00:01<00:05,  5.40it/s]
similar years for 1997 are [2023 2015 2009]
```

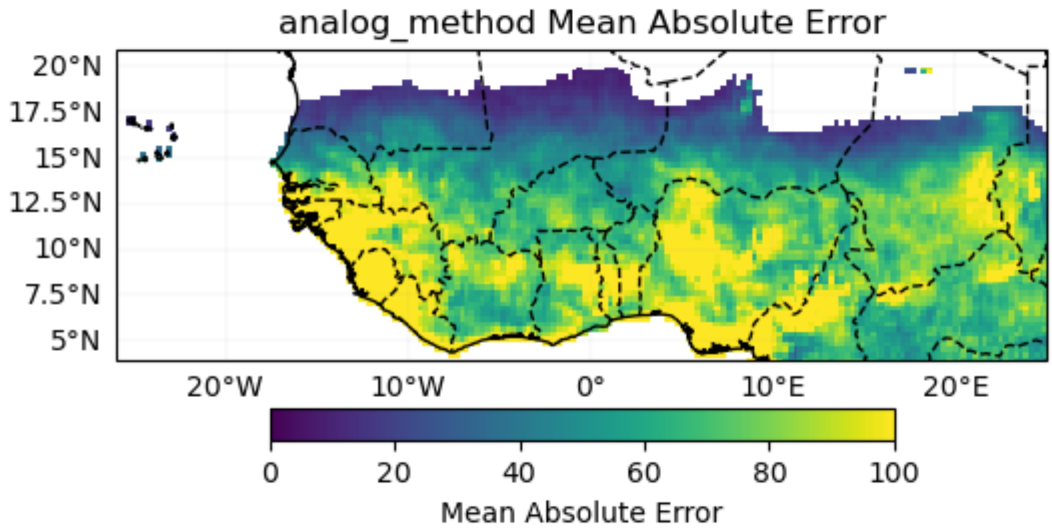
```
 24%|██████████                                 | 8/34 [00:01<00:04,  5.45it/s]
similar years for 1998 are [2010 2016 2024]
 26%|███████████                                | 9/34 [00:01<00:04,  5.42it/s]
similar years for 1999 are [2021 2011 2008]
 29%|████████████                               | 10/34 [00:01<00:04,  5.38it/s]
similar years for 2000 are [2012 2011 2009]
 32%|█████████████                              | 11/34 [00:02<00:04,  5.33it/s]
similar years for 2001 are [2023 2012 2017]
 35%|███████████████                            | 12/34 [00:02<00:04,  5.35it/s]
similar years for 2002 are [2023 1991 2018]
 38%|████████████████                           | 13/34 [00:02<00:03,  5.37it/s]
similar years for 2003 are [2024 1998 2016]
 41%|█████████████████                          | 14/34 [00:02<00:03,  5.37it/s]
similar years for 2004 are [2007 2002 2018]
 44%|██████████████████                         | 15/34 [00:02<00:03,  5.39it/s]
similar years for 2005 are [2010 2016 2020]
 47%|████████████████████                       | 16/34 [00:02<00:03,  5.33it/s]
similar years for 2006 are [2009 2017 2011]
 50%|█████████████████████                      | 17/34 [00:03<00:03,  5.35it/s]
similar years for 2007 are [2016 2010 2024]
 53%|██████████████████████                     | 18/34 [00:03<00:02,  5.38it/s]
similar years for 2008 are [2021 2011 2018]
 56%|████████████████████████                   | 19/34 [00:03<00:02,  5.35it/s]
similar years for 2009 are [2023 1997 2018]
similar years for 2010 are [1998 2007 2016]
 62%|██████████████████████████                 | 21/34 [00:03<00:02,  5.25it/s]
similar years for 2011 are [2008 2023 2021]
 65%|███████████████████████████                | 22/34 [00:04<00:02,  5.34it/s]
similar years for 2012 are [2023 2000 2011]
 68%|█████████████████████████████              | 23/34 [00:04<00:02,  5.39it/s]
similar years for 2013 are [2016 2020 2024]
 71%|██████████████████████████████             | 24/34 [00:04<00:01,  5.43it/s]
similar years for 2014 are [2023 2009 1991]
 74%|███████████████████████████████            | 25/34 [00:04<00:01,  5.44it/s]
similar years for 2015 are [2014 2023 1997]
 76%|████████████████████████████████           | 26/34 [00:04<00:01,  5.40it/s]
similar years for 2016 are [1998 1995 2007]
 79%|██████████████████████████████████         | 27/34 [00:05<00:01,  5.35it/s]
similar years for 2017 are [2001 2023 2011]
 82%|███████████████████████████████████        | 28/34 [00:05<00:01,  5.36it/s]
similar years for 2018 are [2021 2008 2011]
 85%|████████████████████████████████████       | 29/34 [00:05<00:00,  5.44it/s]
similar years for 2019 are [1998 1995 2023]
 88%|█████████████████████████████████████      | 30/34 [00:05<00:00,  5.47it/s]
similar years for 2020 are [1998 2024 2016]
 91%|███████████████████████████████████████    | 31/34 [00:05<00:00,  5.58it/s]
similar years for 2021 are [2008 1999 2018]
 94%|████████████████████████████████████████   | 32/34 [00:05<00:00,  5.64it/s]
similar years for 2022 are [2018 2010 2011]
 97%|█████████████████████████████████████████  | 33/34 [00:06<00:00,  5.67it/s]
similar years for 2023 are [1997 2012 2011]
100%|███████████████████████████████████████████| 34/34 [00:06<00:00,  5.43it/s]
```

```
similar years for 2024 are [1998 2020 2003]
```

Verification

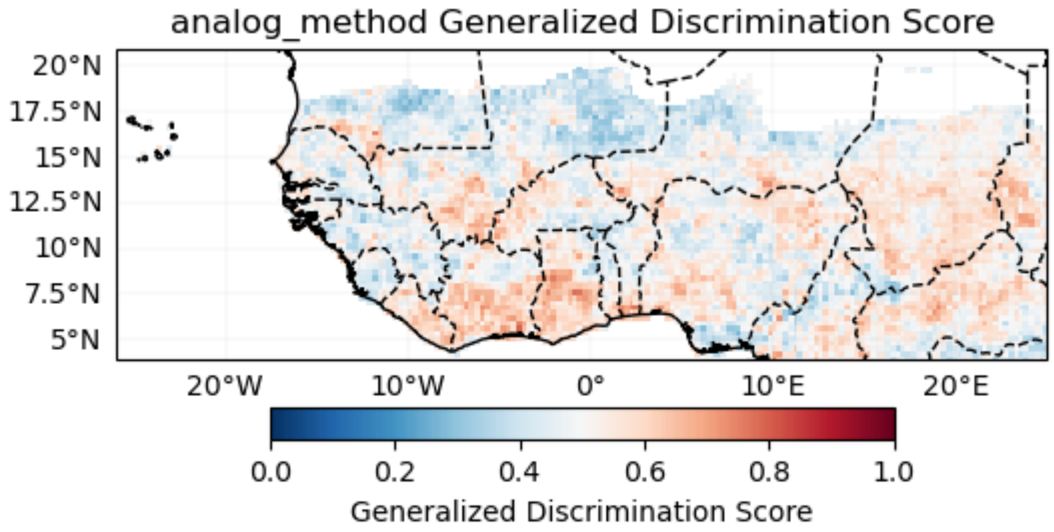
Deterministic





Probabilistic

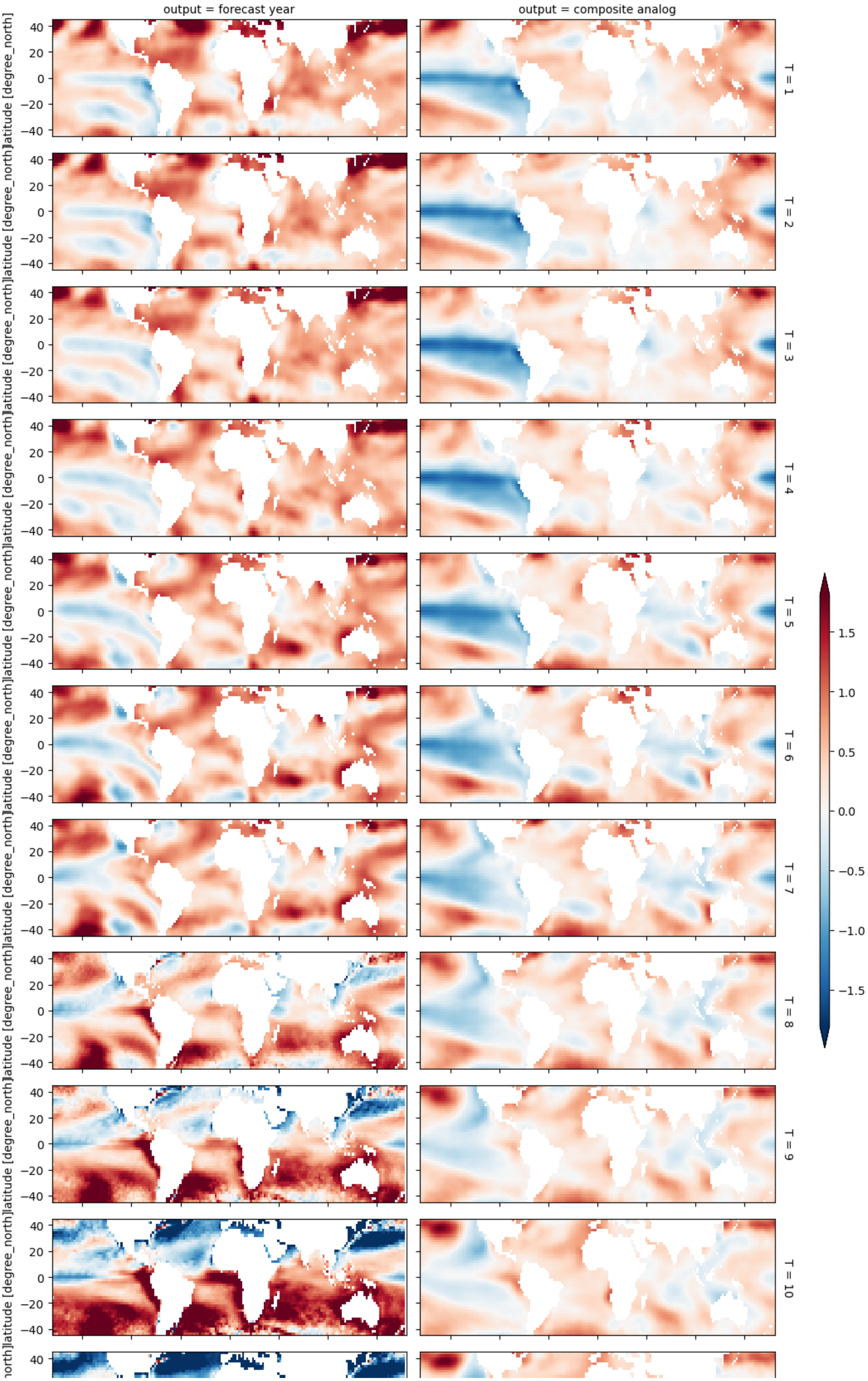
```
Generalized Discrimination Score
```

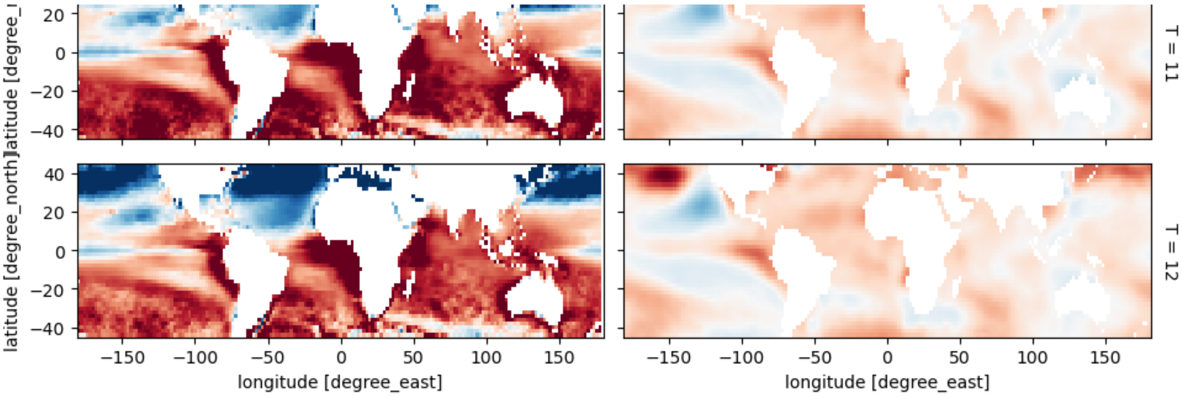


Forecast

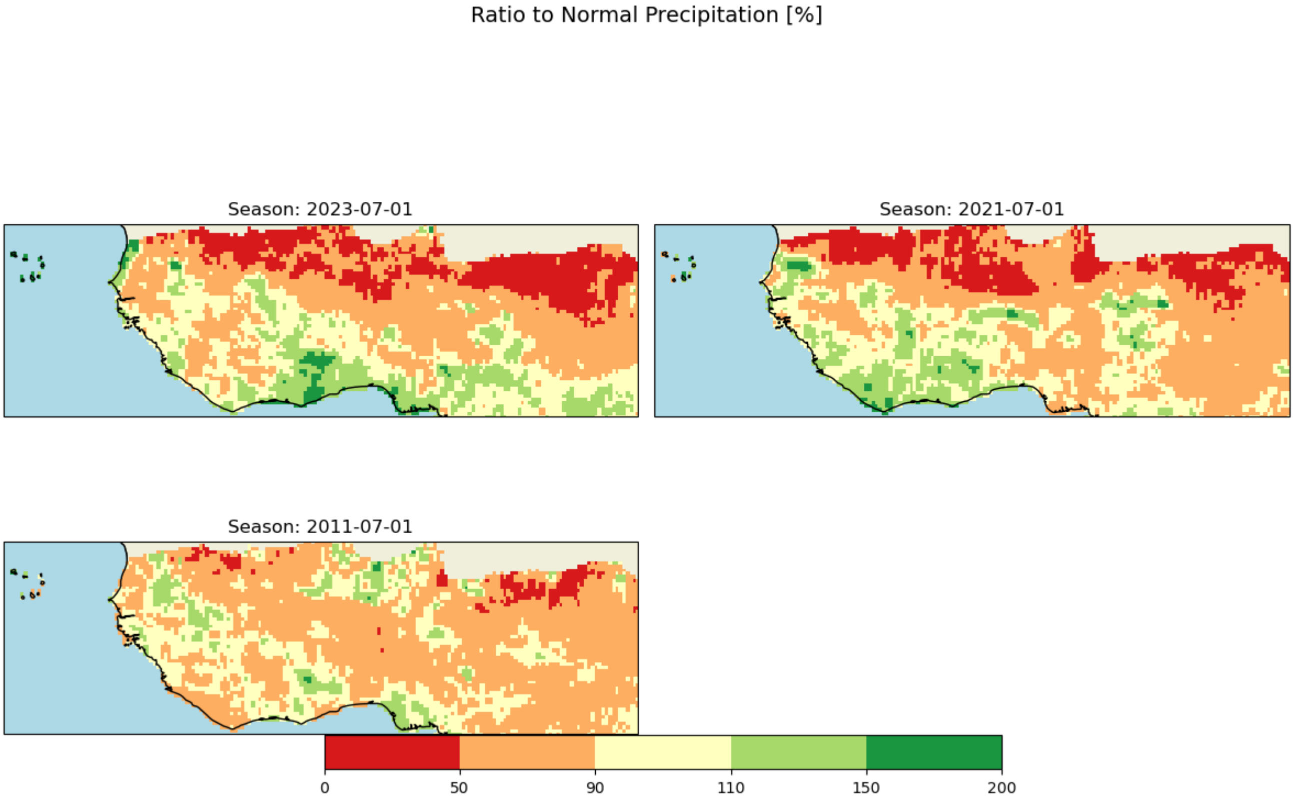
```
JJA_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc alrea
dy exists. Skipping download.
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization m
onth
JJA_2025/analog/forecast_ecmwf51_SST_MarIc_AprMayJunJulAug.nc
File 'JJA_2025/analog/forecast_ecmwf51_SST_MarIc_AprMayJunJulAug.nc' already exist
s. Skipping download.
All requested SST data has been processed.
```

```
array([2023, 2021, 2011])
```


output = forecast year
output = composite analog
latitude [degree_north]
T = 1
T = 2
T = 3
T = 4
T = 5
T = 6
T = 7
T = 8
T = 9
T = 10
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5

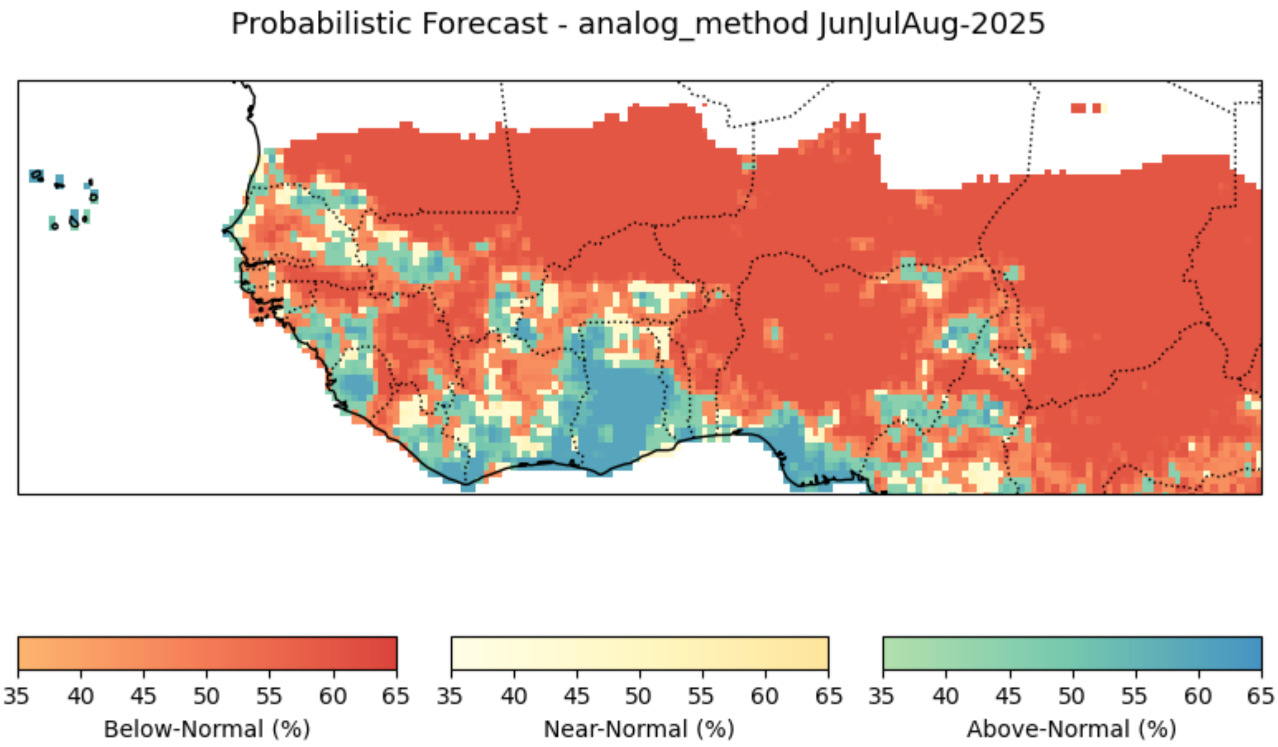



```
JJA_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc alrea
dy exists. Skipping download.
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization m
onth
JJA_2025/analog/forecast_ecmwf51_SST_MarIc_AprMayJunJulAug.nc
File 'JJA_2025/analog/forecast_ecmwf51_SST_MarIc_AprMayJunJulAug.nc' already exist
s. Skipping download.
All requested SST data has been processed.
```




```
array([2023, 2021, 2011])
```

```
JJA_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc alrea
dy exists. Skipping download.
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization m
onth
JJA_2025/analog/forecast_ecmwf51_SST_MarIc_AprMayJunJulAug.nc
File 'JJA_2025/analog/forecast_ecmwf51_SST_MarIc_AprMayJunJulAug.nc' already exist
s. Skipping download.
All requested SST data has been processed.
```

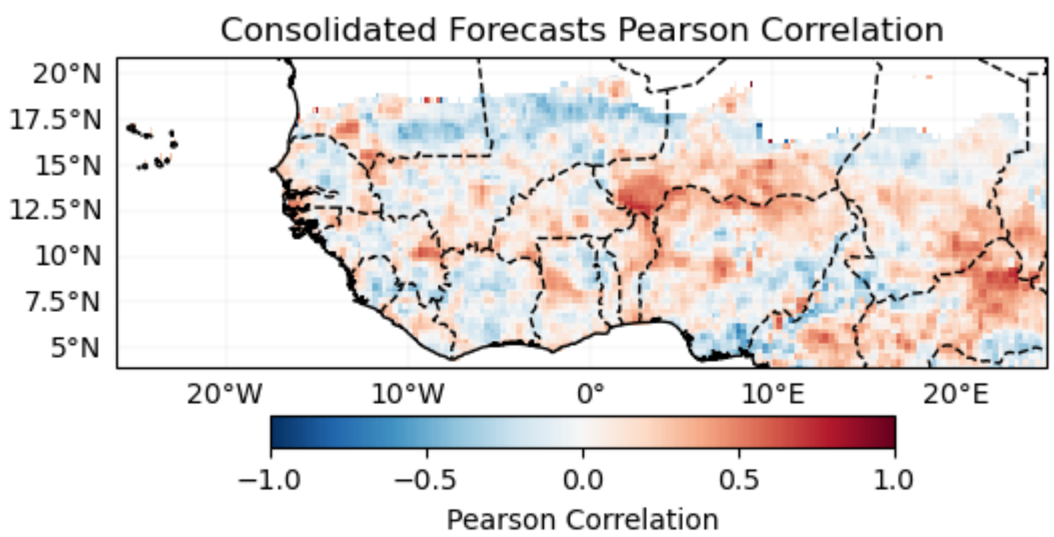


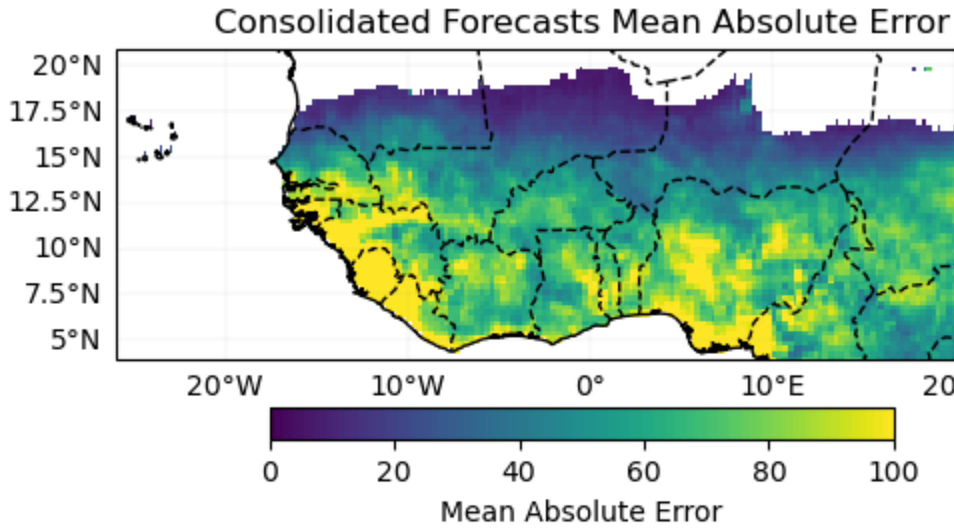
Consolidated Forecasts

Cross-validation

Verification

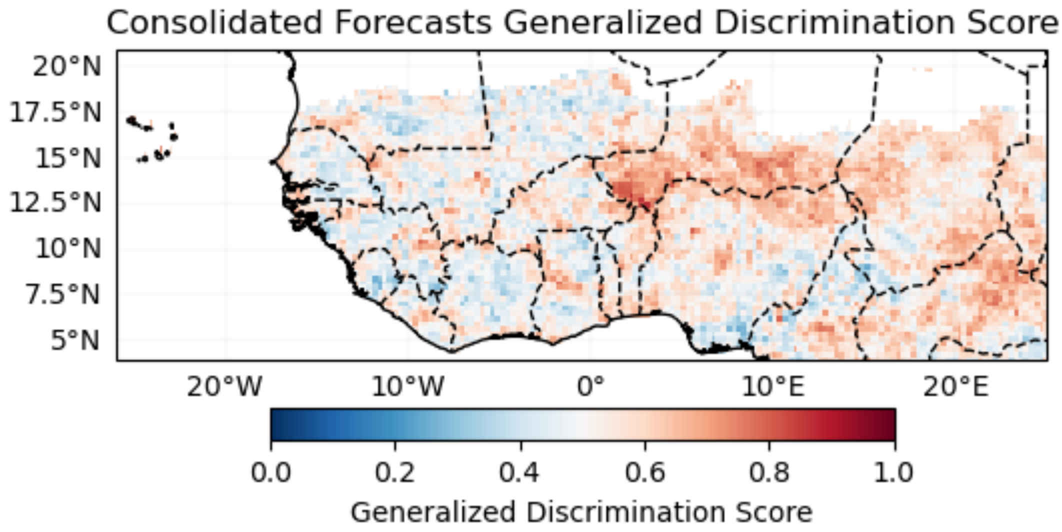
Deterministic



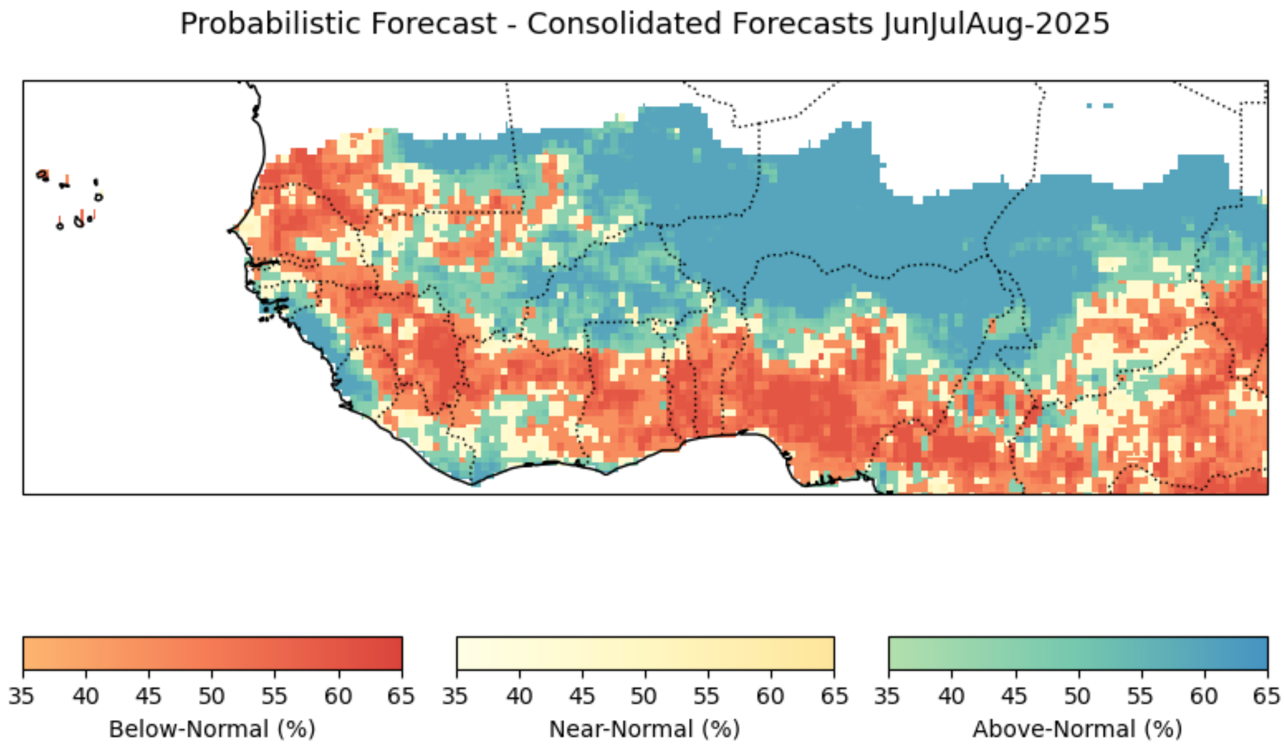


Probabilistic

```
Generalized Discrimination Score
```



Forecast




```
111498
```

Save Forecast for future evaluation

END -------------------------

```
[NbConvertApp] Converting notebook ./01_MME_JJA_2025.ipynb to pdf
[NbConvertApp] Support files will be in 01_MME_JJA_2025_files/
[NbConvertApp] Making directory ./01_MME_JJA_2025_files
[NbConvertApp] Writing 72863 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
```